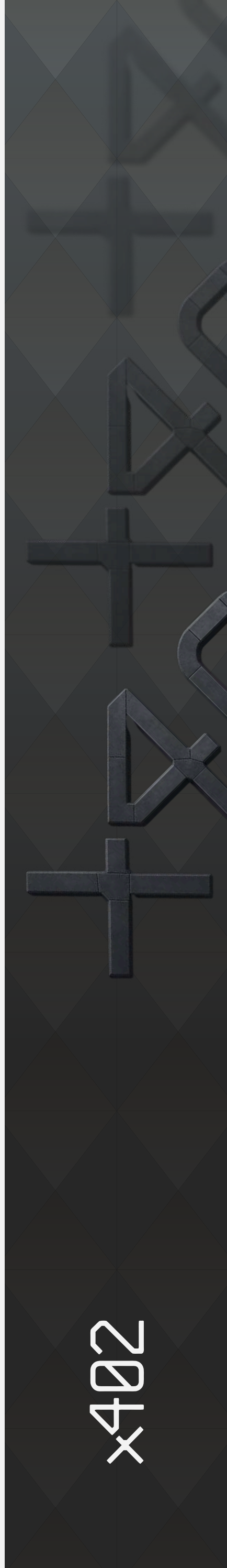
x402

All About x402

Explore the x402 protocol from its core architecture and extensions to the future of agentic commerce. This analysis addresses key market questions, identifies ecosystem value-capture layers, and examines how big tech is strategically responding. Contributions from an x402 co-author and key ecosystem participants offer firsthand perspectives on the opportunities and challenges ahead for x402 and agentic commerce.

Junhyeok Yoo
Four Pillars / Researcher



2026

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x402 is an open payment standard that allows software to check prices, authorize payments, and pay while requesting resources. Sellers present payment terms; buyers interpret them, pay through a supported method, and receive the resources they need. This report has examined what transactions become possible when payment becomes part of the web's request-and-response flow, and who can capture economic value in the process.

In addressing those questions, I deliberately avoided explaining x402's growth primarily through current transaction counts and payment volume. Without sufficiently distinguishing transactions designed to inflate metrics—Gamed Transactions—and test calls from actual service purchases, growing volume does not reveal how much comes from sustained demand. Before treating those metrics as evidence of adoption, I considered it necessary to examine the conditions under which substantive transactions can emerge.

That led to a deliberately ambitious set of scenarios. Some observed data serves as a reference for payment-size distributions and cost conversions, while future market size is estimated through assumptions about delegated payment authority and the adoption of payment networks. Connecting these assumptions to fund holding periods, settlement methods, and provider pricing structures makes it possible to examine how each participant's economic opportunity changes as transactions through x402 grow.

What drew me most strongly to this possibility was x402's native character. If an agent can understand a price and pay while calling a tool, using a resource and paying for it become parts of the same workflow. Software can handle procedures that previously required a person to create an account and register a payment method whenever a new tool was needed. This is where I saw the potential for a market in which agents procure external resources themselves.

Learning about 21.co's 2016 experiment while writing this report put that possibility into a longer historical perspective. The idea of pricing HTTP requests and letting software pay in Bitcoin already existed then. Ten years later, stablecoins and low-cost payment infrastructure have been joined by AI agents that actually discover and call external resources. As both payment methods and the actors using them have developed, I believe x402 has a greater opportunity to realize a vision that did not scale fully at the time.

For this possibility to become real commerce, however, new payment technology must accommodate participants' existing needs. This is why I consider card-network adoption of x402 important. Stablecoins are useful for allowing software to transfer funds directly, but buyers and companies may also need credit, refunds, dispute handling, spending controls, and clear accountability. Once these protections and operating requirements are considered, it is difficult to assume that every transaction will use stablecoins alone.

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I therefore see rail agnosticism as necessary to x402's expansion. Transactions suited to stablecoins should be able to use them; transactions needing card-network protections and operating systems should be able to connect the appropriate payment method. Letting participants choose a method that meets their conditions within a common interface also reflects the broader quality I found increasingly compelling while writing this report: flexibility.

That flexibility extends to service operations. The basic flow in the official documentation verifies payment authorization, performs the work, and returns the result after settlement. Servers can nevertheless adjust the flow to their service requirements. An expensive task, for example, can be implemented to begin only after verification and settlement have completed. The order of work and settlement can thus reflect the risk a service can bear while respecting common verification rules and the buyer's authorization scope.

The same principle applies to extensions. Service discovery, wallet authentication, duplicate prevention, and signed transaction records can be connected to the common payment flow as needed. A basic standard that does not complete every function might initially appear underdeveloped. I see it more as a canvas on which different services can implement their own requirements. An open-source foundation allows participants to build and share the functions they need, then develop them into forms others can use.

The infrastructure companies examined in §6 show how this flexibility can enter actual services. Cloudflare is designing connections between merchant charging and agent spending policies along the request path. AWS has included payment instruments and task-specific authority in its agent runtime. Google connects x402 to common purchase and approval procedures, while Fastly incorporates payment confirmation into edge access policies. Payments enter functions these companies already provide, extending the native character described above into their adoption strategies.

I am particularly optimistic about this top-down path through existing platforms. If platforms provide payment integrations and operating functions, companies and developers can price resources and equip agents to purchase them within the cloud and service environments they already use. Infrastructure providers also have an incentive to advance adoption: helping customers monetize content and purchase external resources expands the uses of their own services.

If open-source communities develop common specifications and capabilities while platforms connect them to existing customers' workflows, technical progress and actual adoption can reinforce each other. I see an opportunity for x402 to become an everyday tool for more companies and developers as these two paths converge.

Views will differ on the form agentic commerce takes and which payment methods or providers become central to it. But if agents are to accomplish more by using external resources, they must be able to understand those resources' prices and pay within their permitted scope.

I believe that as supplier-selection criteria and operating systems for verifying actual purchase outcomes develop together, users will be able to delegate a broader range of purchase decisions that reflect their tastes and goals to agents.

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The possibility I see in x402 is making this process a natural part of the web while leaving room for different trading conditions and services. The next question is which transactions recur on this common standard, and which services create sustained demand.

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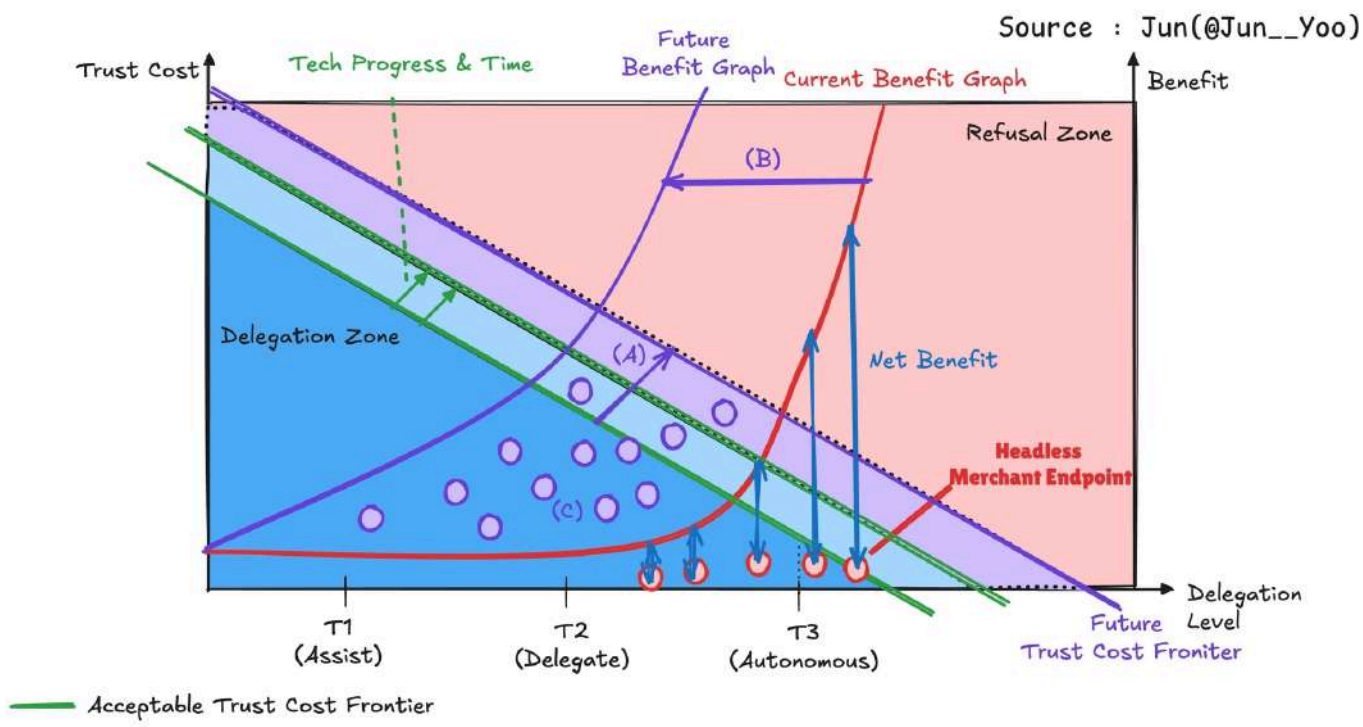
- x402 is an open payment interface that lets agents discover prices and pay within HTTP requests. It is not a new payment asset or blockchain, but a common specification connecting payment requirements, authorization, verification, and settlement.
- Although the idea behind HTTP 402 dates to the 1990s, stablecoins, low-cost blockchains, and AI agents converged in 2025 to make commercialization viable through x402. It has since evolved from a Coinbase product into a joint standardization initiative under the Linux Foundation.
- An x402 payment proceeds through a resource request, a 402 response containing payment requirements, a retry with signed authorization, verification and settlement, and resource delivery. **Separating transport, payment scheme, and network allows the same architecture to support HTTP, MCP, A2A, and multiple settlement rails.**
- x402 accommodates the different signing architectures of the EVM and SVM, with an extensible structure that can also support card networks over time. The facilitator is an optional layer that handles verification, fee sponsorship, and settlement operations rather than a mandatory participant.
- This report distinguishes **T1**, **T2**, and **T3** by the scope of purchase decisions and execution delegated to agents. In **T1**, a person approves the final purchase and payment. In **T2**, the agent selects and purchases products in line with the user's tastes and purchase conditions. In **T3**, the agent also decides which purchases are needed to achieve a goal. Purchases of APIs, data, and computing services are among the areas where this higher autonomy may first become useful.
- As T1 and limited T2 adoption advance, **an early T3 market is also likely to form around headless merchant endpoints selling low-value, high-frequency digital resources**. Payment and trust arrangements validated in this market can subsequently extend to broader agentic commerce.



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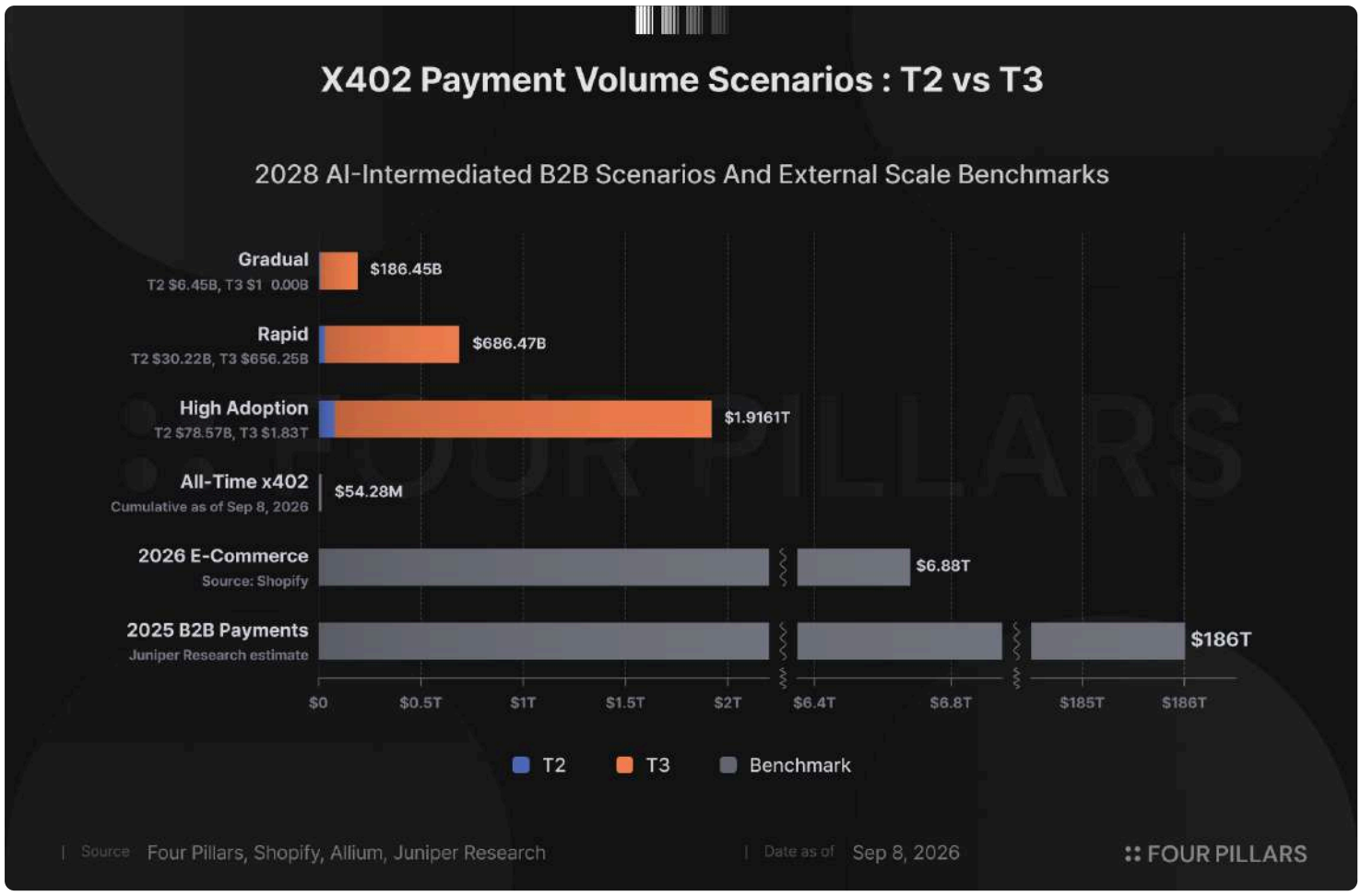
• The author's 2028 AI-intermediated B2B scenarios yield annual x402 payment volumes of **approximately $0.19T–$1.92T**, with **approximately $0.69T** under rapid adoption. Because the model assigns T3 higher shares of request-level payments and x402 payment volume than T2, T3 is calculated to account for approximately 96% of the combined volume in all three scenarios.



• The basis of value differs by layer even at the same payment volume. Issuers can derive value from incremental circulating balances, chains from settlement counts, facilitators from paid processing volume, wallets from funding and settlement routes, and discovery and routing services from supplier selection and order allocation.

• Neutral governance and open code support adoption but do not guarantee safe operations. Conformance testing and audit records must be standardized, with an operational structure in which wallets, merchants, and facilitators are accountable for their authority and the costs of failure.

• Cloudflare connects merchant pricing and agent spending policies, AWS embeds payment instruments and sessions in individual tasks, and Google links purchasing and authorization standards. Fastly connects external payment services to request policies. Agent payments are thus being introduced through the operational capabilities of existing web and cloud infrastructure.

• Open-source development and adoption through established platforms can reinforce each other. The community advances shared specifications and capabilities, while platforms integrate them into their existing customers' workflows. Evidence that this expansion translates into sustained adoption should come from services that generate repeat purchases and lasting demand.

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Chapter 00

Perspectives on x402

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Perspectives on x402

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API keys are an outdated human artifact. A person has to provision them, manage access, and pay the bill, while the language model using the API often has no visibility into what a tool costs or whether it is worth using. x402 changes that by making price and client-side access control part of the agent workflow itself. This improves the LLM experience by enabling agents to discover, reason about, and call external tools, taking actions on the web without human-in-loop configuration hurdles.

The x402 response puts pricing information directly into the agent's context window, so the model can discover an open tool, understand its cost, compare it against alternatives, and decide whether the expected value justifies paying for it.

That enables something fundamentally new: **economic reasoning inside the agent loop.** The agent can reason about marginal cost, operate within a budget, and use the same wallet to both earn and spend in a single runtime. In that sense, x402 turns the open web into a market of discoverable, machine-payable tools.

The x402 community sees alignment as an economic problem. If an agent can see prices, understand resource constraints, and be held to a budget, economic incentives become part of its decision-making. An agent that is economically aligned is, in an important sense, more generally aligned with the goals and constraints of the person or organization it represents.

Kevin Leffew

x402 co-author, Coinbase

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The broader adoption of AI and agentic coding means that nearly every individual can be a developer. But, writing code is only one part of the journey. Deploying it is what will lead to an explosion of new products and services. But this requires provisioning cloud infrastructure which still moves at the speed of human procurement. Connecting open standards like AP2 and x402 with programmable money moves us one step closer to a future where every developer can take ideas from code to cloud at the speed of the model itself.

The shift toward agentic commerce requires cloud infrastructure that is as open as the protocols it supports. x402 is part of the open protocol stack that is creating a common standard for buyers and sellers to enable machine-to-machine payments.

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Richard Widmann

Global Head of Strategy, Web3 and Digital Assets, Google

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On x402

The web has never had a native way for one program to pay another. Every payment rail we use today assumes a human initiates the transaction. That assumption held for fifty years. It breaks the moment the buyer is an autonomous agent transacting at machine speed and frequency.

x402 fixes this at the right layer. It repurposes HTTP's dormant 402 Payment Required status code into a programmable instruction. An agent requests a resource, the server responds with a price, the agent pays in stablecoins onchain, and the resource is delivered. All inside an ordinary web request. No accounts, no API keys, no checkout flow. Settlement in under a second for a fraction of a cent. This unlocks transaction patterns that were never economically viable on card rails: pay-per-API-call, pay-per-crawl, metered compute, agents paying agents with no human in the loop.

The governance matters as much as the protocol. Open standards ultimately win the long term and x402 Foundation powered by the Linux Foundation is an important step to maintain neutrality and enable support for many organizations. That's why Monad Foundation joined as an active contributor. The organizations shaping this standard now will define how value moves between machines for the next decade.

But the vision goes beyond payments. Speed and low fees are table stakes. The real question is what happens to value after it settles. Agents won't just move dollars between wallets. They'll hold balances, earn on idle capital between tasks, swap currencies to pay merchants globally, and pledge collateral against obligations. Payments arise from liquidity, not the other way around. The endgame for x402 isn't a faster checkout. It's the payment layer of a full machine economy, where every web resource is priced, every agent is a market participant, and money is as programmable as the software spending it.

We've seen this pattern before. When the card left the point of sale in the 1990s, it triggered a 25-year chain reaction across every layer of payments. Removing the human entirely is a bigger shift, and it's compressing that timeline into a few years. x402 is the first piece of infrastructure purpose-built for what comes next.

Raj Parekh

Head of Payments and AI, Monad Foundation

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Your next million customers won’t have a face. Put the API behind x402, any agent with a wallet can call it, and you never issue an API key again. That’s Open Agentic Commerce. You implement the standard once. You don’t ask anyone for permission.

shafu

Agentic Commerce, Circle

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Chapter 01

Why HTTP-native payments matter again

1.1 1992: The 402 code built into HTTP

1.2 2016: The machine-payable web proposed by 21.co

1.3 2025: Coinbase announces the x402 protocol

1.4 2026: From a Coinbase product to an industry standardization initiative

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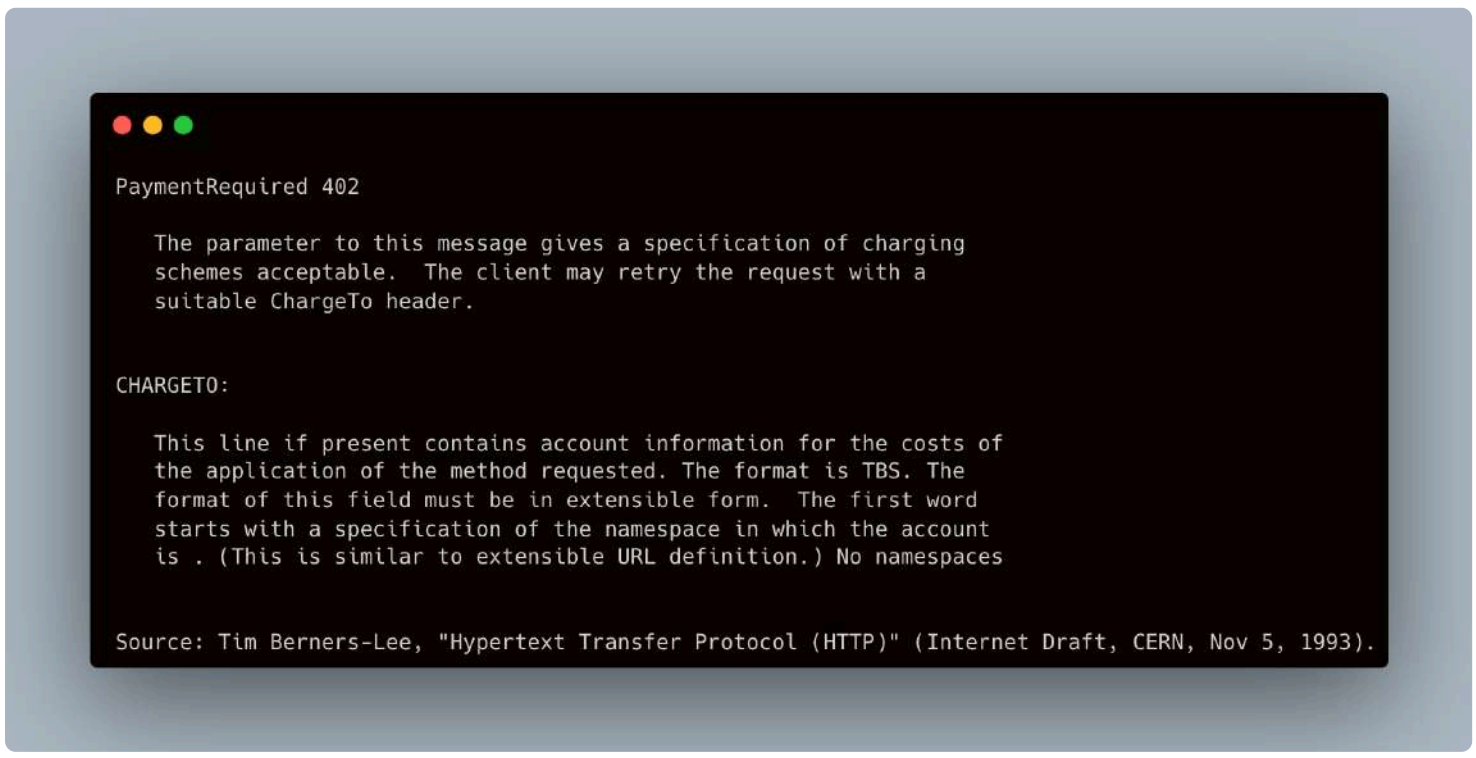
1. Why HTTP-native payments matter again

1.1 1992: The 402 code built into HTTP

HTTP 402 Payment Required first appeared in the 1992 HTTP documentation preserved by the W3C. It described a flow in which the server presents payment requirements and the client retries with account information.



The early documentation did more than name the code: it specified a mechanism in which the server presents acceptable payment methods and the client resends its request with account information in the ChargeTo header. Although the detailed specification was not finalized, the intention to express both a payment-required state and the subsequent action at the HTTP level was already clear.

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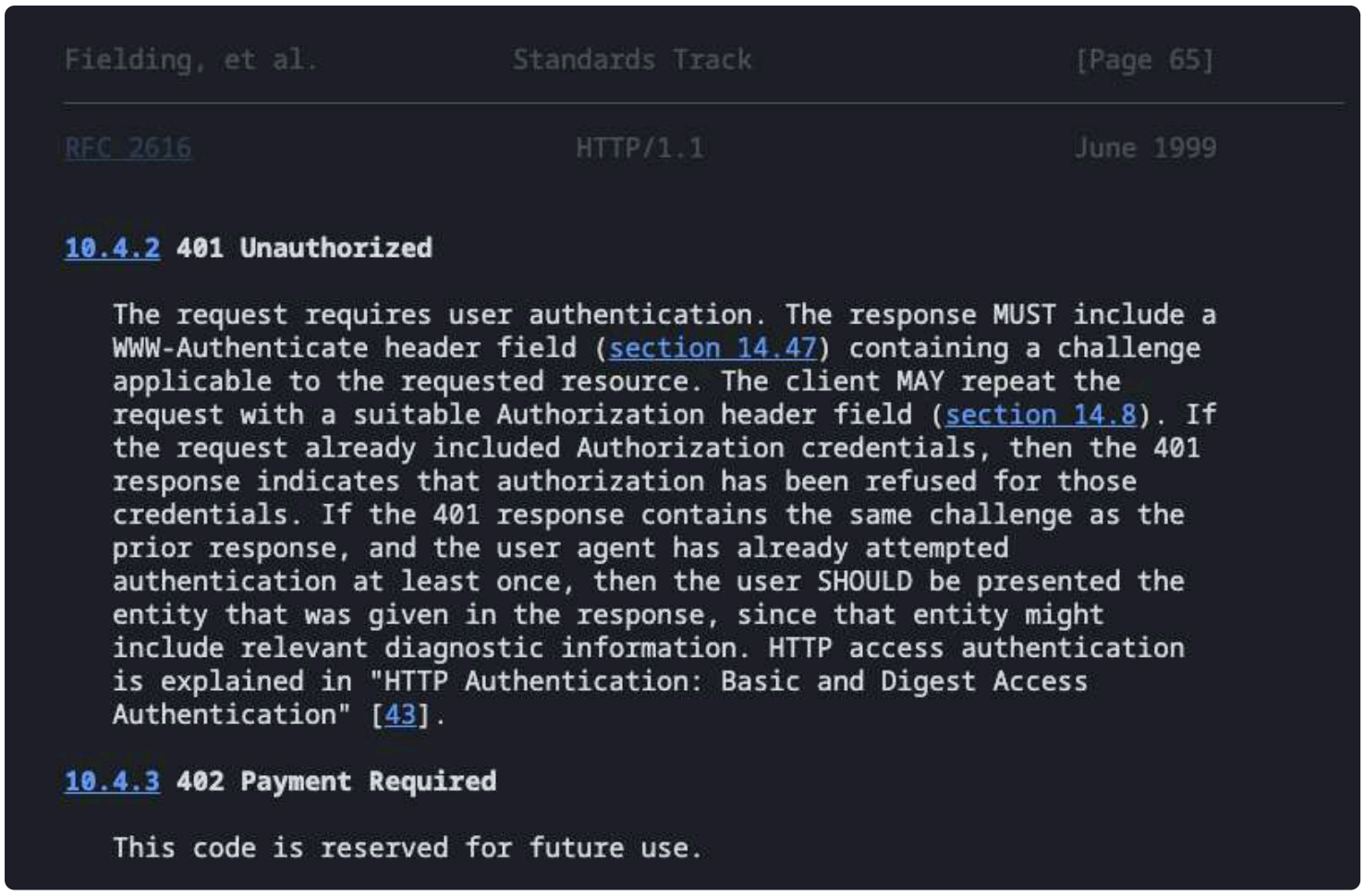

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Fielding, et al.          Standards Track                [Page 65]

RFC 2616                        HTTP/1.1                 June 1999


10.4.2 401 Unauthorized

   The request requires user authentication. The response MUST include a
   WWW-Authenticate header field (section 14.47) containing a challenge
   applicable to the requested resource. The client MAY repeat the
   request with a suitable Authorization header field (section 14.8). If
   the request already included Authorization credentials, then the 401
   response indicates that authorization has been refused for those
   credentials. If the 401 response contains the same challenge as the
   prior response, and the user agent has already attempted
   authentication at least once, then the user SHOULD be presented the
   entity that was given in the response, since that entity might
   include relevant diagnostic information. HTTP access authentication
   is explained in "HTTP Authentication: Basic and Digest Access
   Authentication" [43].

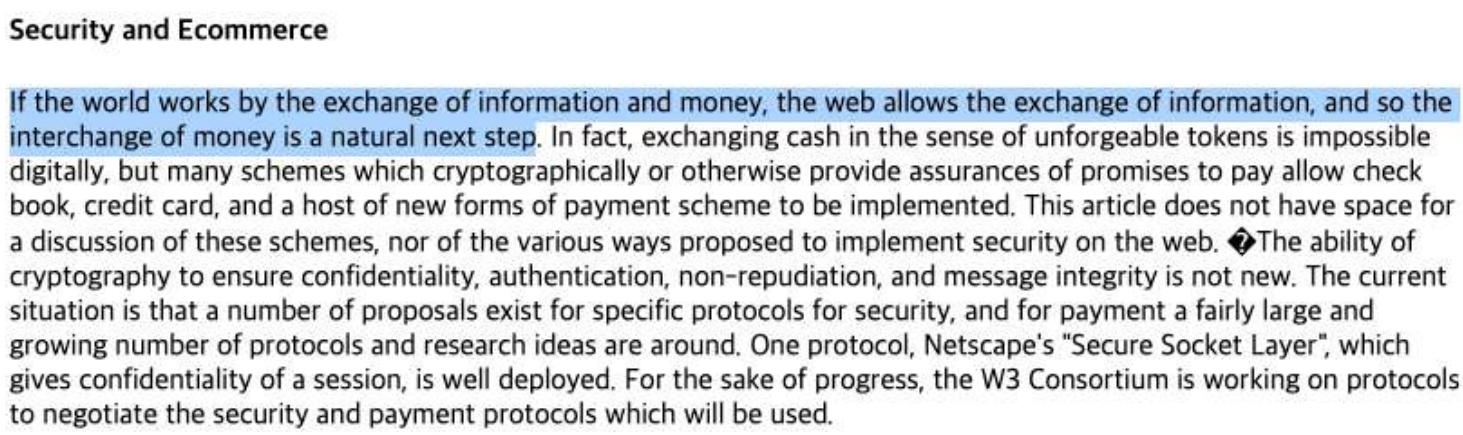
10.4.3 402 Payment Required

   This code is reserved for future use.
```

Source: W3C, Status codes in HTTP (1992)

RFC 9110 defines 402 as a status code reserved for future use. x402 uses it to establish an open specification connecting price discovery, payment authorization, and retries.

In a 1996 essay, World Wide Web creator Tim Berners-Lee argued that, if the world runs on exchanges of information and money, bringing monetary exchange to the web is a natural next step. He also advocated a structure in which the web acts as a neutral carrier for payments rather than mandating a particular payment protocol.



Security and Ecommerce

If the world works by the exchange of information and money, the web allows the exchange of information, and so the interchange of money is a natural next step. In fact, exchanging cash in the sense of unforgeable tokens is impossible digitally, but many schemes which cryptographically or otherwise provide assurances of promises to pay allow check book, credit card, and a host of new forms of payment scheme to be implemented. This article does not have space for a discussion of these schemes, nor of the various ways proposed to implement security on the web. The ability of cryptography to ensure confidentiality, authentication, non-repudiation, and message integrity is not new. The current situation is that a number of proposals exist for specific protocols for security, and for payment a fairly large and growing number of protocols and research ideas are around. One protocol, Netscape's "Secure Socket Layer", which gives confidentiality of a session, is well deployed. For the sake of progress, the W3 Consortium is working on protocols to negotiate the security and payment protocols which will be used.

Source: The World Wide Web: Past, Present and Future (1996)

Roughly three decades later, most web payments still depend on accounts, authentication, and billing outside HTTP. Card and bank-account payments work on the web, but checkout, registration, authorization, and settlement function as external add-ons, separate from individual requests.

Experiments connecting payments to HTTP requests nevertheless predate x402. Although the idea may seem characteristic of the mid-2020s rise of AI agents, 21.co was already pursuing it in 2016. The next section examines the operating conditions of that period.

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1.2 2016: The machine-payable web proposed by 21.co

Source: Medium(Earn.com)

In 2016, 21.co released two1-python, an open-source Python package implementing Bitcoin payments over HTTP. The company later renamed itself Earn.com.

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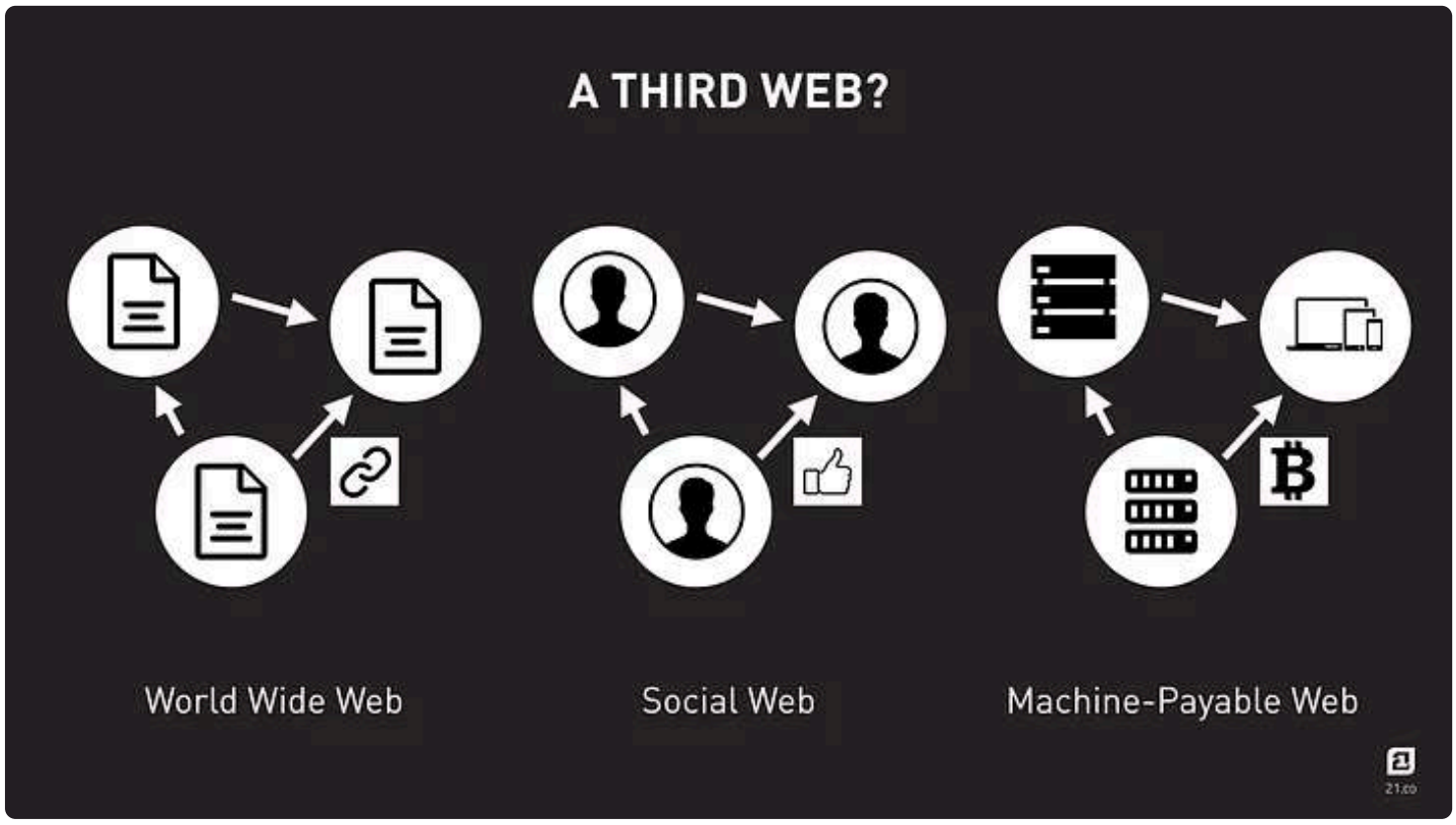
Source: 21.co GitHub

two1-python used the 402 code to initiate payment negotiation. A server wrapped a route with the required(price) decorator and returned a 402 response containing the price and payment method when a request lacked payment information. The client constructed a payment in its wallet and retried with proof attached; on success, the server returned the resource with 200 OK. Payment was embedded in the HTTP request cycle itself. Developers could use the package to integrate Bitcoin payments into applications and charge per API call.



Source: Medium(Earn.com)

21.co called this the “Machine-Payable Web,” a machine-led successor to the World Wide Web and the social web. It envisioned software holding and transferring assets with private keys and purchasing digital goods per call. Credit cards and PayPal assumed human accounts and identity checks, whereas Bitcoin allowed software to execute programmable payments with private keys and no human intervention. This was why Bitcoin was selected.

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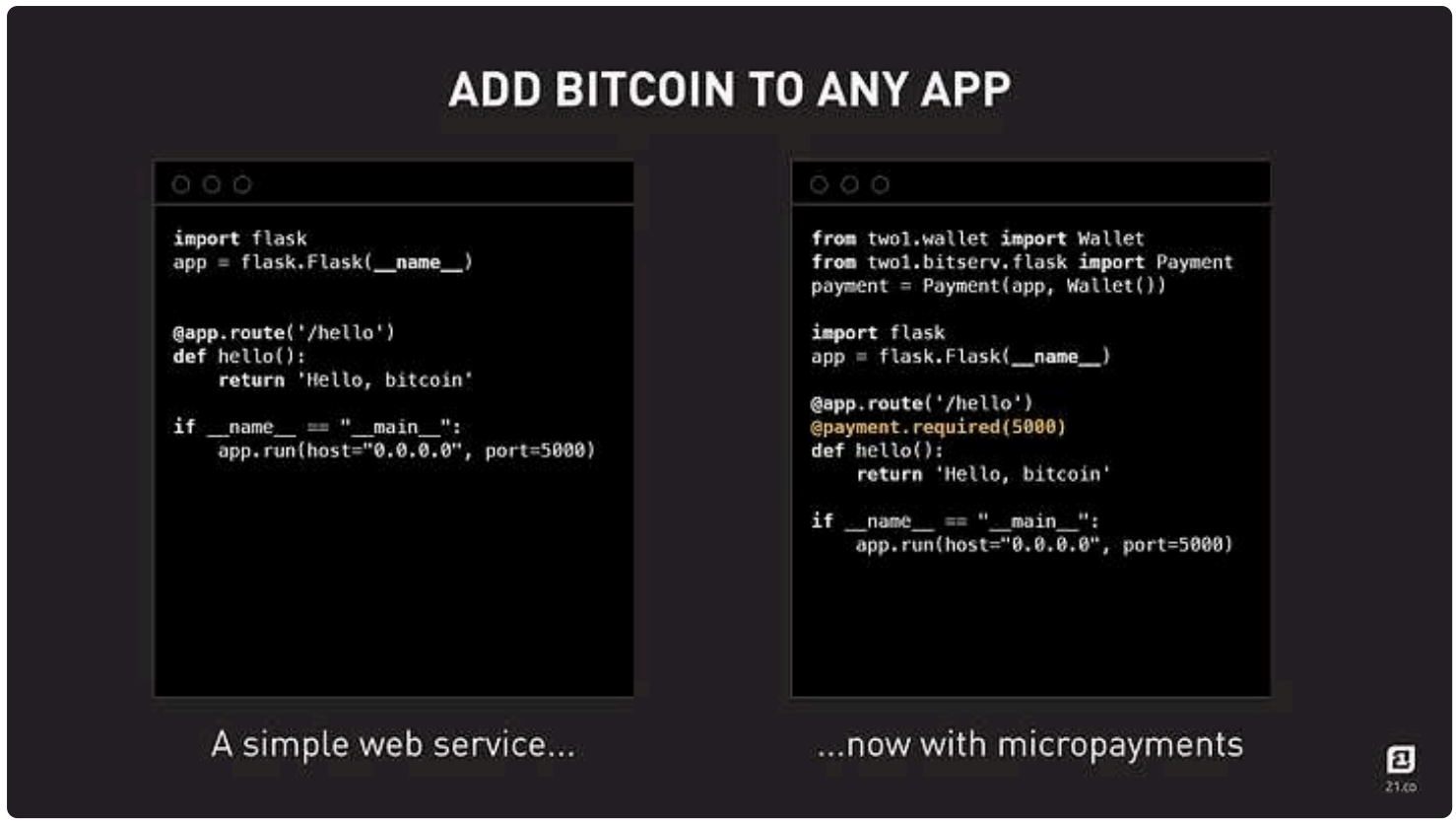
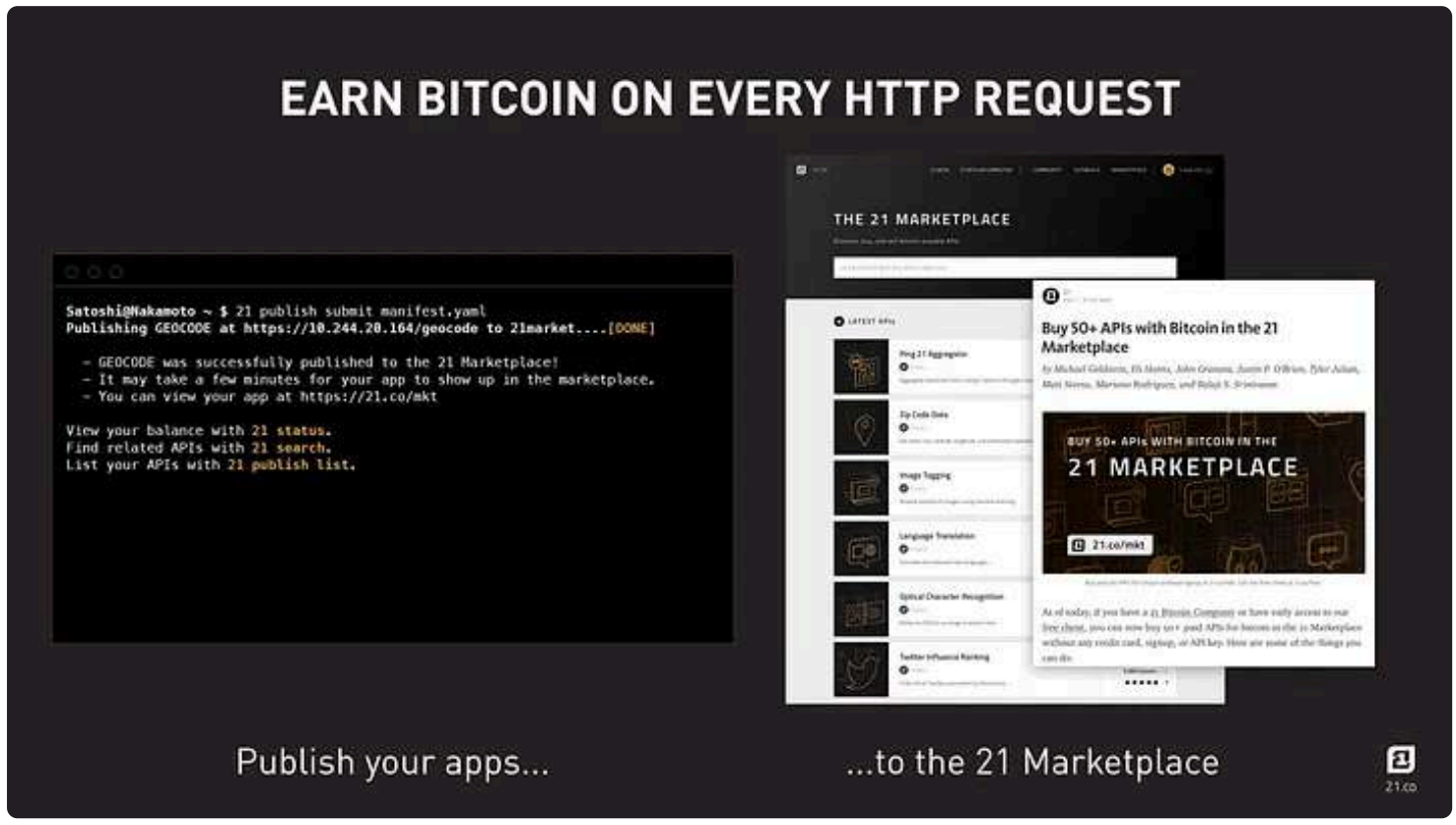


Source: Medium(Earn.com)

The most significant aspect of this Bitcoin experiment was its combination of prices and payment proofs within HTTP requests, an approach that anticipated x402. The 21 library connected Bitcoin to Linux, Python, Django, Flask, and the HTTP stack to support micropayments.



Source: Medium(Earn.com)

To develop HTTP-based monetization further, the company built a programmatically accessible micropayment marketplace. Developers could register API endpoints and earn and monitor revenue in real time.

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The “Machine-Payable Web” name emphasized an early machine-to-machine network in which digital goods were exchanged directly for digital currency, extending beyond social networks organized around likes and bookmarks. Early paid API marketplaces for weather and financial data emerged on this basis.

Commercialization faced two major constraints.

First, onchain fees and Bitcoin price volatility made micropayment operations difficult. 21.co did support payment channels: users deposited Bitcoin onchain to open a channel, updated jointly signed states offchain for individual API payments, and settled only the final result on mainnet when the channel closed. Not every API call therefore incurred a mainnet transaction fee, but opening, funding, and closing channels still required management. On the demand side, there were also too few use cases in which software independently discovered suppliers and purchased resources.

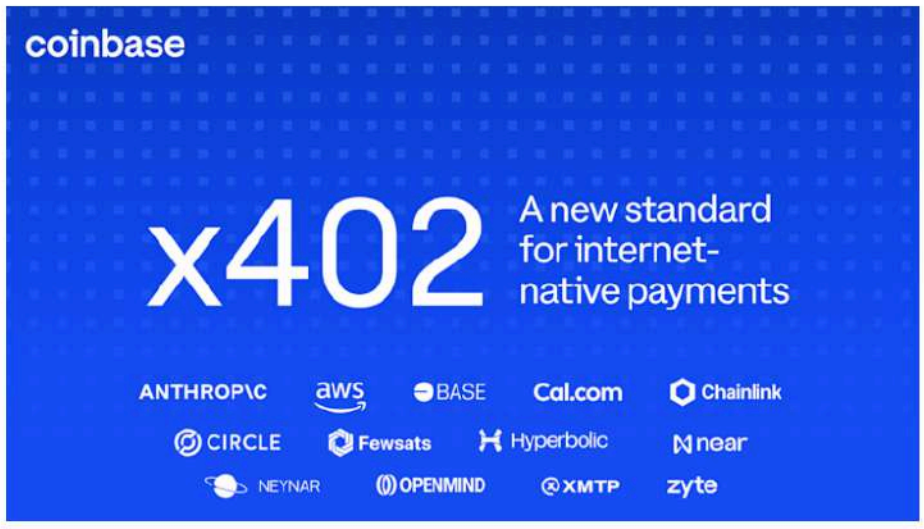
For these reasons, 21.co pivoted to Earn.com, which Coinbase acquired in 2018. The concept of combining prices and payment proofs with HTTP requests nevertheless continued in Coinbase's x402.

1.3 2025: Coinbase announces the x402 protocol

Introducing x402: a new standard for internet-native payments

May 7, 2025

By: Erik Reppel, Nemil Dalal, Dan Kim



TL;DR

Why x402 Matters

How x402 Works

What Developers Can Build

What AI Agents Can Unlock

Who’s Building With x402

Start Building Today

TL;DR: *Coinbase is launching x402, a payment protocol that enables instant stablecoin payments directly over HTTP. It allows APIs, apps, and AI agents to transact seamlessly, unlocking a faster, automated internet economy.*

The internet economy has always struggled with payments.

Create account
It's free and takes 2 minutes or less.
Get started

Source: Coinbase

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On May 7, 2025, eight years after 21.co's experiment ended, Coinbase announced the x402 payment protocol. Whereas the earlier experiment priced HTTP requests in Bitcoin, x402 redesigned the same idea around stablecoins, low-cost blockchains, and external payment-processing infrastructure.

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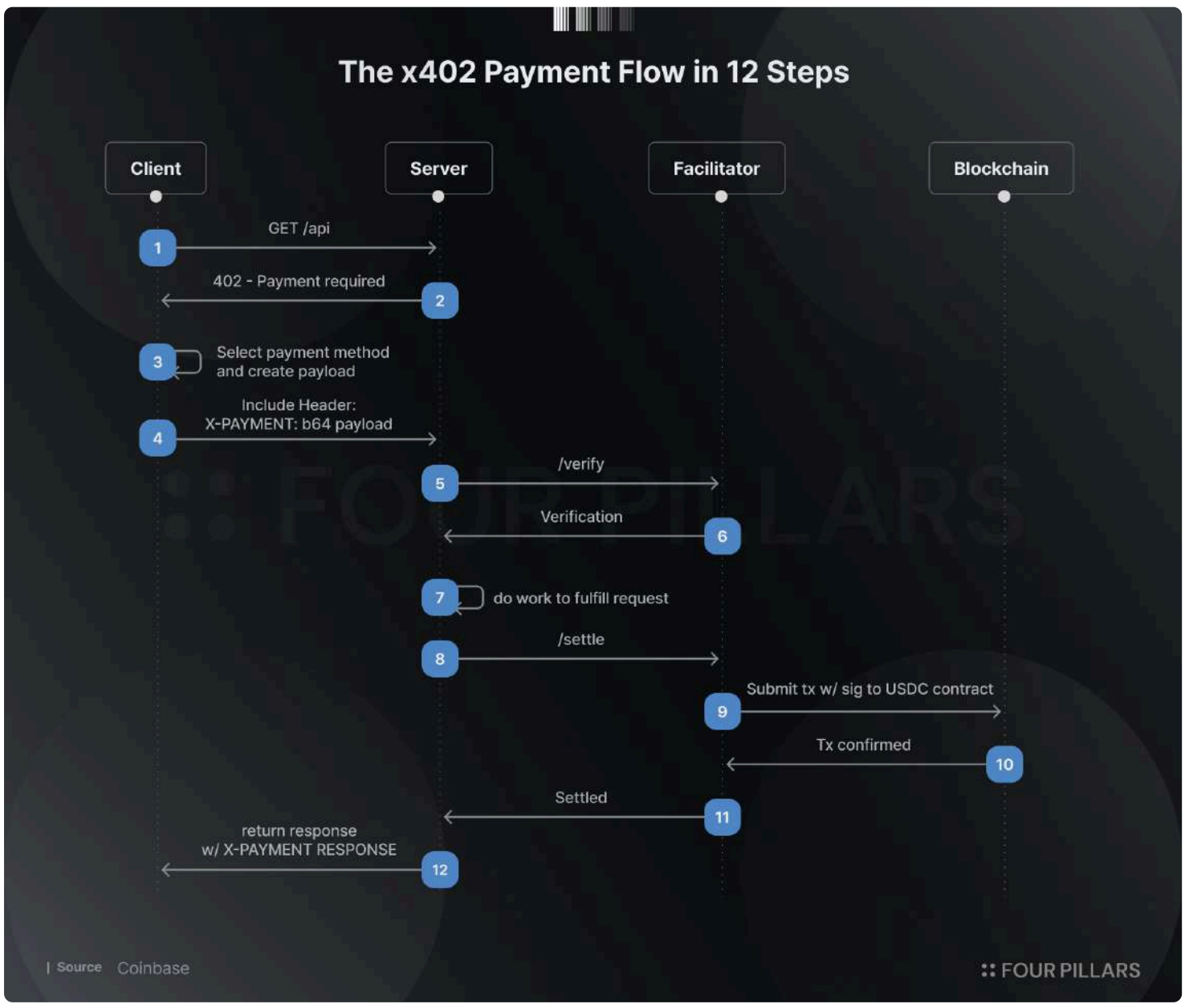

Source: Coinbase

x402 uses HTTP 402 Payment Required to present prices and payment terms, then verifies the authorization attached by the client to provide paid resources. It brings procedures that previously required account creation and API-key issuance into the HTTP request-response flow. §2.1 explains the detailed sequence of signing, verification, and settlement.

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> At Coinbase, we're addressing exactly this challenge by introducing x402: an open standard that leverages the original HTTP "402 Payment Required" status code to embed stablecoin payments directly into web interactions. This protocol draws inspiration from Balaji and team's work on crypto micropayments with http://21.co many years ago. At the time, you could only achieve micropayments with Bitcoin payment channels, which required expensive setup/teardown. But with modern L2s like Base, onchain fees have dropped to 1 cent, so many of the applications they prototyped are becoming possible.

Source: Coinbase

Coinbase has acknowledged direct inspiration from 21.co's Bitcoin micropayment experiment. x402 inherits the basic flow 21.co sought to implement: request → payment requirement → retry with payment proof → resource delivery. The payment asset, wallet and verification infrastructure, machine buyers, and distribution paths differ as follows.

From 21.co to x402:
What Changed for Machine Payments

2016 21.co model	Constraints at the time	Structural changes in the 2025 ×402 model
HTTP micropayment middleware for Django and Flask	Adopters had to open, fund, monitor, and close Bitcoin payment channels themselves.	Low-cost blockchains and facilitator-based verification reduce the operational burden of per-request payments.
Hierarchical deterministic (HD) wallets and payment-channel clients and servers	Developers managed keys, nodes, liquidity, fee balances, and settlement failures.	Managed wallets, policy engines, gas sponsorship, and cloud infrastructure abstract parts of the payment stack.
Machine-payable endpoint marketplace	Demand depended on people, and software struggled to discover, evaluate, and purchase resources independently.	Foundation-model tool use, MCP, and A2A provide the technical basis for autonomous software buyers.
Bitcoin-denominated per-request prices	Volatility and variable fees destabilized small quotes, budgeting, and accounting.	Dollar stablecoins support predictable prices and machine-readable budgets.
Standalone two-sided marketplace	A new platform had to attract buyers and sellers simultaneously.	The API economy, cloud services, edge gateways, SDKs, and tool registries provide existing distribution channels.

Source Four Pillars

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Over those nine years, four conditions for web-based machine payments changed together. Low-cost chains and facilitators reduced settlement overhead, dollar stablecoins stabilized pricing, foundation models with MCP and A2A supported autonomous purchasing, and existing APIs and cloud infrastructure provided distribution. In this changed environment, Coinbase began building an open standard for use across payment networks and businesses.

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1.4 2026: From a Coinbase product to an industry standardization initiative

On April 2, 2026, the Linux Foundation announced the x402 Foundation alongside plans to contribute Coinbase's x402 protocol. The initiative began moving a Coinbase-led protocol toward an industry standard managed through nonprofit and community governance. Its stated goal was a universal payment standard allowing AI agents, APIs, and applications to exchange value as they exchange data.

At the July 14, 2026 operational launch, at least 40 organizations were participating. The membership list compiled for this report as of September 7, 2026 contains 51 organizations, divided into Premier, General, and Associate members.

Premier Members

adyen | aws | AMERICAN EXPRESS | CIRCLE

CLOUDFLARE | coinbase | fiserv. | Google

mastercard | MONAD FOUNDATION | MoonPay | ripple

shopify | SOLANA FOUNDATION | Stellar | stripe

VISA

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- **17 Premier members:** Adyen, AWS, American Express, Circle, Cloudflare, Coinbase, Fiserv, Google, Mastercard, Monad Foundation, MoonPay, Ripple, Shopify, Solana Foundation, Stellar Development Foundation, Stripe, Visa

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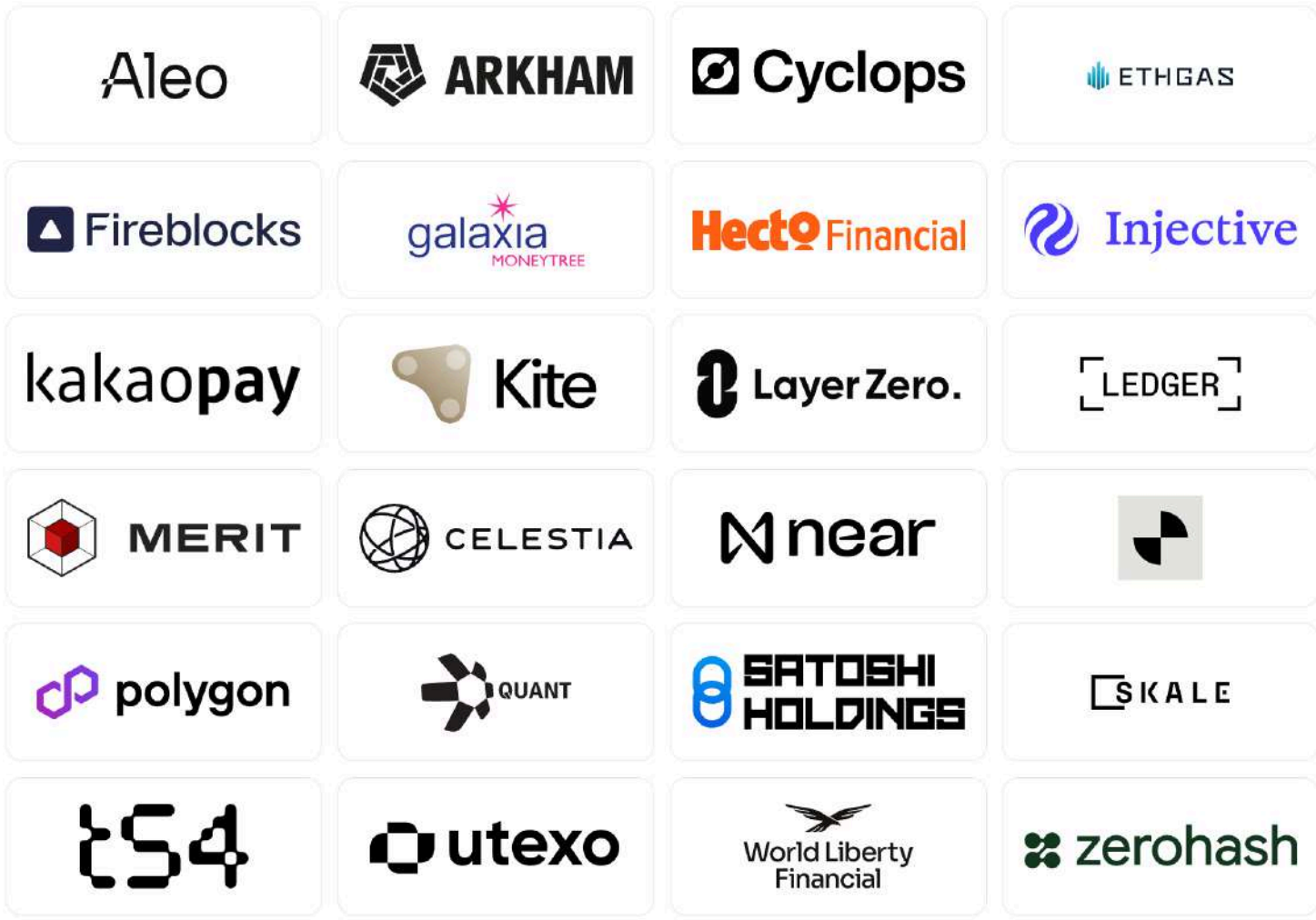
General Members



- **24 General members:** Aleo, Arkham, Fireblocks, Galaxia Moneytree, Hecto Financial, Injective, KakaoPay, Kite AI, LayerZero Labs, Merit Systems, NEAR Foundation, Orthogonal, Polygon Labs, Quant Network, SKALE, t54 labs, utexo, World Liberty Financial, zerohash, Cyclops, ETHGAS, LEDGER, Celestia, Satoshi Holdings

Associate Members

- **10 Associate members:** BSV Association, Cardano Foundation, Casper, Japan Contents Blockchain Initiative (JCBI), OMA3, PPPA, INFTF, Concordium, Hedera, MAG

This membership makes clear that x402 is not intended to remain limited to one payment asset or blockchain. Card networks, payment providers, cloud and commerce businesses, stablecoin issuers, and blockchain foundations participate. The significance lies in governing a shared specification through which different rails can use the same x402 payment interface.

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Foundation membership alone does not guarantee product integration or transaction volume. Even so, the presence of major financial, cloud, and blockchain organizations within one governance structure shows how x402's position has changed. What Coinbase proposed as an individual company in 2025 has become a joint standardization initiative in 2026, with participants establishing common rules for connecting their payment networks. The next chapter examines how those rules operate within HTTP requests.

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Chapter 02

How x402 works: Processing payments within HTTP requests

2.1 Payments proceed through request, authorization, verification, and settlement

2.2 Transport, scheme, and network define a payment flow

2.3 Extensions: Adding operating functions outside the payment core

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2. How x402 works: Processing payments within HTTP requests

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x402 specifies how a server's payment requirements, a client's authorization, and verification and settlement results are exchanged within the same request-response flow. This chapter covers the basic payment flow using an external facilitator, combinations of transports and payment schemes, and optional extensions.

We first examine the mechanism that takes a single payment from request to settlement (§2.1). We then consider how that flow travels through HTTP, MCP, and A2A, and which rules control the amount and timing of settlement (§2.2). Finally, we examine how extensions add operational capabilities such as discovery, authentication, and retries outside the core protocol (§2.3).

2.1 Payments proceed through request, authorization, verification, and settlement

In the x402 v2 HTTP transport, payment objects are Base64-encoded and exchanged through three headers:

The Three HTTP Headers in x402 v2

Header	Direction	Role
PAYMENT-REQUIRED	Server → Client	Returned with the 402 response; contains the PaymentRequired object.
PAYMENT-SIGNATURE	Client → Server	Attached to the retried request; contains signed payment authorization in PaymentPayload.
PAYMENT-RESPONSE	Server → Client	Returned with the final response; contains verification and settlement results in SettlementResponse.

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> v1 uses X-PAYMENT and X-PAYMENT-RESPONSE, with payment requirements in the response body. If those names appear, check whether the implementation is v1. v1 is not inherently an error; the important point is not to mix objects and headers from different versions in one flow.

Execution depends on whether the server handles payments itself or delegates to a facilitator, and which scheme and network it selects. Following the official repository's typical flow, this section uses the x402 v2 HTTP transport, an external facilitator, and the exact scheme with EIP-3009.

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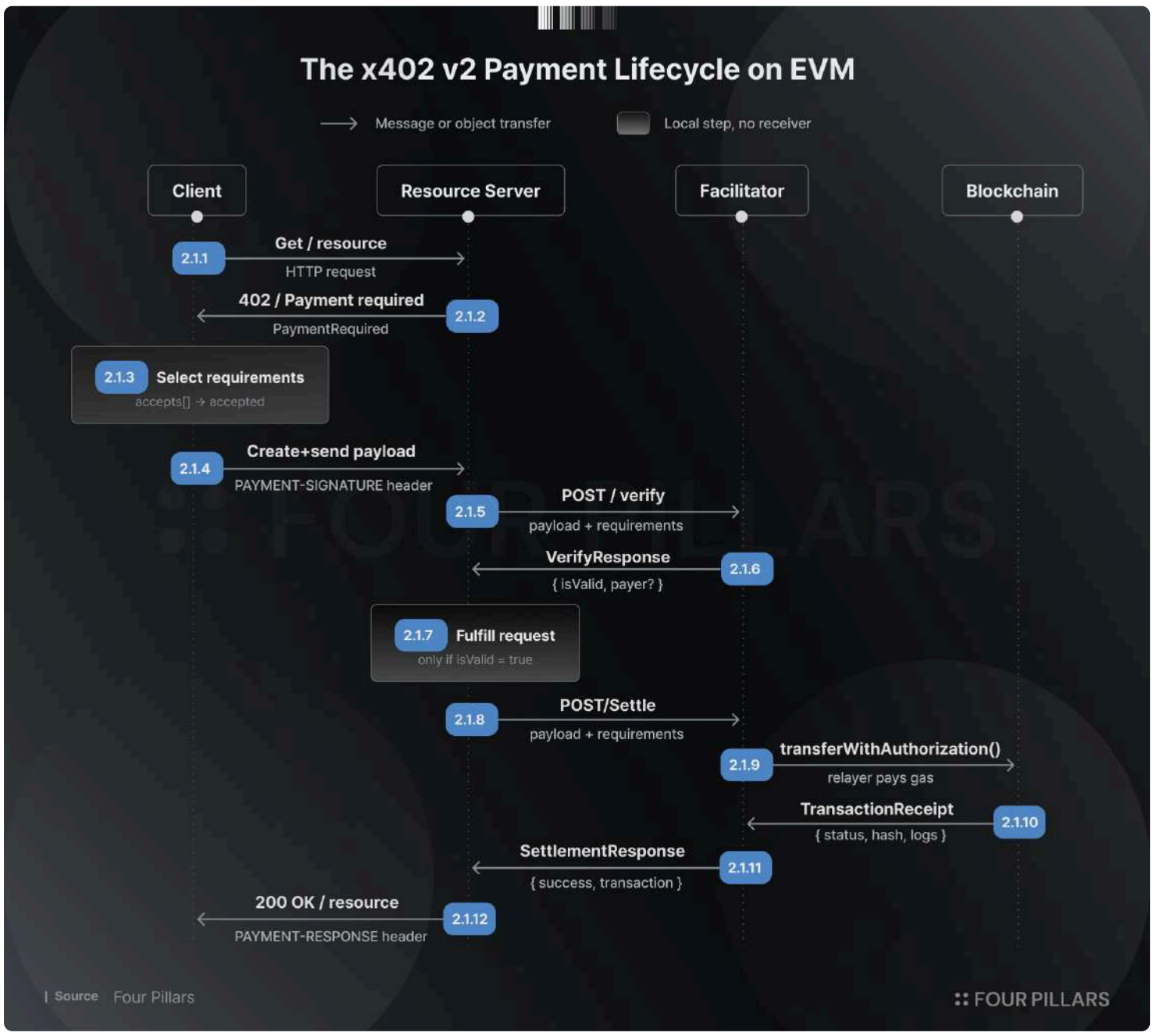
Consider a client requesting San Francisco weather or a real-time asset price. If the service provider delegates blockchain payment verification and transaction submission to an external facilitator, the process returns both the resource and payment result through the following 12 steps (§2.1.1–2.1.12).



The payment flow can be summarized as follows:

- **Steps 1–3 (§2.1.1–2.1.3), price and payment options:** The server quotes $0.01 for one day's San Francisco weather data, payable in USDC on Base or Solana.
- **Step 4 (§2.1.4), wallet authorization:** The user authorizes a $0.01 USDC payment on Base.
- **Steps 5–6 (§2.1.5–2.1.6), authorization feasibility checks:** The server checks whether the wallet-signed transfer authorization meets its requirements and can execute in the current state.
- **Steps 7–12 (§2.1.7–2.1.12), service execution, settlement, and response:** The server performs the requested work, and the facilitator submits the settlement transaction to the blockchain. Once settlement is complete, the server returns the data and payment result to the client.

We now examine the objects exchanged and the checks performed at each step.

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2.1.1 The Client Requests a Resource

The client sends a GET or POST request to access an API endpoint, paid content, data, or computation. It can obtain the price, payment asset, and network from the server's first response.

If the client has cached a previous response or obtained requirements through a discovery API, it can skip the initial request and 402 response. The central design, however, is that the client requests data, the server presents requirements (§2.1.2), and the client accepts an option (§2.1.3), without prior bilateral agreement.

2.1.2 The Server Presents PaymentRequired

If the requested service is paid and no valid payment authorization is attached, the server returns HTTP 402 Payment Required and a PaymentRequired object. The object contains the requested resource, price, and acceptable payment methods. In x402 v2, it is Base64-encoded and sent in the PAYMENT-REQUIRED header.

```
// Explanatory comments only — remove them before sending this data.
{
  "x402Version": 2,
  // The seller identifies the paid service and expected result.
  "resource": {
    "url": "https://api.example.com/weather",
    "description": "Tomorrow's weather"
  },
  // The seller lists acceptable payment methods and settlement terms.
  "accepts": [{
    "scheme": "exact",
    "network": "eip155:84532",
    // These fields fix the amount, asset, recipient, and time limit.
    "amount": "10000",
    "asset": "0x036CbD53842c5426634e7929541eC2318f3dCF7e",
    "payTo": "0x209693Bc6afc0C5328bA36FaF03C514EF312287C",
    "maxTimeoutSeconds": 60,
    "extra": { "name": "USDC", "version": "2" }
  }],
  "extensions": {}
}
```

PaymentRequired comprises the protocol version x402Version, the requested resource, acceptable payment conditions in accepts, and optional capabilities in extensions. error is optional.

The key is the accepts array. Each entry is a PaymentRequirements object representing an accepted combination of scheme, network, and asset. Because clients support different assets and networks, the server can offer multiple alternatives.

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- amount and asset specify the payment amount and asset.
- scheme defines authorization, verification, and settlement rules.
- network identifies the payment rail in CAIP-2 format.
- payTo is the recipient address or a role identifier defined by that network.
- maxTimeoutSeconds specifies the maximum time condition used for the chosen scheme's validity and settlement-wait handling. It must be interpreted alongside the signed expiry and the network's transaction-validity period.
- extra contains additional values required by the selected scheme and network.

The example authorizes $0.01 in USDC to the designated wallet (0×2096...) on Base Sepolia, identified by eip155:84532, with a signature valid for 60 seconds from that point.

> CAIP-2 identifies networks as namespace:reference. For example, eip155:84532 denotes Base Sepolia in the EVM ecosystem. Standard identifiers reduce naming collisions and let one field represent multiple networks.

2.1.3 The Client Chooses Payment Requirements

The client finds a supported scheme and network combination in the accepts array, then selects an option consistent with its assets, amount limits, and permitted networks. This step applies the buyer's wallet state and spending policies to the paths offered by the server.

2.1.4 The Client Creates Authorization and Retries

- **The wallet signs limited payment authority under the selected conditions. No funds are transferred at this stage.**

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```
// Explanatory comments only — remove them before sending this data.
{
  "x402Version": 2,
  "resource": {
    "url": "https://api.example.com/weather",
    "description": "Tomorrow's weather"
  },
  // The buyer repeats the payment option selected from the seller's offer.
  "accepted": {
    "scheme": "exact",
    "network": "eip155:84532",
    "amount": "10000",
    "asset": "0x036CbD53842c5426634e7929541eC2318f3dCF7e",
    "payTo": "0x209693Bc6afc0C5328bA36FaF03C514EF312287C",
    "maxTimeoutSeconds": 60,
    "extra": {
      "name": "USDC",
      "version": "2"
    }
  },
  // The payload carries the buyer's signed authorization to move funds.
  "payload": {
    "signature": "0x2d6a...b571c",
    // The authorization limits the sender, recipient, amount, and time window.
    "authorization": {
      "from": "0x857b06519E91e3A54538791bDbb0E22373e36b66",
      "to": "0x209693Bc6afc0C5328bA36FaF03C514EF312287C",
      "value": "10000",
      "validAfter": "1740672089",
      "validBefore": "1740672154",
      "nonce": "0xf3746613c2d920b5fdabc0856f2aeb2d4f88ee6037b8cc5d04a71a4462f13480"
    }
  },
  "extensions": {}
}
```

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The client creates payment authorization and places it in PaymentPayload. This object contains x402Version, accepted (a copy of the selected PaymentRequirements), and payload (authorization data such as a signature or transaction). resource and extensions are optional.

The selected scheme and network determine the authorization format. For the exact and EVM combination used here, eip3009 is the default unless the server specifies PaymentRequirements.extra.assetTransferMethod. The server should therefore omit that field only when the payment asset supports transferWithAuthorization().

EIP-3009 relieves the payer of submitting an onchain transaction directly. Instead, the payer signs an EIP-712 message offchain to delegate transfer authority to a submitting party—the facilitator—for an ERC-20 token such as USDC that implements transferWithAuthorization().

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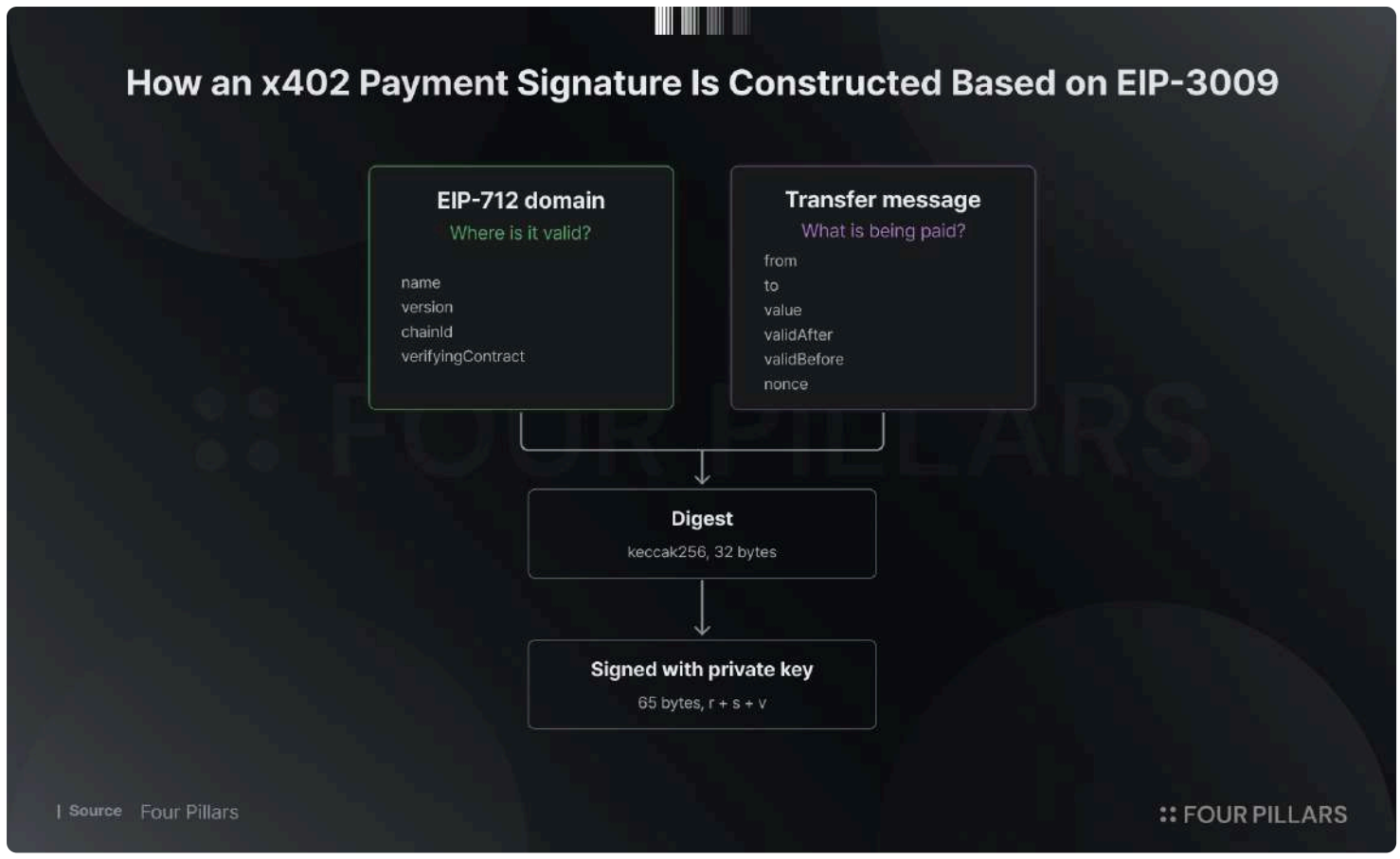

The signed data includes the payer from, recipient to, amount value, validity bounds validAfter and validBefore, and a 32-byte nonce. The token contract records authorization usage by payer address and nonce, rejecting expired, premature, or previously used authorizations.

EIP-3009 uses random rather than sequential nonces, allowing multiple applications to create authorizations concurrently without coordinating a next number. A nonce prevents reuse; it does not guarantee execution order. If order matters, the relayer must confirm an earlier transaction before submitting the next.

The EIP-712 domain—token name and version, chainId, and token address verifyingContract—binds the signature to a particular chain and token. PaymentPayload.payload includes both the signature and the authorization required to reconstruct the signed message. If the facilitator changes the recipient or amount, the reconstructed message changes and signature verification fails.

> Signing does not immediately move funds through onchain settlement. The payer grants permission to transfer a specified amount to a specified recipient within a validity window. Because the wallet submits no onchain transaction at this stage, it spends no gas. The balance moves only when the facilitator later calls transferWithAuthorization() and the transaction succeeds (§2.1.9).

The completed PaymentPayload is Base64-encoded in PAYMENT-SIGNATURE, and the client retries the same request with that header.

2.1.5 The Server Requests Verification

The server sends the client's PaymentPayload and its own PaymentRequirements to the facilitator's POST /verify endpoint. The facilitator compares the two objects to check whether the buyer's authorization matches the merchant's payment requirements.

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2.1.6 The Facilitator Checks Authorization and Executability

- **The facilitator checks the payment payload sent by the client in §2.1.4, including its signature format, balance, expiry, recipient, and network conditions, to identify potential execution failures before settlement.**

For exact with EIP-3009, it checks:

1. The address recovered from the signature matches authorization.from.

2. The payer's balance covers the payment.

3. The amount and validity window satisfy the server's requirements.

4. The token, network, and recipient match the offered values.

5. The nonce is unused and the contract state permits execution.

6. A simulation of transferWithAuthorization() succeeds.

Simulation catches failures that field comparisons alone may miss, including a used nonce, paused token contract, or address restrictions. The facilitator returns the result as VerifyResponse.

Success

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Failure

2.1.7 The Server Performs the Requested Work

- **Once the facilitator confirms that the payment payload can execute, the server performs the buyer's requested work.**

If VerifyResponse.isValid is true in §2.1.6, the server performs the API call, inference, or other requested operation.

Executing work and delivering its result are separate stages: the basic flow prepares the result, confirms successful settlement, and then returns the final response. Although performing work before settlement may seem counterintuitive, the official repository's typical flow recommends completing work, settling, and then returning the result.

A server may nevertheless settle earlier depending on cost and risk. Verification only establishes execution feasibility, not actual movement of funds. The server should choose the sequence according to both service-delivery risk and payment-failure risk.

> The payer's balance or authorization state can change after verification, so successful verification does not guarantee settlement. High-cost work or other situations requiring payment security may call for settlement before execution. The ordering of work, verification, and settlement also depends on trust in the service provider: a provider with an established reputation can more readily require advance settlement than an untrusted one.

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A flexible payment workflow must therefore reflect both technical settlement risk and trust in the participants.

The upto scheme, where the final charge is determined after work, allows a different flow at this stage. An AI text-generation service cannot know token usage in advance, so it authorizes a maximum amount first, measures actual usage after completion, and settles within that approved limit.

2.1.8 The Server Requests Settlement

After completing the work and calculating the charge, the server sends PaymentPayload and PaymentRequirements to the facilitator's POST /settle endpoint. This step uses the payment authorization checked earlier to transfer funds.

2.1.9 The Facilitator Submits the Transaction

- **After offchain authorization and verification, the facilitator executes an onchain transfer based on the authorization in the payment payload.**

It creates an execution request for the selected scheme and network. For an onchain payment, the relayer constructs the transaction, pays gas, and broadcasts it.

With exact and EIP-3009, the client's signature and authorization are passed to transferWithAuthorization(from, to, value, validAfter, validBefore, nonce, v, r, s) on the token contract. The limited transfer authority created in §2.1.4 becomes an actual contract call.

2.1.10 The Facilitator Confirms the Transaction

- **The facilitator checks whether the submitted payment has actually settled on the network.**

It waits for inclusion and the network's required success condition, then checks the execution status. During execution, the EIP-3009 token contract rechecks the signature, validity window, and nonce. On success it marks the nonce as used, transfers tokens, and emits AuthorizationUsed and Transfer. A revert leaves the payer's balance and nonce unchanged, although the facilitator still pays gas. On SVM, processed means a node has processed the transaction, confirmed means it has received votes from a supermajority of stake, and finalized means final confirmation. This report uses confirmed as the default SVM settlement criterion absent a separate policy; finalized may be appropriate for high-value payments or resources that are difficult to recover.

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2.1.11 The Facilitator Returns the Settlement Result

The facilitator returns the network's processing result to the server in a SettlementResponse. The main fields are success, transaction, and network. Depending on the selected scheme and processing result, it may also include payer, the actual settled amount, and an errorReason.

```
// Explanatory comments only — remove them before sending this data.
{
  // Settlement succeeded; transaction identifies the resulting network transfer.
  "success": true,
  "payer": "0x857b...e36b66",
  "transaction": "0x1234...cdef",
  "network": "eip155:84532"
}
```

The server considers the payment executed only after checking success: true. A transaction hash alone is insufficient: a submitted transaction can still fail.

2.1.12 The Server Returns the Resource and Payment Result

On successful settlement, the server returns HTTP 200 OK and the requested resource, with the Base64-encoded SettlementResponse in the PAYMENT-RESPONSE header. The client receives the service result and payment record in a single response.

```
// Explanatory comments only — these lines are not transmitted.
// The seller returns the settlement result with the purchased data.
HTTP/1.1 200 OK
PAYMENT-RESPONSE: <base64-encoded SettlementResponse>
Content-Type: application/json

// The response body contains the purchased weather result.
{ "temperature": 18, "unit": "celsius" }
```

The client can record the service and payment results together by associating the response body with PAYMENT-RESPONSE. If settlement fails, the server ends the request with an error without returning the resource.

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These 12 steps standardize a common flow: the server presents conditions, the client provides limited authorization, and verification and settlement results remain tied to the same request context. This is the representative scenario. The next section examines variations across transports, authorization rules, and settlement rails.

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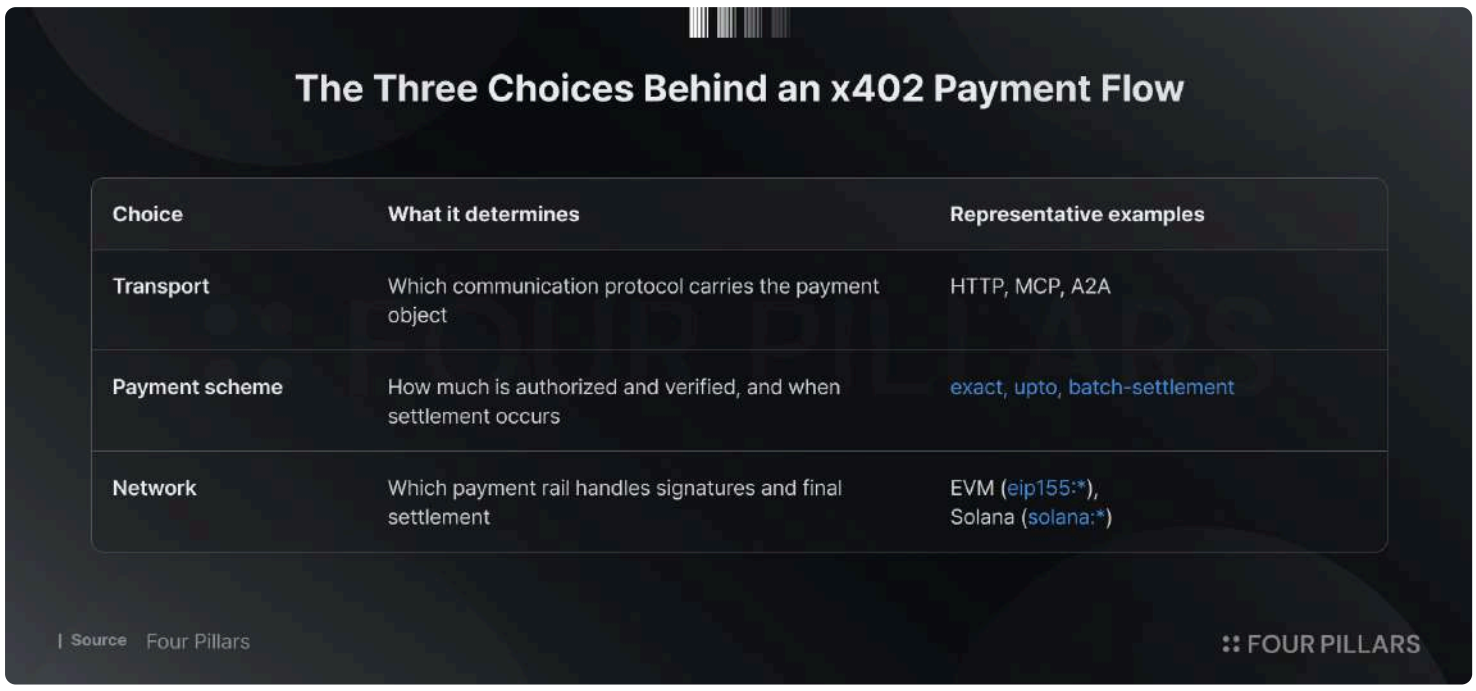
2.2 Transport, scheme, and network define a payment flow

The request, authorization, verification, and settlement mechanism in §2.1 forms x402's basic framework. A concrete implementation emerges through three choices:

- Where to carry the payment object (§2.2.1).
- How much to authorize and under which terms (§2.2.2).
- Which payment rail performs final settlement (§2.2.3).

x402 classifies these choices as the transport, payment scheme, and network.



In the HTTP × exact × Base Sepolia (eip155:84532) example from §2.1, x402 objects move through HTTP headers, the buyer authorizes an exact amount, and Base Sepolia records the final settlement result. With MCP × exact × Solana, the same payment objects are placed in an MCP tool call and settled under Solana's signature and token-transfer rules.

x402 v2 separates the common object types, the logic that executes a scheme and network combination, and the representation layer that carries the objects through each transport. This design preserves a common syntax for payment negotiation while allowing implementations to change with the communication environment and settlement rail. Support must be checked for the actual combination of all three elements, rather than for a transport or network alone.

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2.2.1 Transport: Connecting payments to API requests, tool calls, and agent tasks

The transport determines how payment terms, payment authorizations, and settlement results fit into a service's communication format. HTTP connects x402 payments to resource requests; the Model Context Protocol (MCP) connects them to tool calls; and the Agent-to-Agent protocol (A2A) connects them to delegated tasks.

These three paths are the integration points where developers add payments. MCP and A2A messages can also travel over HTTP. Developers choose whether to price a resource URL, a tool call, or an agent task.

- **HTTP charges for a resource URL.** The server quotes a price through 402 and PAYMENT-REQUIRED. The client puts its payment authorization in PAYMENT-SIGNATURE and requests the same resource again. The server returns the resource and PAYMENT-RESPONSE. This path fits paid APIs, files, and web content.

- **MCP charges for a tool call.** When a server responds with payment terms for a paid tool, the client keeps the tool name and arguments and repeats the call with a payment authorization in _meta["x402/payment"]. The tool result includes the settlement result in _meta["x402/payment-response"].

- **A2A charges for delegated work.** The seller agent returns a Task in the input-required state with payment terms. The buyer submits a payment authorization under the same taskId, and the seller processes the payment and continues the task. Payment progress is recorded in x402.payment.status, and settlement history in x402.payment.receipts.

A single service can use all three paths. A research agent might delegate specialized analysis through A2A; the agent performing that work might call a search tool through MCP; and that tool might purchase an external HTTP API. Once the price and payer are defined at each integration point, x402 can be added wherever it is needed. The following examples examine its use in MCP and A2A in greater detail.

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MCP's request-level design fits naturally with x402

The statelessness principle in the MCP development draft requires each request to carry the context needed to process it. Rather than implicitly reusing information agreed upon in an earlier connection, protocol versions and capabilities travel in request metadata, while work spanning multiple calls uses explicit identifiers.

The basic x402 flow also carries a payment authorization in the repeated request to verify that call. Because the tool's execution arguments and payment authorization travel together, developers face less work designing a separate, long-lived payment session for every paid tool. Server instances can process requests against shared policies and processing records, making this structure suitable for per-tool charging and horizontal scaling.

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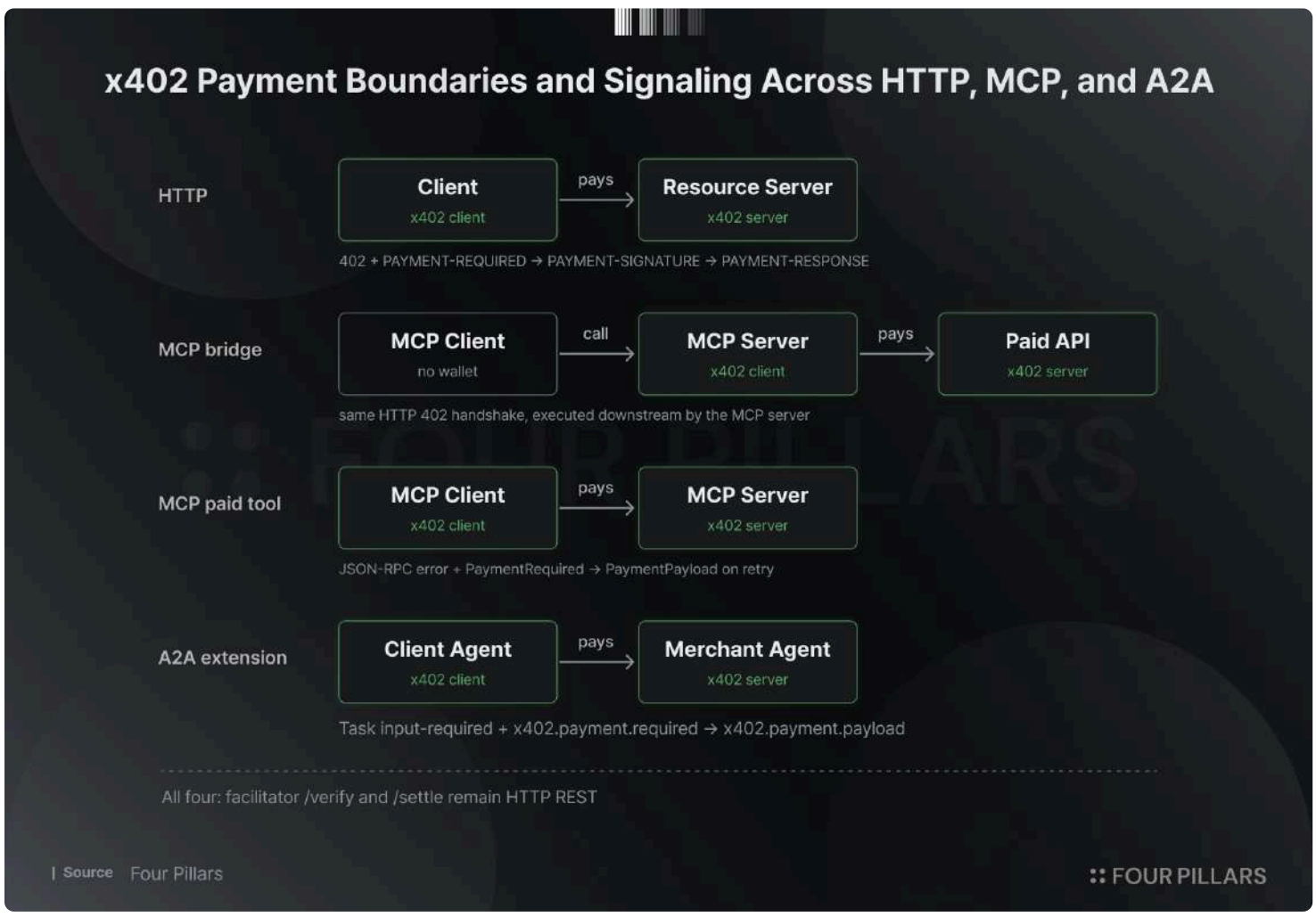

Two MCP implementations: A bridge and direct payment

In the bridge structure, the MCP server buys an external paid API. When an agent calls a search tool, the MCP server purchases external data through an HTTP client such as @x402/axios, processes it, and returns a tool result. The x402 payment takes place between the MCP server and the external API.

In the direct-payment structure, the MCP server sells the tool itself. The official @x402/mcp package connects the return of payment terms, a repeated call carrying authorization in _meta, verification, and settlement. The buyer is the MCP client, and the seller is the MCP server operating the tool.

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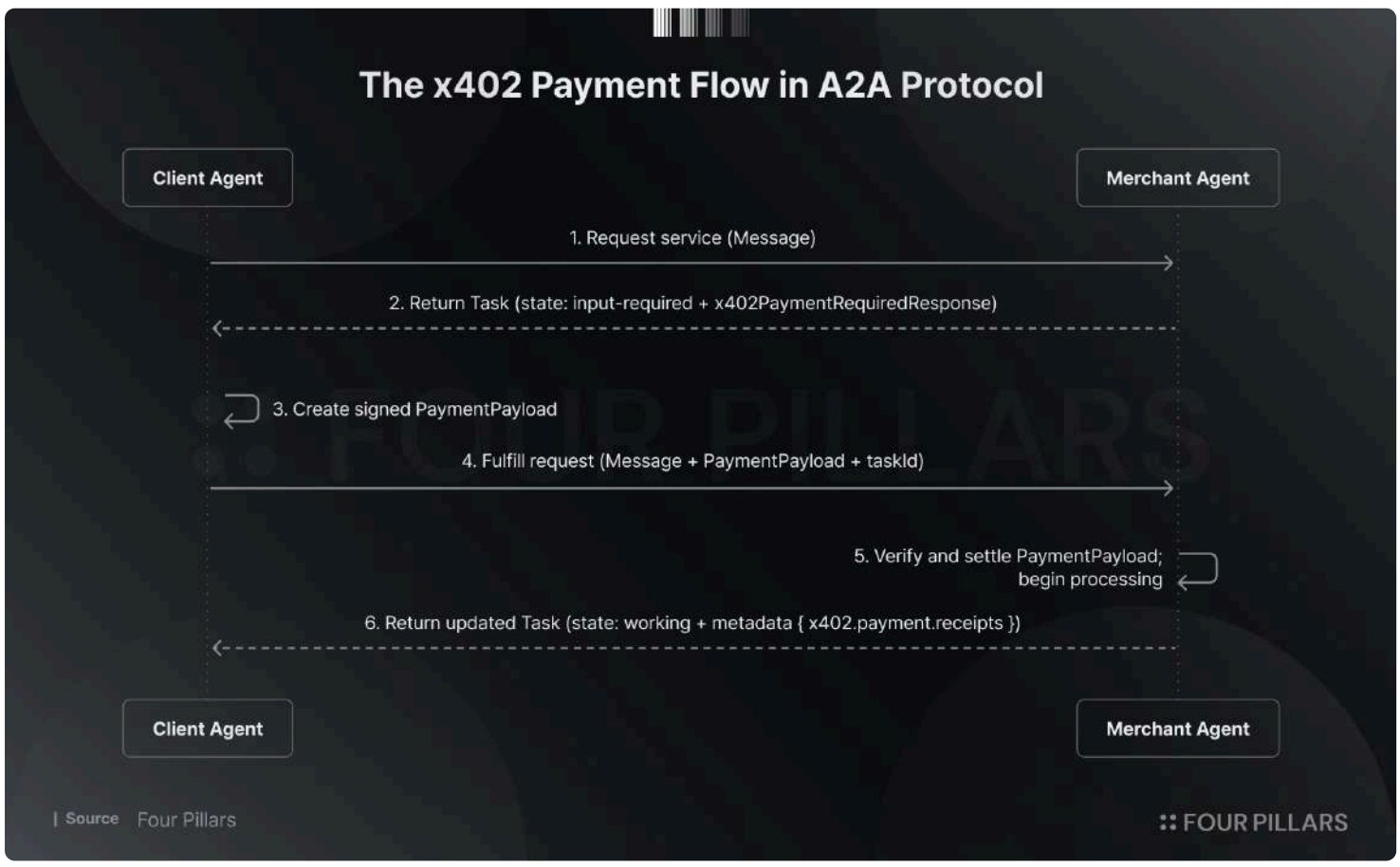
The current package also provides a compatibility path that conveys a payment requirement as a structured error inside the tool result. The client reads PaymentRequired from this result and constructs the payment authorization. Implementations should use the object format returned by the relevant SDK when handling a tool call's payment requirement.



A2A connects payment and deliverables through a task ID

In A2A x402 Extension v0.2, the seller agent uses taskId to connect a quoted price with the submitted payment authorization. The payment status progresses through values such as payment-required, payment-submitted, and payment-completed, while the task itself may continue in the working state. This suits services that continue after payment, such as specialized analysis or long-running data processing.

Version 0.2 distinguishes standalone and embedded modes. Standalone mode places the payment terms and payload directly in metadata. Embedded mode uses x402 as a payment method under a higher-level commerce protocol, placing the same objects inside AP2 CartMandate and PaymentMandate and carrying them through task.artifacts and message.parts. Both flows require x402.payment.status; clients distinguish them by the presence of x402.payment.required.

The transport connects a payment to an API request, tool call, or task. The payment scheme in the next section determines the amount authorized at that integration point and when funds are settled.

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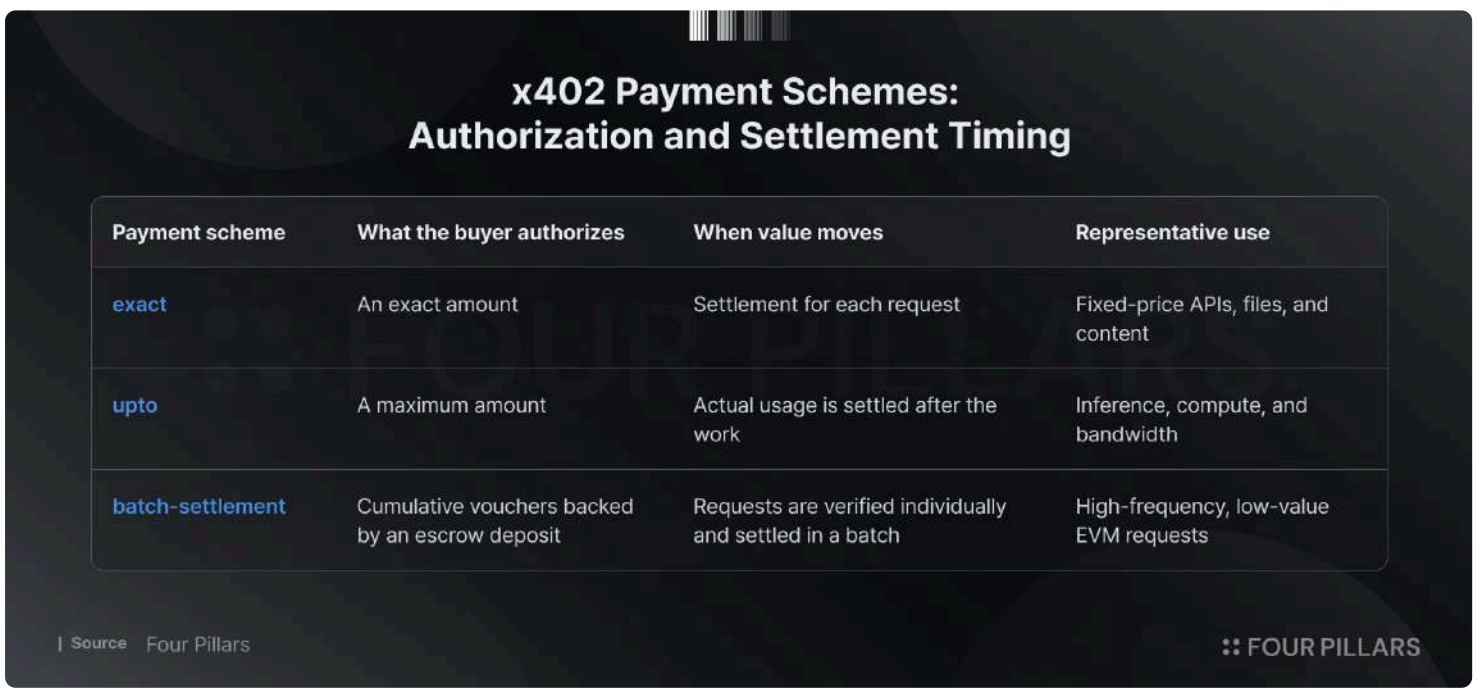
2.2.2 Scheme: Setting the authorization scope and settlement timing

The payment scheme defines what the buyer authorizes, how much the server can charge, and when value moves. The official Payment Schemes documentation supports exact, upto, and batch-settlement.



exact settles the exact amount presented by the server for each request. It fits services whose price can be fixed before payment, such as fixed-price APIs, files, and content.

With upto, the buyer first authorizes a maximum amount, and the server settles only actual usage after completing the work. It fits services whose usage cannot be determined before a request ends, such as model inference, compute, and bandwidth.

With batch-settlement, the client submits a cryptographic payment commitment or usage right for each request, but funds do not move immediately. The server verifies the commitment and provides the resource. The network implementation accumulates these commitments in vouchers, channel state, or an account ledger. Several requests are settled together later through a contract call, channel closure, or scheduled billing.

Under batch-settlement, the buyer may authorize a maximum amount for each request, and the server sets actual usage at or below that amount and reports it in PAYMENT-RESPONSE. The buyer's capital can secure the commitment through prepaid escrow or a payment channel. A network operator may also guarantee it based on identity and a billing account. A successful response may therefore contain a commitment identifier for later settlement instead of an immediately executed transaction hash.

The signature format, verification target, and finality test vary with the network even under the same payment scheme. The scheme sets the economic rules, and the network implements those rules on a payment rail.

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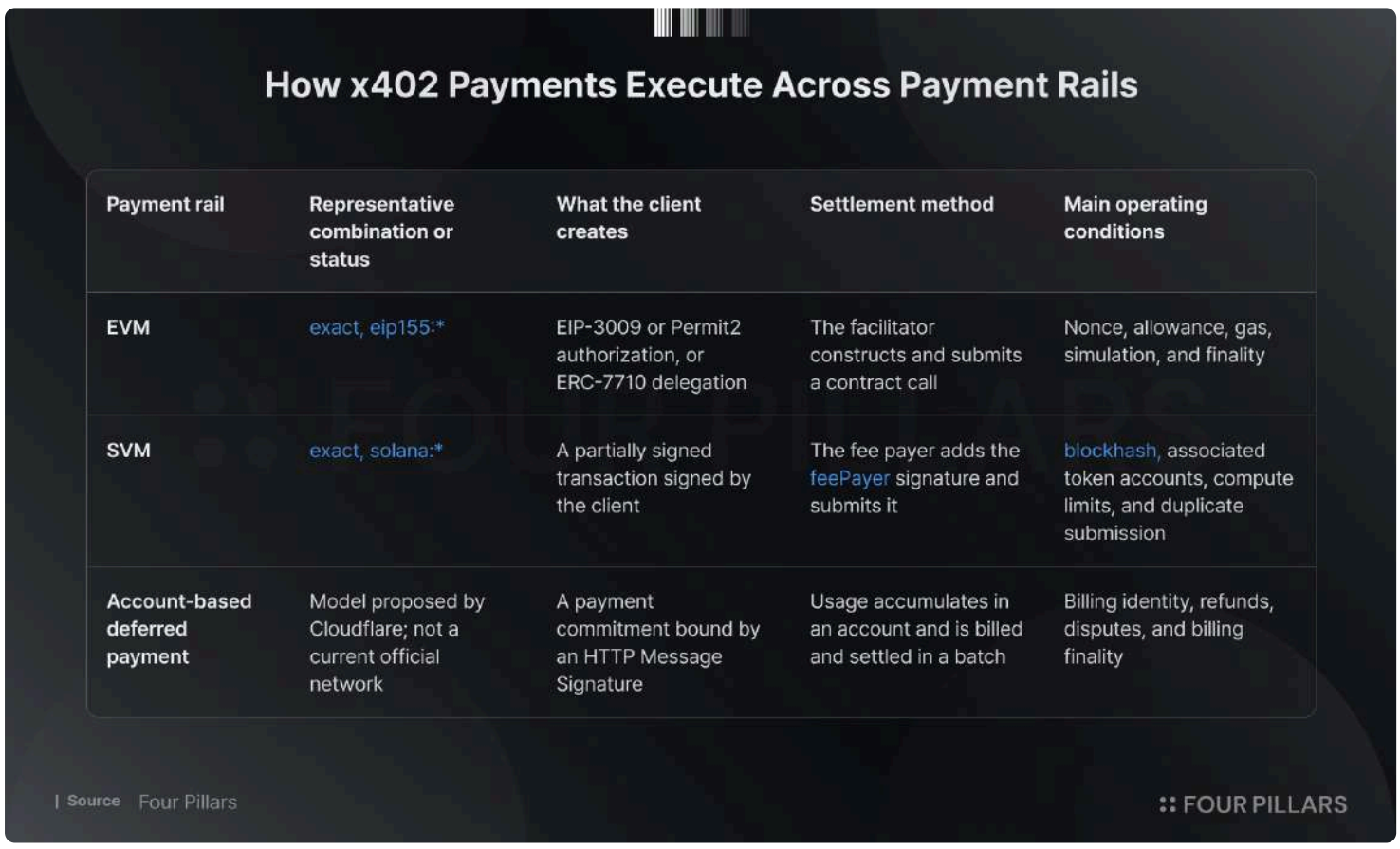
2.2.3 Network: Executing a payment scheme under each payment rail's rules

The network determines which signatures, accounts, transactions, and finality rules implement the selected payment scheme. x402 support must therefore be evaluated by the combination of scheme and network. Even under exact, EVM and SVM use different client authorization data and facilitator transactions.



An EVM exact payment may use EIP-3009, Permit2, or ERC-7710 delegation, depending on the transfer method supported by the asset. EIP-3009 calls transferWithAuthorization() on the token contract directly. Permit2 transfers an approved token through a proxy contract, while ERC-7710 executes authority delegated by a smart account in advance. All three bind the amount and recipient within a signature or delegation scope, but the facilitator checks different items: nonce, allowance, or delegation terms.

For an SVM exact payment, the client constructs the payment transaction itself. Using the feePayer presented by the server, it creates a transaction containing an SPL Token or Token-2022 transfer and signs first. Acting as the fee payer, the facilitator or seller checks the mint, amount, recipient's associated token account, and compute cost, then adds the final feePayer signature and submits the transaction to the network.

On EVM, the facilitator receives signed authority and constructs a contract call. On Solana, the fee payer completes a partially signed transaction created by the client.

The network field can also extend to payment paths beyond blockchains. The table's batch-settlement × cloudflare:402 example represents a proposed account-based deferred-payment model. Under this proposal, HTTP message signatures bind requests to payment commitments. The operator verifies each request's signature, accumulates usage against a billing account, and settles with sellers on a defined billing cycle. Service agreements and implementations determine accumulated spending limits, billing-failure handling, and the criteria for completed settlement.

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Separating transport, scheme, and network reduces the need for a resource server to implement every difference among payment rails. The server presents the scheme and network combinations it accepts and can delegate verification and settlement to a facilitator supporting that combination. x402 therefore does not depend on a specific payment network, but actual compatibility depends on the implementation quality and support scope of each combination.

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2.3 Extensions: Adding operating functions outside the payment core

The scope of the x402 core is clear. The server presents payment terms, the client submits authorization data, and the facilitator or server verifies and settles the payment. Keeping this scope small allows implementations in different languages and payment networks to share the same payment flow.

Running a service requires functions beyond payment execution. An agent must find APIs to buy and determine whether a request repeated after a network error belongs to the same payment. A wallet must be authenticated when it returns to a purchased resource. The price offered by the server and its claim of delivery may also need to be recorded. Placing all these functions in the core specification would require every implementation to restore compatibility whenever one operating function changed.

Extensions separate these functions into optional modules. A server declares the required extension keys for each route and executes only the functions it has registered. If an extension requires the client to create an ID or signature, or the facilitator to perform additional verification or settlement logic, the counterparty implementation must support the same key.

- The resource server places extension data in the 402 response and settlement response and executes application logic before and after payment.
- The client creates the ID or signature requested by the server and returns it in PaymentPayload.extensions.
- The facilitator processes the required metadata and additional settlement logic during verify and settle.
- An extension does not alter core payment terms such as the amount, asset, or recipient.

As of August 2026, the official specification lists seven extensions. They are divided into resource-server extensions and facilitator extensions according to the implementing party and can be classified further by their point of action in the payment lifecycle. The figure below shows each extension's classification and main role.

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2.3.1 Bazaar helps agents find payable resources

- **Role:** Connects descriptions, input and output schemas, and payment terms for x402-enabled paid APIs and MCP tools to a searchable catalog.
- **Scope:** Suitable when an agent must discover and call external services without prior registration.

x402 defines how to pay for a discovered service but does not identify which services exist. People can find APIs through search engines or developer documentation. For an agent to purchase an external service autonomously, descriptions, input formats, output examples, and prices must be available in a machine-searchable form. Bazaar addresses this discovery problem before payment.

A service seller uses the Bazaar extension to publish a service description, input schema, output examples, and payment terms. A facilitator that supports the extension collects these declarations and operates a searchable catalog. Buyers use the catalog to find the services they need.

Bazaar is not a single global registry. Each facilitator operates its own catalog and registration policy, so search coverage and ranking may differ. A listing shows that a service exists and how to call it, but does not guarantee availability or output quality. The Coinbase CDP Facilitator, a representative implementation, shows the boundary between registration and search.

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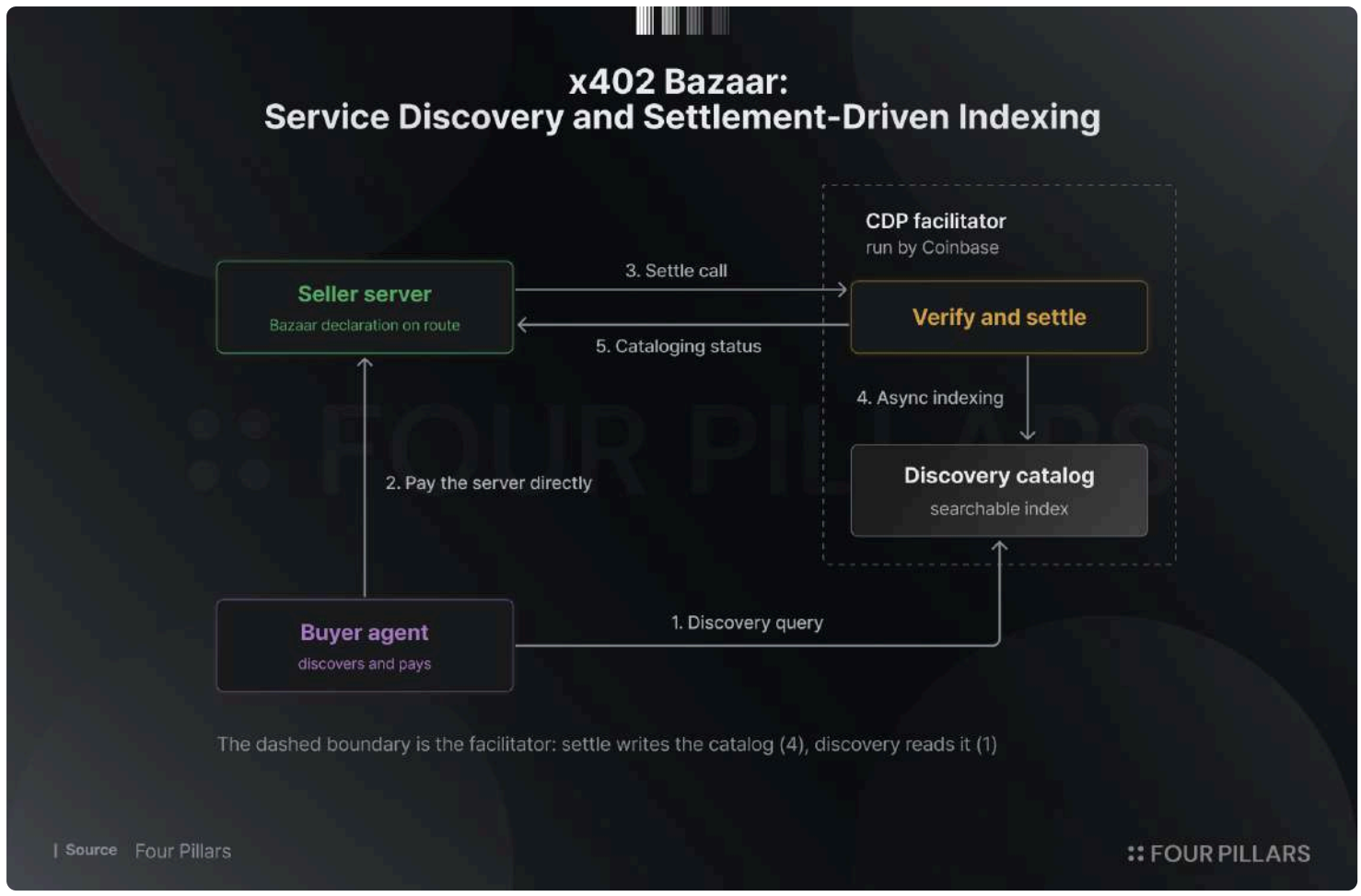

When configuring a route, the server uses declareDiscoveryExtension to declare the service description, input schema, and output examples. When a client requests the resource, the server includes the payment terms and Bazaar declaration in the 402 response. The client includes the declaration in PaymentPayload.extensions and submits the request again.

During verify and settle, the CDP Facilitator validates the server data to be registered in Bazaar and reports registration status to the server through the EXTENSION-RESPONSES header. There is no separate registration request. After the first successful settlement on a Bazaar-enabled endpoint, the facilitator adds the metadata to the catalog asynchronously.

After registration, search and usage data determine ranking. Agents use semantic search in CDP Bazaar to find services related to a query. The ranking system considers relevance, the number of unique buyers and successful payments observed by the CDP Facilitator, recency of use, and metadata completeness.

CDP's Bazaar MCP server also connects discovery to the subsequent call. An agent finds a service with search_resources and calls the endpoint with proxy_tool_call. If payment is required, the x402 client creates a payload and repeats the same request. An agent that has not registered each API in advance can proceed from search to checking terms, payment, and invocation in one flow.

Bazaar's search and usage records are a starting point for finding suppliers that can accept payment. For an actual order, the agent must compare candidates against the user's conditions and select a supplier to call. Section 4.4.5 examines these discovery and order-allocation functions through discovery and routing.

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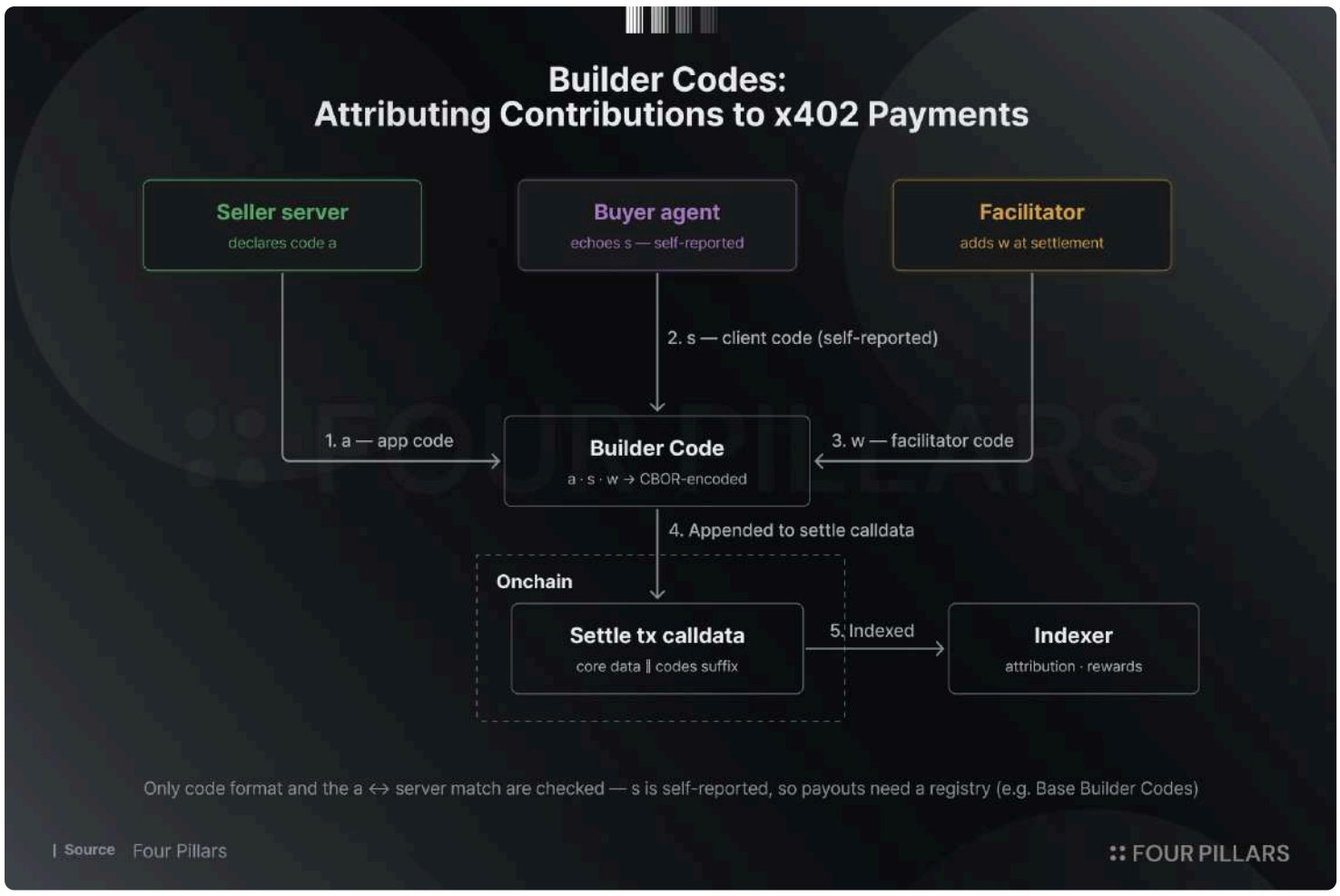
2.3.2 Builder Code identifies the parties that contributed to a transaction

- **Role:** Records contribution codes for the application, client, and facilitator in an ERC-8021 settlement transaction.
- **Scope:** Can be used to analyze distribution paths and allocate referral rewards or rebates.

An x402 transaction may involve the application that provides the paid endpoint, the client that created the purchase request, and the facilitator that executed onchain settlement. A record of fund movement alone does not show which party brought in and processed the transaction. Builder Code adds this participation data to the settlement transaction.

Builder Code uses ERC-8021 Schema 2. It places the application, client, and facilitator codes in the a, s, and w fields, respectively, encodes them in CBOR, and appends them to the settlement transaction's calldata.



An indexer can link these codes to settlement data such as the payer address and payment amount. This supports analysis of facilitator market share, routing paths, partner combinations, contribution by distribution channel, cross-platform usage, and customer retention. Linking a, s, and w to a separate accounting system can also support referral rewards, facilitator rebates, and MCP distribution rewards.

Builder Code checks the code format and whether the value returned by the client (a) matches the server declaration. The s value submitted directly by the client can contain false attribution. Using these codes for rewards therefore requires checking registry records, code ownership, and actual contributions to a transaction together.

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Base Builder Codes addresses part of this limitation. It issues each code as a unique ERC-721 and links the code owner and reward address in an onchain registry. An indexer can extract the code from calldata, check its registration and payment address, exclude unregistered codes, and assign rewards to the registered address.

The registry proves ownership of the code. It does not prove that the owner generated the payment or that transaction volume was not inflated through self-dealing. A reward system must also apply policies such as payment-level signatures and self-dealing detection.

Even when the participants are identified correctly, a temporary network error can cause the same request to be sent again. The Payment-Identifier extension described below can provide an additional control against duplicate execution of the payment and subsequent work.

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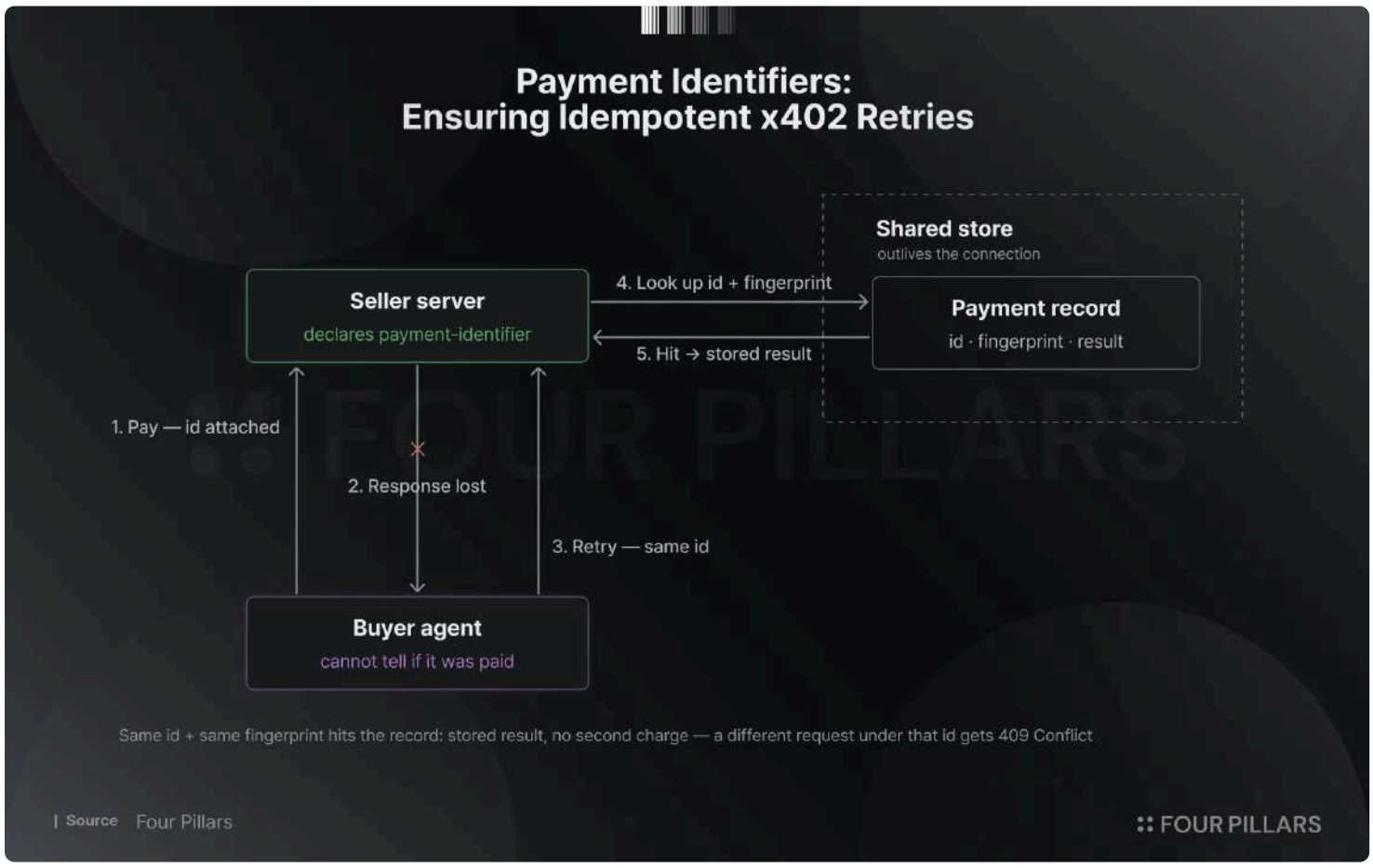
2.3.3 Payment-Identifier groups retries under the same payment

- **Role:** Identifies requests repeated after a network error as one logical payment and task.
- **Scope:** Suitable for services where duplicate payments and execution are costly, such as automatic retries, long-running tasks, and expensive inference.



Consider a connection failure between the server and client immediately after onchain settlement. The client cannot easily tell whether payment failed or succeeded without a response. If it repeats the request with new payment information, the same payment and work may execute twice.

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Payment-Identifier is an idempotency key that groups retries under one logical request. When the server declares the extension in a 402 response, the client creates a task-level ID and includes the same ID in PaymentPayload on every retry. The operator stores the payment ID and request fingerprint, and retrieves the existing processing state and result when the same request returns. An implementation should reject a different request associated with the same ID as a conflict; an HTTP server can use 409 Conflict.

Preventing duplicates requires storing the state of both the payment and the work. An environment with multiple servers needs a shared store and atomic state updates. It must also recover when settlement has completed but the response has not been recorded. The payment ID connects these parts of the implementation.

Payment-Identifier standardizes only the format used to carry the ID. The server must implement the store, locks, and idempotency of subsequent work. It should bind the ID to a request fingerprint that includes the payment terms, request path, HTTP method, and a jobId or orderId, then control concurrent requests with a unique constraint or distributed lock. It must also store the payment state, transaction hash, and work result at each stage to recover from a failure that occurs after settlement but before the response is recorded.

2.3.4 Sign-In-With-X authenticates repeat access by the paying wallet

- **Role:** Proves control of a wallet address and links it to purchase records held by the server.
- **Scope:** Suitable for returning to purchased content, restoring a session, or checking continued access rights.

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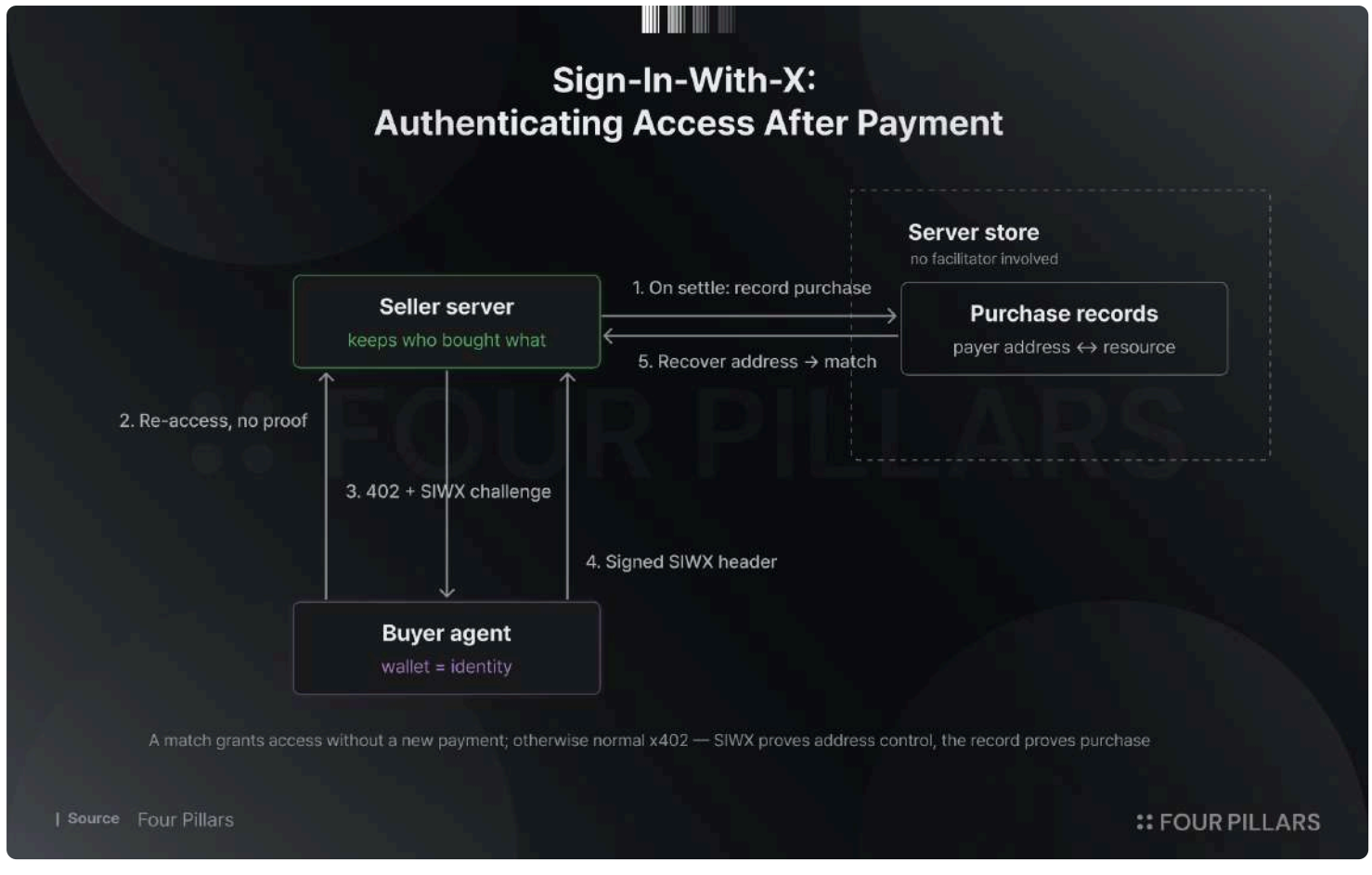

A signature under the x402 exact scheme is a one-time payment instrument that authorizes a specific token transfer. It does not act as a login session for later access to content. Allowing a buyer to return therefore requires wallet authentication separate from payment and a server-side record that maps the payer address to the purchased resource.

Sign-In-With-X, or SIWX, is a chain-neutral wallet-authentication method based on CAIP-122. It uses a wallet signature to verify control of an address and compares the recovered address with purchase records stored by the server. An SIWX signature does not prove a purchase by itself. The server's records determine whether a purchase occurred.

After successful settlement of the first payment, the resource server stores the payer's wallet address and the purchased resource. If the client later accesses the same resource without valid SIWX proof, the server returns a challenge in the extension data of a 402 response. The challenge contains the domain, URI, nonce, issue time, expiry time, and supported chains.

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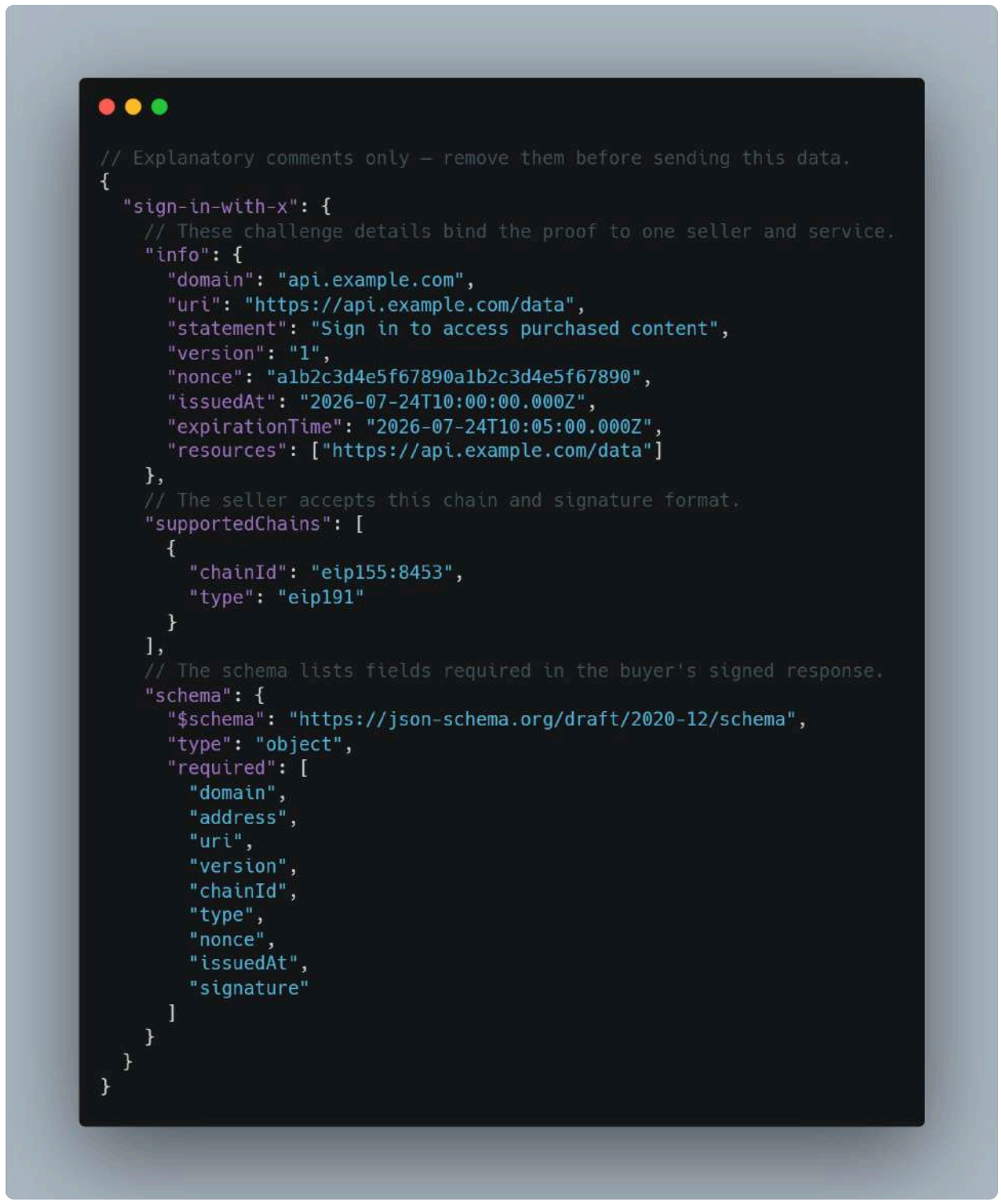

```
// Explanatory comments only — remove them before sending this data.
{
  "sign-in-with-x": {
    // These challenge details bind the proof to one seller and service.
    "info": {
      "domain": "api.example.com",
      "uri": "https://api.example.com/data",
      "statement": "Sign in to access purchased content",
      "version": "1",
      "nonce": "a1b2c3d4e5f67890a1b2c3d4e5f67890",
      "issuedAt": "2026-07-24T10:00:00.000Z",
      "expirationTime": "2026-07-24T10:05:00.000Z",
      "resources": ["https://api.example.com/data"]
    },
    // The seller accepts this chain and signature format.
    "supportedChains": [
      {
        "chainId": "eip155:8453",
        "type": "eip191"
      }
    ],
    // The schema lists fields required in the buyer's signed response.
    "schema": {
      "$schema": "https://json-schema.org/draft/2020-12/schema",
      "type": "object",
      "required": [
        "domain",
        "address",
        "uri",
        "version",
        "chainId",
        "type",
        "nonce",
        "issuedAt",
        "signature"
      ]
    }
  }
}
```

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The client selects the chain that matches its wallet and signs the challenge. It Base64-encodes proof containing the signature and address and sends it in the SIWX header. The server compares the domain and URI with its configured public origin, then verifies the nonce, issue and expiry times, chain ID, and signature.

If the recovered address matches the purchase record for the requested resource, the server permits access without a new payment. If no purchase record exists or authentication fails, the server switches to the standard x402 payment flow.

The facilitator does not participate in this authentication process. In production, the resource and payer address should be stored in persistent storage such as a database, rather than in memory. Used nonces must also be stored to prevent the same signature from being reused.

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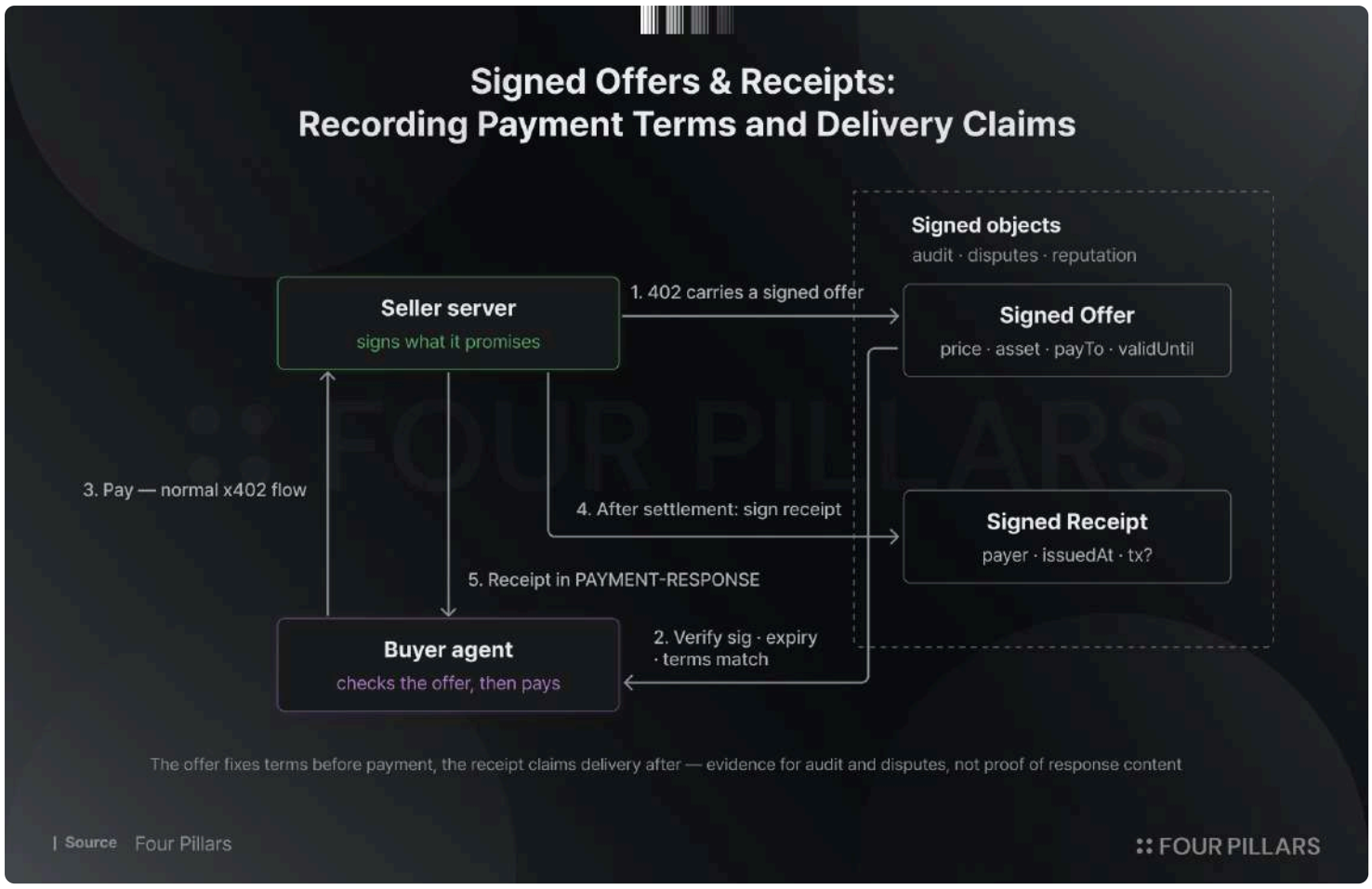
2.3.5 Signed Offers & Receipts records prices and delivery results with signatures

- **Role:** Records the terms presented by the server before payment and its claim of completed delivery after payment as signed objects.
- **Scope:** Suitable for marketplaces and B2B APIs that require audits, dispute handling, and verification of purchase history.



The basic x402 flow verifies the client's payment authorization and onchain settlement. It does not prove which terms a resource server presented or what the server claims to have delivered after payment. An onchain transaction records the movement of funds, not the delivery of a specific API response or digital resource.

Signed Offers & Receipts adds two signed objects without changing the existing payment flow. The server issues a Signed Offer before payment and a Signed Receipt after settlement.

When creating a 402 response, the server creates and signs an OfferPayload for each payment term in accepts[]. A Signed Offer records the server's commitment to provide a specific resource under specified terms.

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```
// The seller signs these fields to bind one offer.
interface OfferPayload {
  version: number;
  // These fields identify the service and settlement route.
  resourceUrl: string;
  scheme: string;
  network: string;
  asset: string;
  payTo: string;
  // These fields set the price and expiration time.
  amount: string;
  validUntil: number;
}
```

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Before payment, the client checks the Signed Offer's signature and expiry time. It then compares scheme, network, asset, payTo, and amount with the payment terms it selected.

After successful settlement, the resource server creates a Signed Receipt from the result returned by the facilitator. The object contains resourceUrl, network, payer, and issuedAt, and may include transaction. A Signed Receipt records the server's claim that it confirmed the payer's payment and completed delivery of the resource. It is added to extensions in the final settlement response and delivered to the client through the encoded PAYMENT-RESPONSE header over HTTP.

A Signed Offer records the server's signature over its price and terms. A Signed Receipt records its signature confirming payment and claiming completed delivery. Including request and response hashes can bind the receipt to a particular output and reveal subsequent changes.

Operators can use these records for audits and dispute handling, while evaluating the accuracy and usefulness of an output against service-specific criteria. Long-term verification also requires preserving the attribution of signing keys and their rotation and revocation history.

2.3.6 EIP-2612 Gas Sponsoring replaces the initial approval with an offchain signature

- **Role:** Replaces the initial Permit2 approve() with an EIP-2612 permit() signature and processes approval and payment in one transaction.
- **Scope:** Suitable for ERC-20 payments when the token supports EIP-2612 but not EIP-3009.

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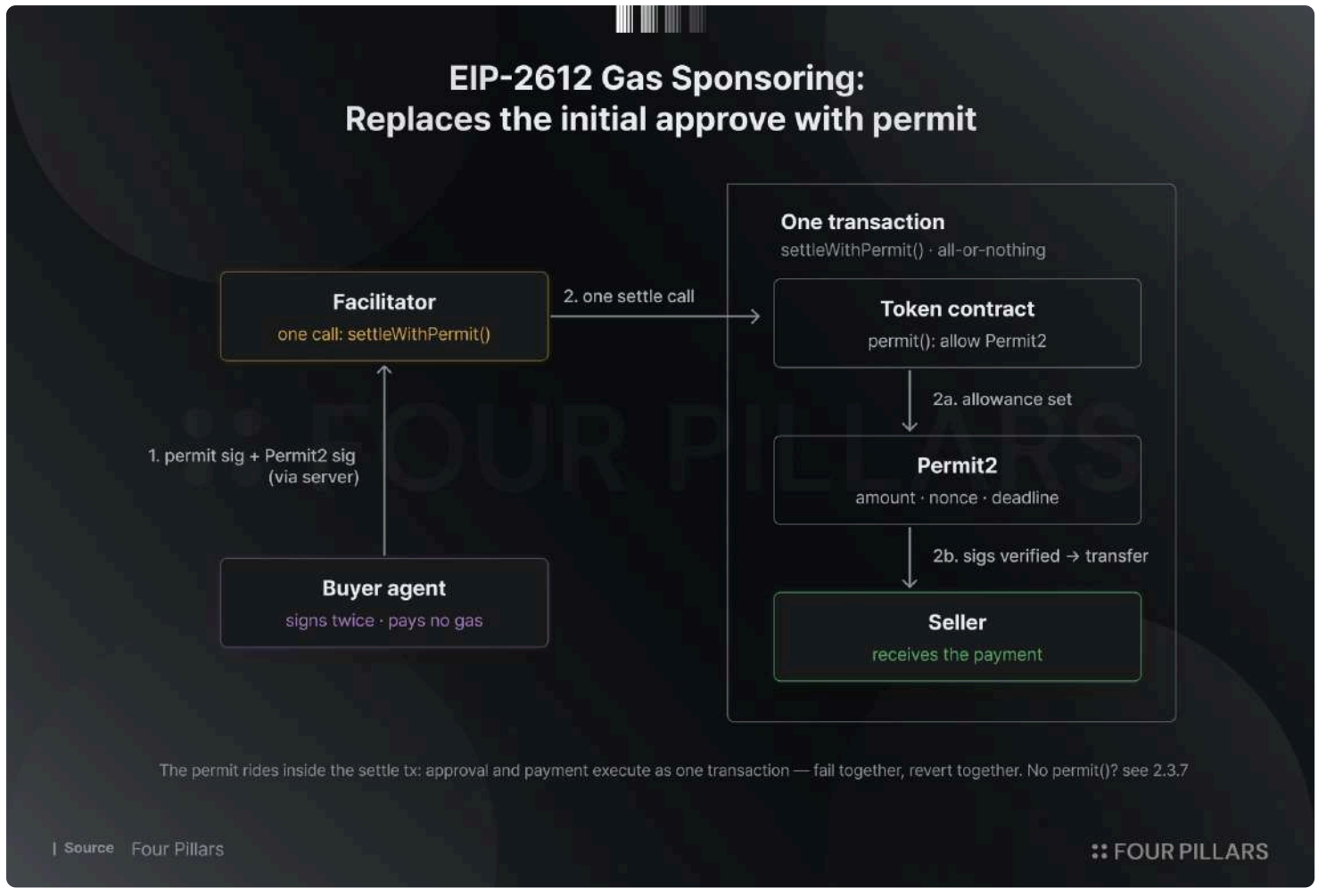

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USDC supports transferWithAuthorization(), which allows a payer to sign transfer authority for each payment without prior approval. An ERC-20 token that does not implement EIP-3009 can use Uniswap's Permit2 as a general transfer path.

After a buyer grants the Permit2 contract spending authority in the token contract once, Permit2 transfers tokens for later payments with a signature bound to the amount, nonce, and deadline. Setting this initial authority requires an onchain approve(). An agent without the native gas token cannot proceed to its first payment.

EIP-2612 Gas Sponsoring replaces the initial approval with an offchain signature for a token that supports permit(). The buyer creates an EIP-2612 signature that grants Permit2 authority to use the token and a separate Permit2 signature that authorizes the payment.

When the server declares eip2612GasSponsoring in the extension data of a 402 response, the client's ExactEvmScheme queries allowance(owner, Permit2) on the token contract. If no allowance exists, it signs a permit bound to the token's EIP-712 domain and the owner, spender, value, nonce, and deadline. The spender is the Permit2 contract.

The client places the permit and Permit2 payment signature together in PaymentPayload. The facilitator calls settleWithPermit(), submits the permit to the token contract, verifies the Permit2 signature in the same transaction, and sends the tokens to the seller.

Because approval and payment occur in one transaction, a payment failure also reverts the permit state change. Anyone can submit a permit, so it requires a short deadline and the correct nonce. The Permit2 signature and witness must restrict the recipient and amount for each payment.

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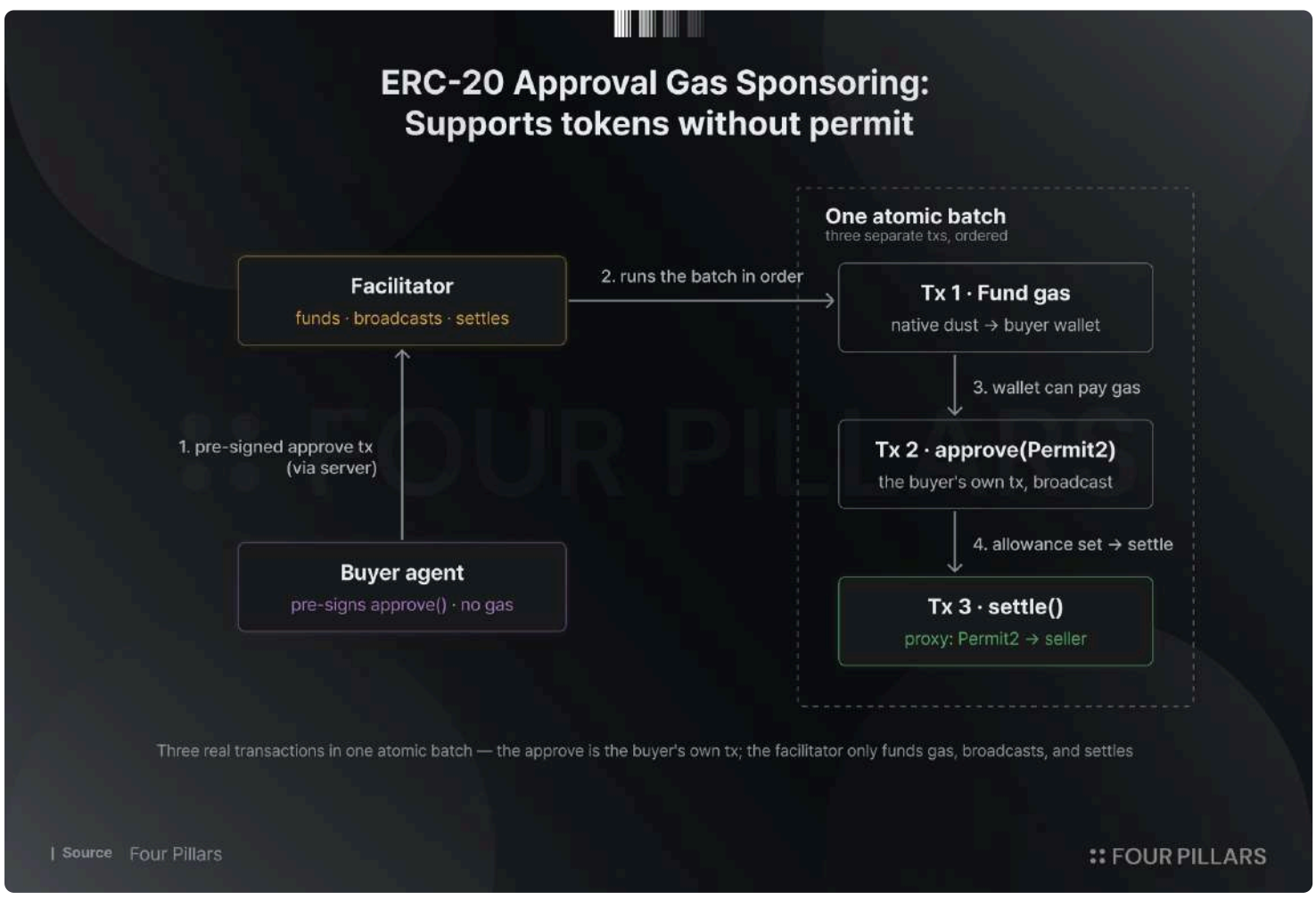
The client does not pay the approval gas directly. The facilitator bears the settlement gas and failure risk. If a Permit2 allowance is already set, the standard settle() path is used without a permit. A token that does not support EIP-2612 requires another method that submits the onchain approve() on the buyer's behalf.

2.3.7 ERC-20 Approval Gas Sponsoring submits approval for tokens without permit support

- **Role:** The facilitator pays gas for and submits an approve() transaction signed by the buyer.
- **Scope:** Can be used for an ERC-20 token that supports neither EIP-3009 nor EIP-2612.



A token without EIP-2612 support cannot set a Permit2 allowance with a signature alone. It requires a transaction that calls approve() on the token contract.

ERC-20 Approval Gas Sponsoring does not remove the approve() step. The buyer signs the transaction, and the facilitator supplies gas and submits it to the network. The buyer can complete the initial Permit2 approval without holding the native gas token in advance.

When the server declares erc20ApprovalGasSponsoring in the extension data of a 402 response, the client's ExactEvmScheme queries the Permit2 allowance. If no allowance exists and EIP-2612 is unavailable, it creates and signs a raw transaction containing approve(Permit2, MaxUint256) and places it in PaymentPayload.extensions.

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After receiving this value through the server, the facilitator sends a small amount of the native token to the buyer's wallet if it lacks gas. It then broadcasts the presigned approval and executes payment with x402ExactPermit2Proxy.settle(). The EIP-2612 path processes approval and payment in one transaction. The official documentation for this extension describes a path that executes separate transactions as an atomic batch.

The facilitator must manage the buyer account's nonce, gas fee, approval inclusion order, and allowance state. It also needs a revocation policy for users who do not want a long-lived approval. The method supports a wider range of assets, while transferring operational work and failure risk to the facilitator.

These extensions address different operating problems but follow the same principle.

> Preserve the core flow of payment terms, authorization, verification, and settlement, and add discovery, attribution, idempotency, authentication, evidence, and gas sponsorship only where needed.

This separation allows x402 to extend across communication environments and payment rails, but the protocol does not automatically guarantee each function's reliability or its operational accountability. The benefits depend on how clients, servers, facilitators, and networks divide these responsibilities. That division deserves careful consideration: different allocations of responsibility inevitably involve trade-offs.

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Chapter 03

Main questions for x402: Network expansion and facilitators

3.1 SVM payments use joint signatures on the full transaction

3.2 Can card networks become x402 networks?

3.3 Is a facilitator necessary?

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3. Main questions for x402: Network expansion and facilitators

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This section examines three questions frequently raised about x402.

- How does SVM, or Solana Virtual Machine, process x402? (§3.1)
- Can card networks become x402 payment networks? (§3.2)
- Is a facilitator that handles verification and settlement required for every transaction? (§3.3)

For questions without an established standard answer while the protocol remains under development (§3.2 and §3.3), the section also presents the author's view.

3.1 SVM payments use joint signatures on the full transaction

> EVM buyers sign a message that specifies the transfer terms and delegate execution. SVM buyers first sign the transaction to be executed, after which the facilitator adds the fee payer signature.

Consider the EVM-based x402 payment described earlier. The buyer signs a single transfer authorization under EIP-3009, with the recipient, amount, and validity window fixed. After verifying the signature, the facilitator creates a separate onchain transaction and executes the transfer. The buyer does not sign a completed transaction. Instead, the buyer signs an authorization message that permits a transfer under specified conditions.

This comparison concerns the basic path in which an externally owned account (EOA) uses an EIP-3009 token. In a conventional transaction, the sending account pays gas and acts as the caller of the top-level call. Separating the payer from the gas sponsor therefore requires a distinct execution path, such as a token-authorization signature. This constraint should not be generalized to other EVM paths, including account abstraction and delegated accounts.

> EIP-3009 separates authorization of the funding source from the onchain transaction signature and moves it to an application-layer EIP-712 signature. The buyer grants only transfer authority; the facilitator uses it to submit the transaction and sponsor gas.

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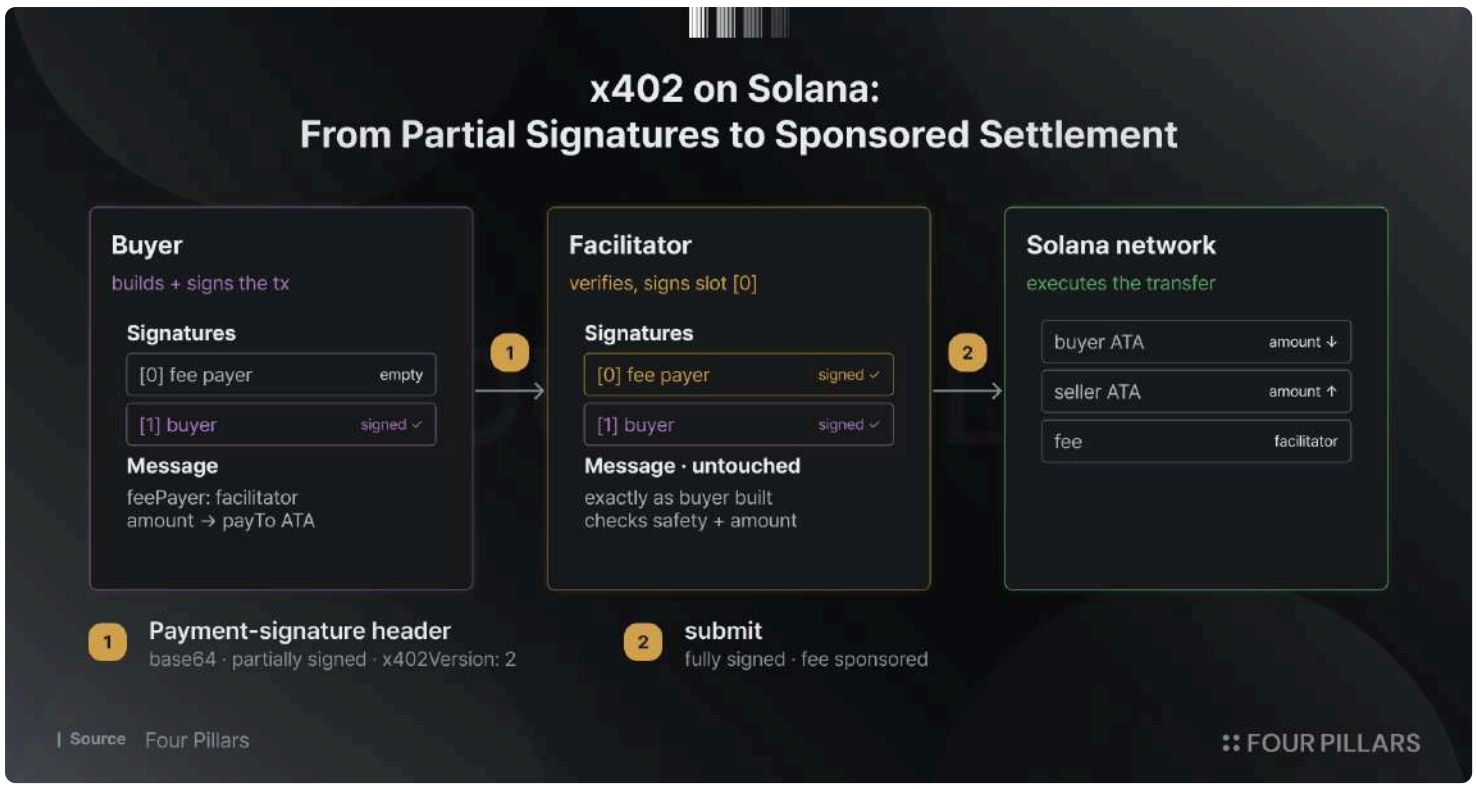

Solana supports this separation at the protocol level. A Solana transaction contains an array of signatures, and the first account in the message's account list is designated as the fee payer. Different parties can fill separate signature slots at different times. This enables partial signing, where the account transferring funds and the account paying the transaction fee sign the same transaction in sequence. x402 SVM payments use this structure directly.

> On Solana, SPL tokens are held in separate token accounts owned by a wallet, rather than at the wallet address itself. An Associated Token Account (ATA) is the default token account derived deterministically from the wallet address, token program, and mint. An owner can use another token account for the same mint, but x402 SVM payments designate the ATA of the seller's wallet as the destination account.

The buyer and facilitator complete a single transaction in the following order:

- **Review the payment requirements:** The buyer checks the recipient, amount, and other payment terms in the HTTP 402 response. Under the exact payment scheme, the buyer identifies the fee sponsor, either the seller or a third-party facilitator, through the PaymentRequirements.extra.feePayer field.
- **Create and send a partial signature:** The buyer sets that address as the fee payer, creates a partially signed transaction, and signs only the buyer's signature slot. The buyer then serializes the transaction and sends it in the payment header.
- **Verify and submit the transaction:** The facilitator simulates the transaction and verifies that the recipient and amount match the proposed terms. Once verification is complete, the facilitator fills the empty fee payer signature slot and submits the transaction to the network.

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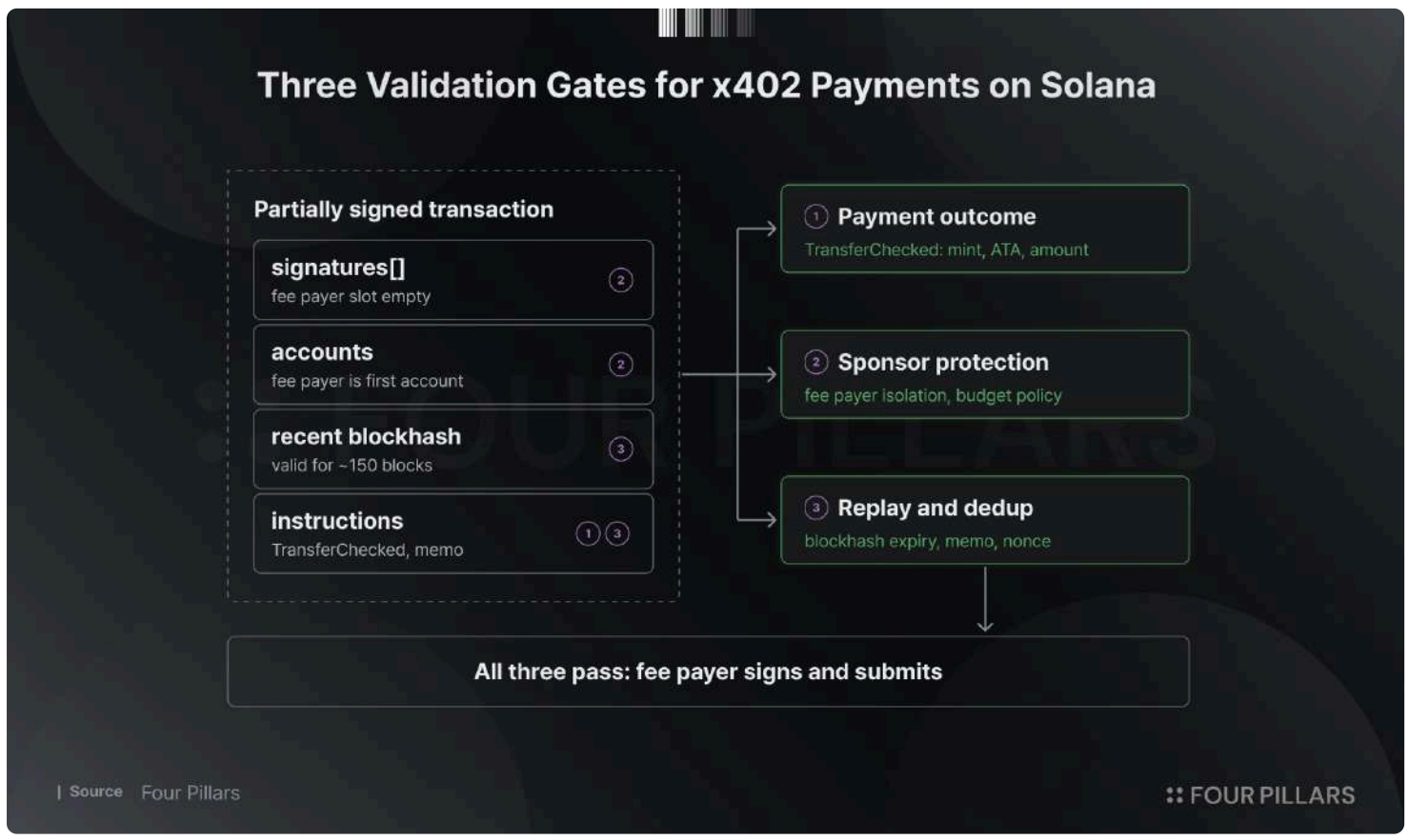
```jsonc
// Explanatory comments only — remove them before sending this data.
{
  // The seller defines exact payment terms on Solana.
  "scheme": "exact",
  "network": "solana:5eykt4UsFv8P8NJdTREpY1vzqKqZKvdp",
  "amount": "1000",
  "asset": "EPjFWdd5AufqSSqeM2qN1xzybapC8G4wEGGkZwyTDt1v",
  "payTo": "2wKupLR9q6wXYppw8Gr2NvWxKBUqm4PPJKkQfoxHDBg4",
  "maxTimeoutSeconds": 60,
  // These fields name the fee sponsor and constrain transaction validity.
  "extra": {
    "feePayer": "EwWqGE4ZFKLofuestmU4LDdK7XM1N4ALgdZccwYugwGd",
    "memo": "pi_3abc123def456",
    "recentBlockhash": "EZ3rST5dvHmbanh75jc4PuLfV96vp9fEYBVeNk4FfM1k",
    "lastValidBlockHeight": "291470237"
  }
}
```

The third step, verification, is central to SVM payments. The facilitator does more than compare the transaction against a fixed instruction array. It performs three sets of checks: whether the required payment result is achieved, whether the facilitator key is protected from misuse, and whether the same payment can be replayed or executed more than once.



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First, the facilitator verifies the payment result. It examines all top-level and inner instructions to confirm that TransferChecked from the SPL Token Program or Token-2022 Program executes exactly once for the required mint and destination ATA. For a standard SPL token without a transfer fee, the nominal transfer amount must also match the required amount. If the buyer uses a smart wallet, a TransferChecked instruction executed by the wallet program through a Cross-Program Invocation (CPI) may satisfy the same conditions. This remains subject to the sponsor's acceptance policy and post-settlement verification.

A Token-2022 mint may include a transfer fee extension that reduces the amount the seller receives below the nominal transfer amount, or a transfer hook extension that invokes another program during the transfer. The facilitator must therefore do more than read the instruction amount. It must inspect the mint's extension set and fee rules, verify that the seller's net received amount is at least the required amount, and confirm that any transfer hook and invoked program fall within policy. An implementation that cannot perform these checks must restrict acceptance to a verified mint list.

Second, the facilitator protects the sponsor. It checks that the fee payer account is not used as a token source account or transfer authority. It also confirms that the compute budget and invoked programs remain within policy limits. For the same reason, the current x402 SVM exact profile requires the destination ATA to exist and be usable when the transfer occurs. The standard transaction produced by the reference client does not create the ATA, and the sponsor isolation rules prohibit the sponsor's fee payer key from funding ATA creation. The facilitator therefore does not automatically cover this cost. The seller must create the destination ATA in advance.

> On Solana, token accounts, including ATAs, occupy separate onchain storage. Creating an account requires a rent-exempt minimum deposit proportional to the number of bytes it occupies.

Third, the facilitator prevents replay and duplicate execution by checking the Recent Blockhash's validity. The official Solana documentation explains expiry in terms of processing age and block progression; implementations use the lastValidBlockHeight returned by getLatestBlockhash. This expiry acts as a practical upper limit on maxTimeoutSeconds on Solana. A Durable Nonce Account, a general Solana feature, may be considered when a longer authorization window is needed.

> This is not a drop-in feature of the current x402 reference transaction and requires separate client and facilitator implementations.

While the blockhash remains valid, validators retain recently processed transaction signatures in a status cache and reject signatures they have already seen. Identical payment terms and messages can produce identical signatures, so a seller memo or random nonce must make each payment message unique. The facilitator must also maintain a short-lived duplicate-settlement cache to prevent repeated /settle requests before the first submission is confirmed.

In an SVM environment, the client signs the serialized transaction message itself. Any change to an account, recent blockhash, or instruction after signing invalidates the signature. The sponsor can only add the signature of the fee payer designated in advance. This gives the client full transaction-level visibility while requiring it to construct the correct transaction within a short validity window.

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x402 accepts these trade-offs as properties of each network layer while keeping the higher-level payment flow consistent across EVM and SVM.

3.2 Can card networks become x402 networks?

Cards already have broad merchant acceptance, consumer protection, fraud detection, and dispute-handling systems. Their role in global e-commerce includes both direct payments and use as the underlying payment method for digital wallets. Cards are therefore unlikely to become peripheral as agentic commerce expands. If existing acceptance and risk-management systems can be adapted for automated payments, this report expects card networks to remain a major payment method in agentic commerce.

Visa and Mastercard are Premier members of the x402 Foundation. Visa's Trusted Agent Protocol (TAP) and Visa Intelligent Commerce (VIC), together with Mastercard's Agent Pay, are preparing systems that connect agent identity, user delegation, and payment authorization to existing card networks.

Visa also launched stablecoin settlement in the United States in December 2025, allowing issuers and acquirers to meet VisaNet settlement obligations in USDC on Solana. In April 2026, it expanded support to nine blockchain networks, including Base, and recorded an annualized settlement volume of $7 billion. Mastercard is also expanding systems that settle card transactions onchain in regulated stablecoins. As card networks incorporate stablecoins and blockchains into their own settlement layers, integration in the opposite direction is also worth examining.

> **"Could x402, a payment protocol close to being blockchain-native that has primarily used stablecoins for settlement, use card networks as another settlement rail?"**

Nevermined's announcement on April 8, 2026 presented a demonstration connecting Visa Intelligent Commerce, VGS, and existing payment service providers (PSPs) to an x402 request flow. A cardholder delegates payment authority under conditions such as a total budget, a per-transaction limit, permitted merchants, and an expiry time. Nevermined then issues the agent a signed token proving that authority.

The virtual-card examples in §4.1.2 illustrate how spending limits and usage conditions can be set on existing card networks. In addition to these spending controls, this section examines how funding and card credentials can be connected to the x402 request and verification flow.

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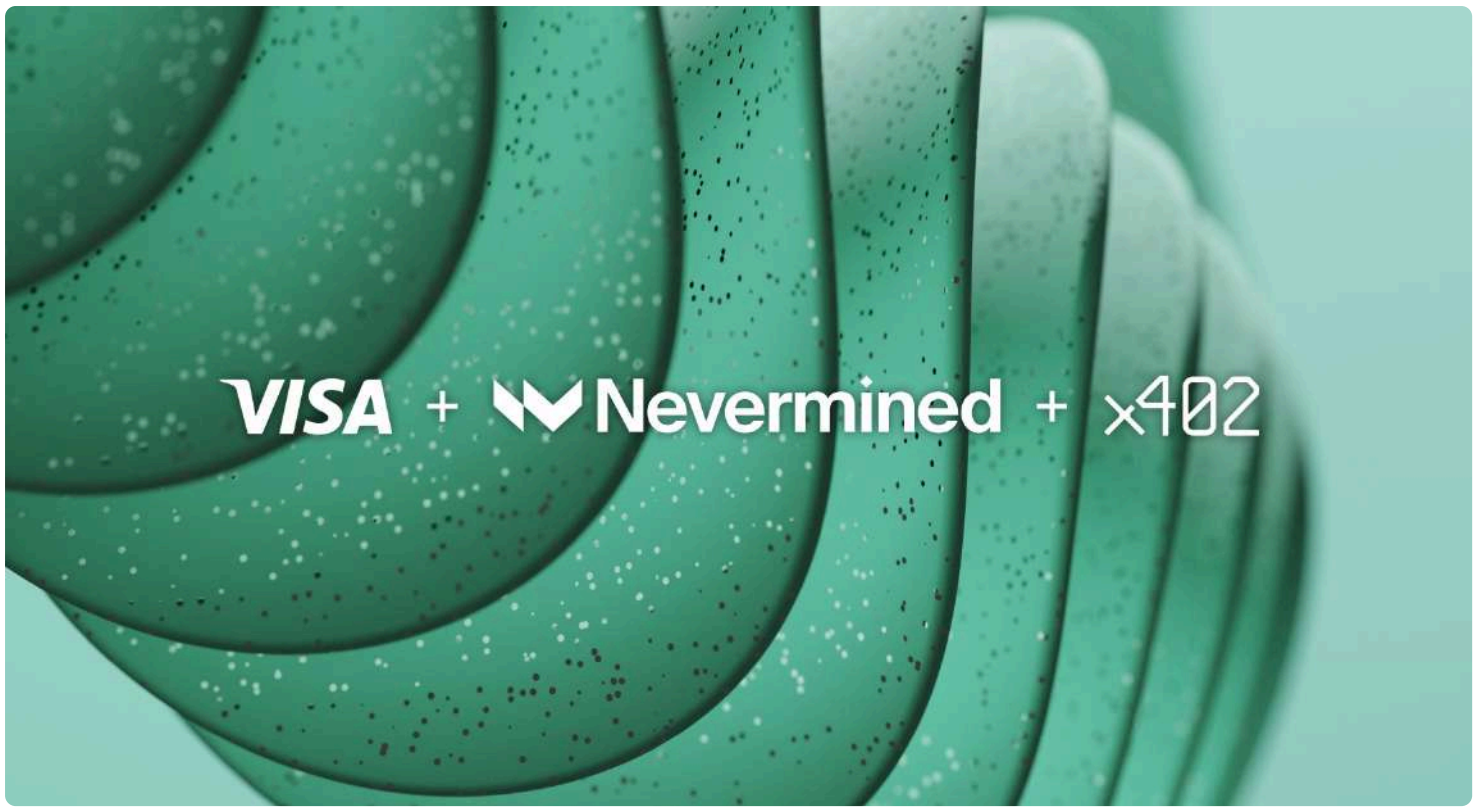

Source: Nevermined

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When an agent requests a paid resource such as an article or API and receives 402 Payment Required, it repeats the request with this token as payment proof. Nevermined verifies the token's signature and expiry, checks that the amount and merchant fall within the delegated scope, and processes the card payment through Visa and the PSP. The merchant receives fiat through its existing card-processing infrastructure, and the agent receives the resource. A separate card authorization does not necessarily occur for every small API call: the system also uses credits funded by card and deducted per call.

Here, x402 acts as an interface through which agents and merchants exchange prices, payment proofs, and processing results. Nevermined performs a facilitator-like role connecting that interface to the existing card network. This implementation is Nevermined's nvm:card-delegation extension, rather than a standard card-payment scheme shared by multiple providers.

Making card payments a common x402 path therefore requires further standardization of credential formats, verification and settlement results, expiry, and retry rules. The following card-payment draft proposes functions needed for these shared rules.

3.2.1 The card-payment draft carries network tokens in HTTP requests

The Machine Payments Protocol (MPP) uses Payment HTTP Authentication to exchange pricing terms and payment credentials. The server identifies the payment method through method and the processing approach through intent; the client returns proof satisfying those terms.

“Card Network Charge Intent for HTTP Payment Authentication,” authored by Visa's J. Brans, defines a one-time card-payment procedure using the card and charge combination. An agent delivers an encrypted network token, and the merchant-side payment provider decrypts it and requests authorization through the existing card network.

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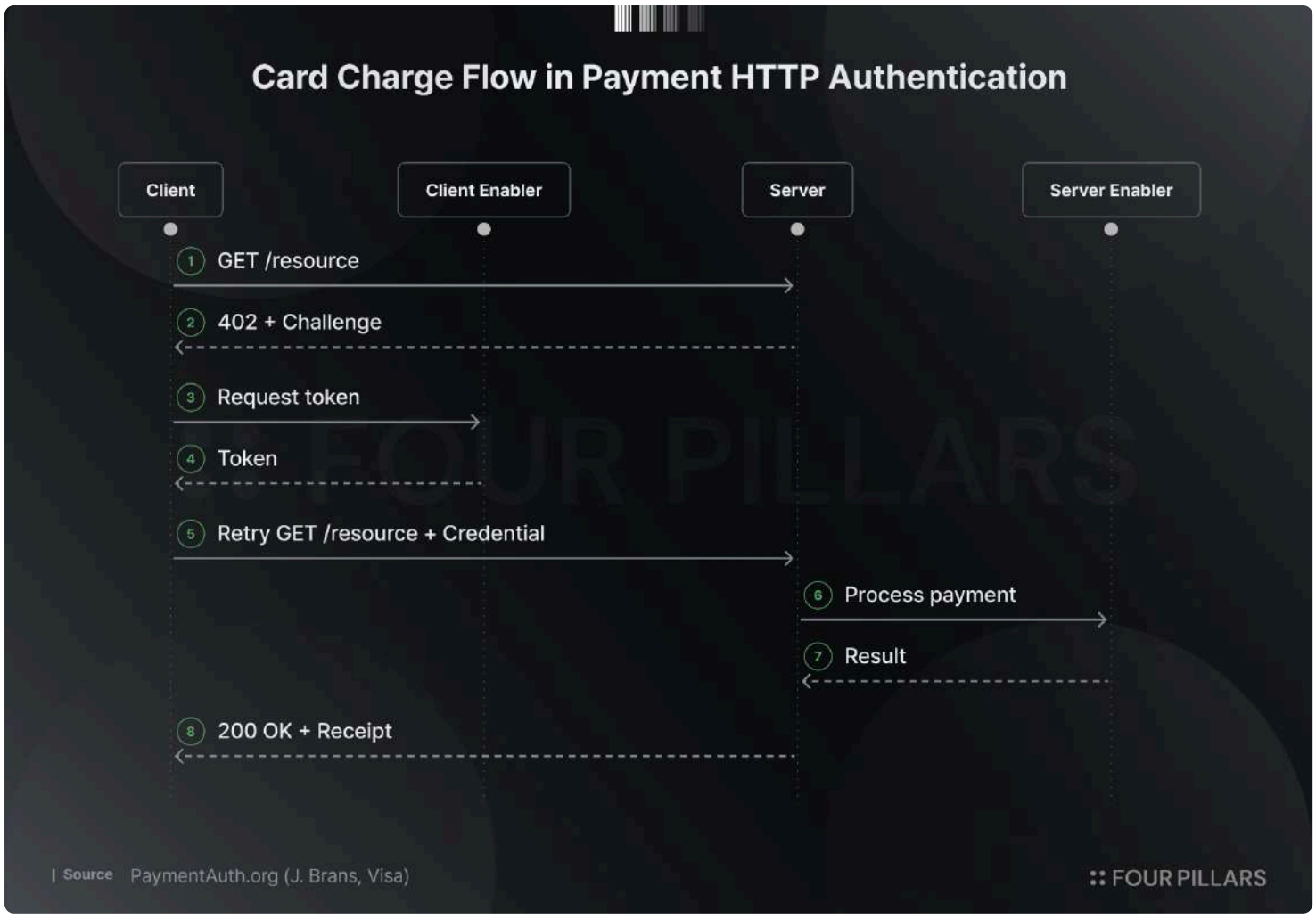

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When a buyer calls a paid API, the card payment proceeds as follows:

1. **The server quotes a price.** The WWW-Authenticate: Payment header in the 402 response contains a challenge identifier and expiry. The request object specifies the amount, currency, permitted networks, merchant details, and encryption public key.

2. **The buyer-side payment module creates a credential.** The client sends a registered card identifier and the challenge to the Client Enabler. This module obtains a network token and transaction-specific dynamic authentication data from a Token Service Provider (TSP).

3. **The token is encrypted and the request repeated.** The buyer-side module wraps the token data in JSON Web Encryption (JWE) using the merchant-side public key. The client places the credential containing this ciphertext in Authorization: Payment and requests the same resource again.

4. **The merchant side verifies the request and returns the authorization result.** The server compares the submitted terms with its issued challenge, uses challenge.id as an idempotency key, and passes the request to the Server Enabler. This module handles decryption and card authorization, after which the server returns Payment-Receipt and the resource.

The application handles pricing and resource access, while payment providers handle token creation, decryption, and card authorization. Merchants retain their existing acquiring path while incorporating challenge verification and result recording into API operations. The payment integration point expands from the checkout screen to the resource request. See the specification for further details.

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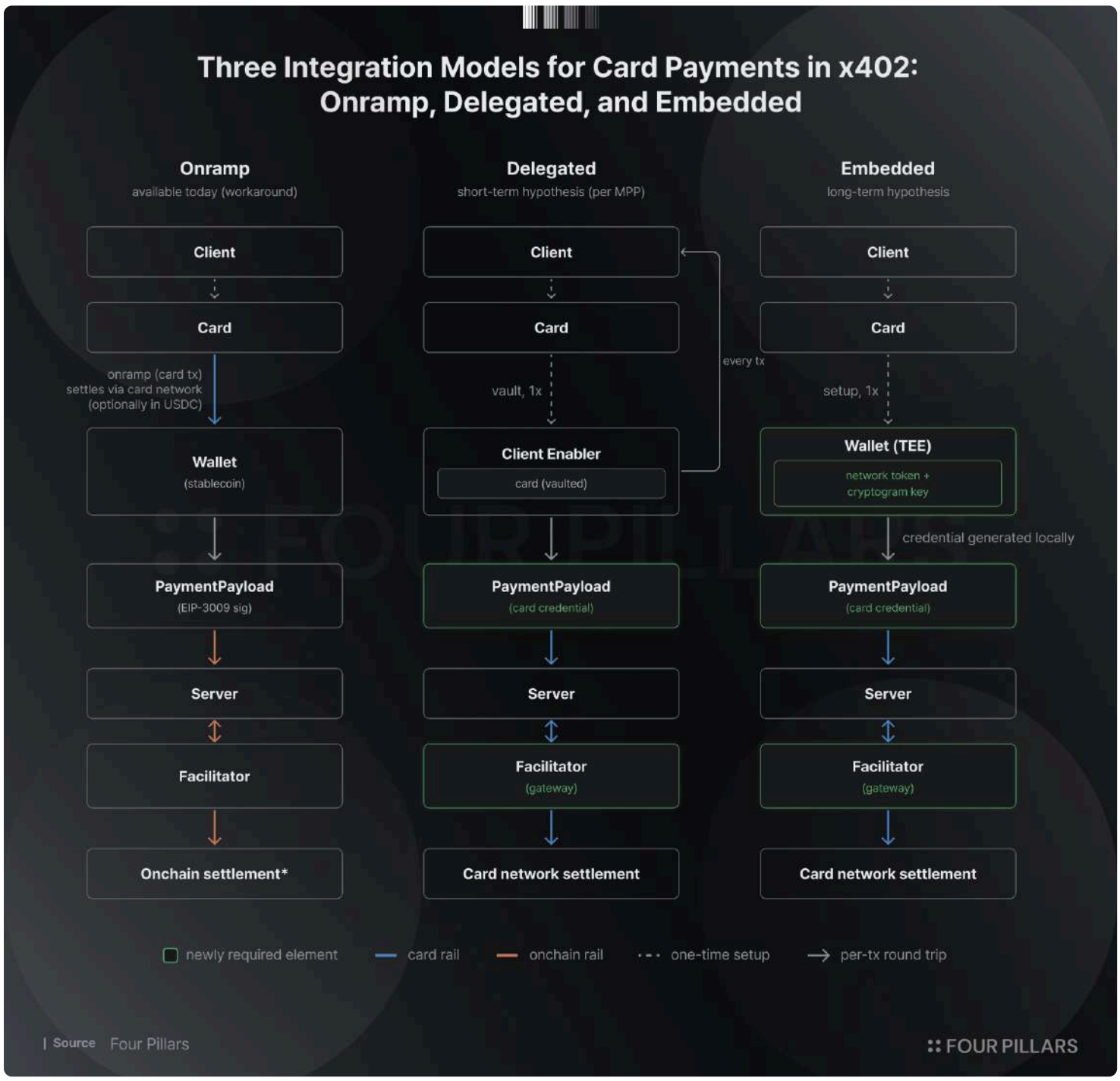
3.2.2 Three analytical paths: Onramp, delegated, and embedded

This section proposes three paths based on how funds are prepared and who creates the credential for each transaction. The onramp path uses existing onramps. The delegated and embedded paths are models proposed by the author, drawing on public card-payment structures to examine the conditions for applying them to x402.



- **Onramp:** A user purchases stablecoins with a card and funds a wallet; the agent uses that balance for x402 payments. The onramp provider is the merchant in the card transaction, while the resource provider is the merchant in the subsequent API transaction. Operators manage authorization, settlement, refunds, and dispute records for the two transactions separately.

- **Delegated:** This direct card-payment path reuses existing tokenization infrastructure. It assumes an external payment module creates transaction-specific card credentials and the client carries them in the x402 payload. Implementations must define a card-specific (scheme, network) combination together with verification fields, authorization results, and retry rules. The facilitator routes requests to a card processor and returns outcomes such as declined authorizations or duplicate requests to the resource server.

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- **Embedded:** This longer-term configuration places transaction-specific credential generation inside the wallet's trusted execution environment or secure element. The wallet can apply the same budget policy to card and onchain payments. It requires card-network-approved token environments and issuance procedures, key management and revocation, and execution-environment verification.

The integration scope expands from funding a wallet, to external tokenization services, to credential generation within the wallet. Developers can select a path based on the merchant's accepted payment methods, how users prepare their funds, and the support offered by card-processing partners. The embedded model has the broadest integration scope: wallet providers may expand their services to include policy enforcement and credential generation. Token issuance uses the network and TSP systems, while wallet providers bear the operating costs of key protection, authority revocation, and error handling.

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3.2.3 Card acceptance in x402 and the conditions for wider adoption

x402 is actively pursuing card acceptance. At its launch in July 2026, the x402 Foundation adopted support for the card-payment ecosystem as a central roadmap item. Global payment networks including Visa, Mastercard, and American Express participate as Premier members. The Foundation has also established a Card Acceptance Working Group to develop card-payment infrastructure for x402. Four areas of infrastructure must nevertheless be established before x402 card payments can become widely usable:

- **Common specifications:** Standardize card-payment objects and verification rules.
- **Reference implementations:** Establish consistent authorization and error handling between clients and facilitators.
- **Provider integration:** Secure technical support from tokenization and card-processing partners.
- **Operating rules:** Standardize transaction retries and dispute-resolution procedures.

With this infrastructure in place, agents could submit credentials through a common interface, while merchants accept transactions through their existing card networks. Three developments merit attention:

1. Whether the Card Acceptance Working Group's specifications extend beyond payment objects and verification to authorization, acquiring, and disputes.

2. Whether Visa and Mastercard release working, general-purpose implementations on top of those specifications.

3. Where x402's structure diverges from MPP's approach to card payments.

Combining x402's blockchain identity with traditional card payments may initially seem unfamiliar, but this is a necessary direction if agentic commerce is to evolve into a broadly usable payment system. Adoption remains early, with no complete standard or broadly commercialized implementation yet in place. As global payment organizations begin integration work in earnest, card acceptance could become a critical turning point for the ecosystem's expansion.

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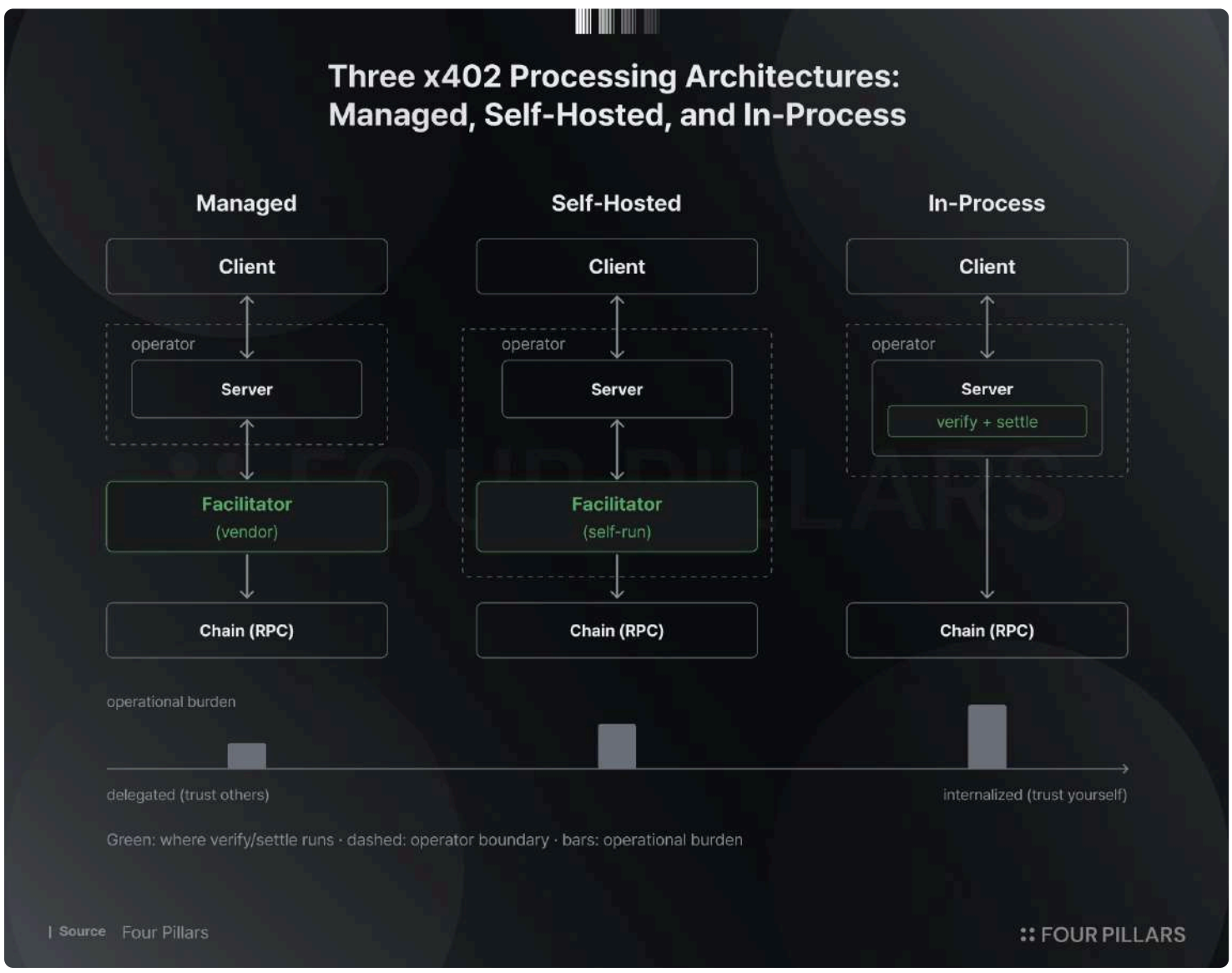
3.3 Is a facilitator necessary?

The payment flows discussed so far assume that a facilitator handles verification and settlement. The x402 protocol does not require a facilitator. A resource server that implements chain-specific signature verification, RPC connections, transaction submission, and confirmation logic can perform verify and settle locally.

Facilitators have become a common operating model because of operating costs. A resource provider that implements x402 directly must manage the full payment infrastructure, including network-specific verification rules, node connections, transaction-fee sponsorship, key management, and settlement confirmation. A facilitator groups these functions behind a common interface. The resource provider can begin by connecting payment endpoints, and the client does not have to handle operating differences among networks.

The need for a facilitator is therefore not a binary question. The choice is whether responsibility for verification and settlement sits outside or inside the organization. There are three operating models.



- **Managed facilitator:** An external provider supplies contracted operating functions such as verification, RPC access, fee sponsorship, and settlement confirmation. Compliance support or loss compensation falls under a separate service scope; using a managed provider does not transfer all legal responsibility. This model is the fastest to adopt, but depends on the provider's supported assets, policies, incident response, and concentration risk.

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- **Self-hosted facilitator:** The organization operates the facilitator while retaining the same API boundary. It gains more control over policies and data, while assuming responsibility for chain operations, key management, fee budgets, and security patches.
- **In-process settlement:** The server application performs verification and settlement directly. This is simple with a narrow support scope. As networks and payment schemes increase, the application code becomes tightly coupled to payment infrastructure.

Moving from a managed facilitator toward in-process settlement brings verification and settlement inside the organization. Dependence on an external provider decreases, while the operating burden and responsibility for incidents increase. This spectrum describes the placement of payment-operating responsibility rather than the custody of assets.

The choice depends on the trust and operating capacity an organization already has. A new resource provider that handles verification and settlement directly must prove payment accuracy and fund safety on its own. A managed facilitator with an operating history and chain infrastructure may be more cost-effective. A large provider with its own security team and chain-operating capacity may choose a self-hosted facilitator or in-process settlement to reduce external dependence and control its policies and data.

A facilitator is an optional layer that separates operational complexity from the protocol. Each organization should select a deployment model that balances the cost of trust with the cost of operation.

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Chapter 04

The position of x402 in agentic commerce

4.1 T1–T3: How far are purchase decisions and execution delegated?

4.2 How will agentic commerce develop?

4.3 A scenario model for x402 payment volume in the 2028 AI-intermediated B2B market

4.4 Which layers convert x402 payment flows into economic value?

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Agentic commerce is a form of commerce in which AI agents receive authority from people or companies to discover and compare goods and services, then perform part or all of ordering and payment. As agents enter enterprise workflows, their role can expand from supporting purchase decisions to directly procuring external resources needed to complete a task. This section focuses on **B2B transactions, including enterprise purchases and resource procurement by software**.

> Agents can also assist with purchases and execute payments in B2C commerce. This report nevertheless assumes that approval of individual purchases and existing checkout flows will remain prevalent through 2028, limiting the extent to which agent involvement in consumer purchases translates into x402 payment volume. The quantitative analysis therefore focuses on B2B transactions, where purchase conditions and budgets can be specified in advance and recurring resource procurement can be delegated.

Enterprise agents can procure external resources in several ways. An agent might compare suppliers, propose a purchase, and wait for an employee's approval. It might instead select and purchase APIs, data, or compute within a defined budget and authority scope. AI participates in both transactions, but its responsibilities and the payment requirements differ. Assessing x402's opportunity therefore requires looking beyond total AI-intermediated spending to the transactions agents actually execute and pay for.

The key distinctions are the extent of delegated purchase authority and the existing trading relationship. Using resources from a contracted supplier and settling the bill afterward does not require a new payment method simply because an agent is involved. By contrast, when an agent selects an external service during a task and must pay at the point of use, integrating resource access with payment becomes more useful. x402's applicability depends on the role payment plays in this procurement process.

The scope of delegable payment authority also depends on the transaction. Companies must define what resources an agent may purchase and how much it may spend, while considering the cost of verifying counterparties and outcomes. As more transactions meet these conditions, the range agents can execute directly expands. Potential x402 payment volume ultimately depends on which AI-intermediated transactions can receive delegated payment authority and which of those adopt x402.

The following subsections distinguish T1, T2, and T3 by the scope of purchase decisions and execution delegated to agents. They then examine each stage's trust costs and delegation conditions. Section 4.3 uses AI-intermediated B2B spending in 2028 as a starting point and narrows the scope of transactions to which x402 can apply to estimate potential payment volume.

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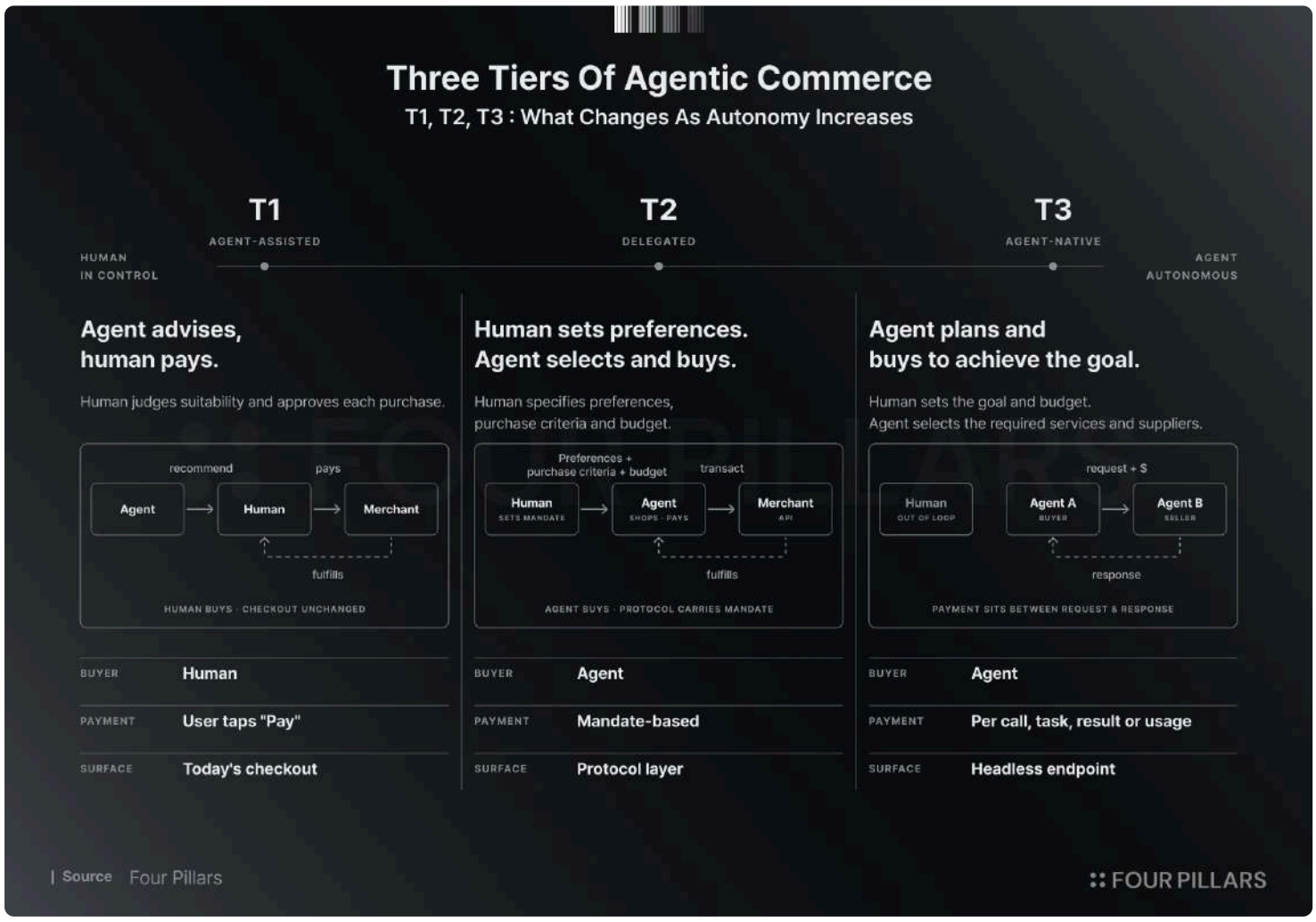
4.1 T1–T3: How far are purchase decisions and execution delegated?

Institutions classify agentic commerce in different ways. Stripe divides it into five levels based on the degree of automation. A single product or service is also difficult to classify under one fixed criterion. The same service is closer to purchase assistance when the user approves each payment directly, and becomes a delegated transaction when an agent pays autonomously within a preset limit.



This report distinguishes T1, T2, and T3 by the scope of purchase decisions and execution delegated to agents. Autonomy here includes not only the ability to process payments automatically, but also the authority to interpret the user's tastes and requirements and decide what to buy and from whom. The fewer purchase decisions remain with a person, the greater the agent's autonomy.

In T1, the agent assists with discovery and comparison, while a person approves the final purchase and payment. In T2, the user specifies tastes and preferences, a budget, and purchase conditions, and the agent selects and purchases products within that scope. In T3, a person provides a goal and budget, and the agent plans the purchases needed to achieve that goal and coordinates supplier selection and execution.

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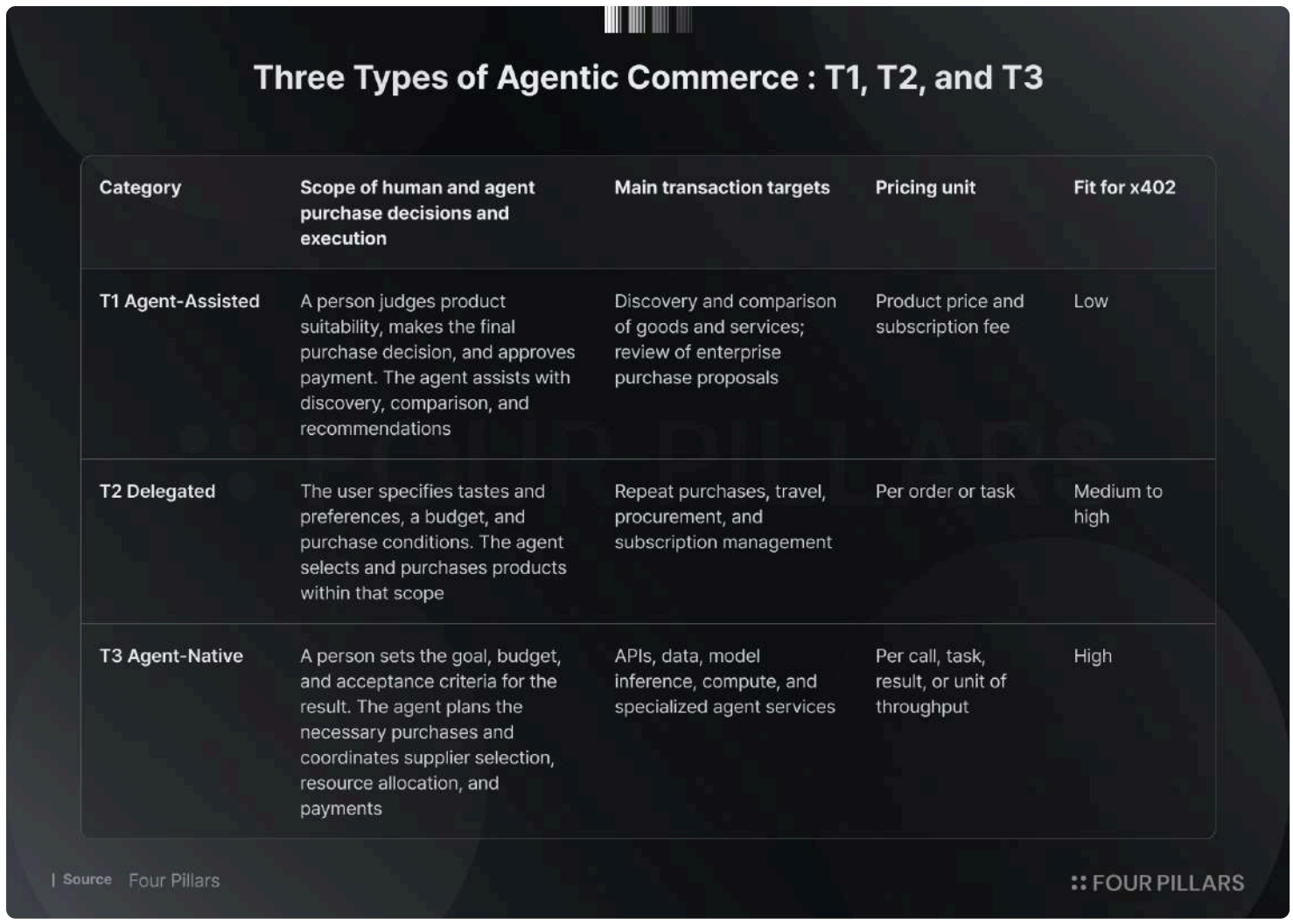

Even a purchase of data can fall into different stages depending on the scope of delegation. If a person reviews the candidates found by an agent and approves the purchase, it is T1. If the agent automatically purchases the data specified by a person under defined conditions, it is T2. If a person provides an analytical goal and total budget, and the agent independently assembles and purchases the data, inference, and verification services it needs, it is T3.

4.1.1 T1: The agent discovers and the person pays

In T1, the agent finds and compares products that meet a person's or company's needs and presents purchase candidates. A person judges their suitability and approves each final purchase and payment. Existing checkout, card, and account payments are likely to remain relevant even with conversational interfaces. At this stage, the agent's economic effect appears first in product discovery and purchase conversion, before it appears in a new payment rail.

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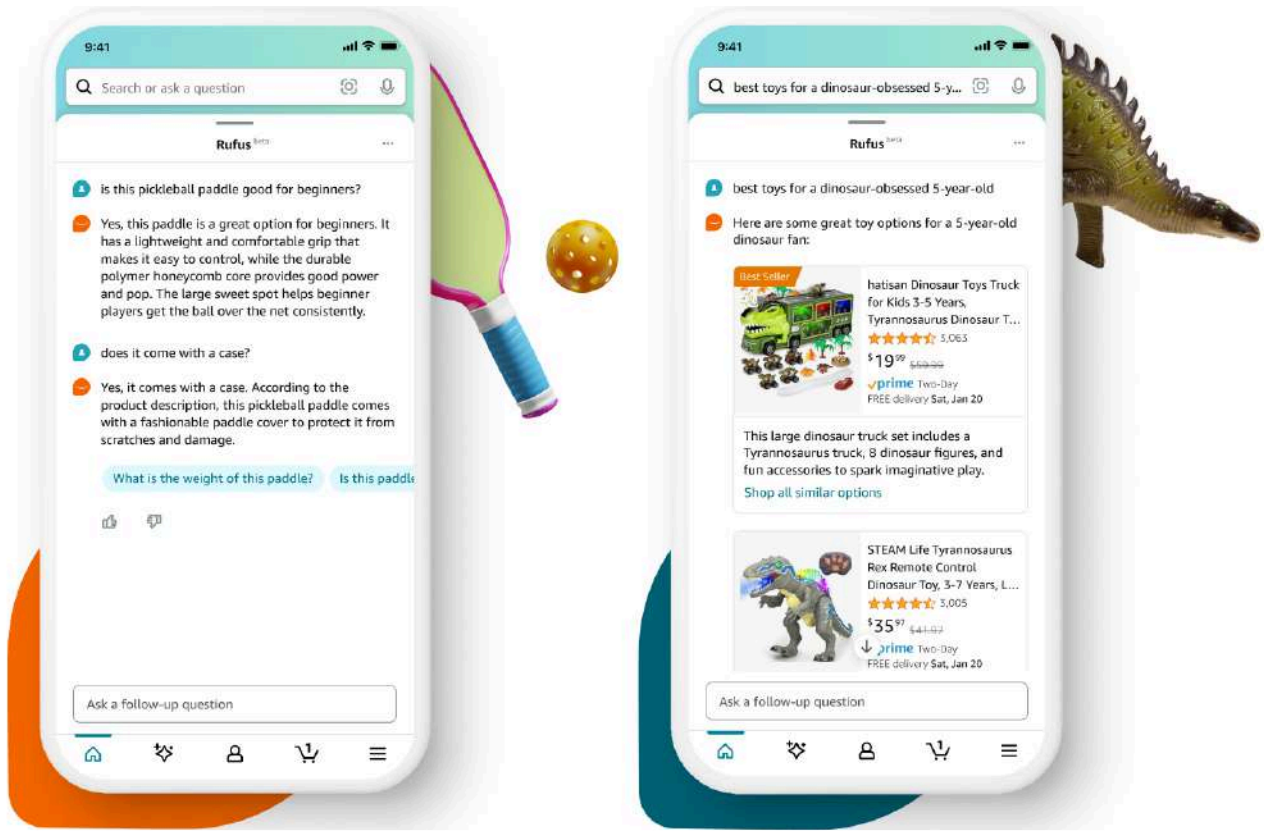

Source: Amazon

Amazon's Rufus is a representative T1 case. Rufus answers questions and compares products using product catalogs, customer reviews, and external information. Its Buy For Me function also assists with purchases from external stores. According to Amazon's fourth-quarter 2025 results, more than 300 million customers used Rufus, and the company estimated that it contributed about $12B in annualized incremental sales. An agent can therefore affect substantial transaction volume at the discovery and recommendation stage without executing payment directly.

4.1.2 T2: Delegated transactions reflecting the user's tastes and purchase conditions

In T2, the user specifies tastes and preferences, a budget, and purchase conditions, and the agent reflects them when selecting and purchasing products. A person provides the main criteria for judging what constitutes a suitable purchase, and the agent acts autonomously within that scope. How specifically the person must define conditions to capture their tastes and context determines the range of judgment that can be delegated.

Spending authority is specified through a total budget, per-transaction limits, permitted sellers and items, and a validity period. Who granted authority and under what conditions, whether the agent stayed within that scope, and which records establish responsibility after a problem are as important as the choice of payment method.

Google's Agent Payments Protocol, or AP2, records the user's intent and payment authority through electronically signed mandates. AP2 is a higher layer that connects to payment methods such as cards or x402 and provides authority management and audit records, rather than a rail that directly processes payment assets. A user who defines specific purchase conditions, records authority through AP2, and delegates execution to an agent creates a T2 configuration. As §6.3 discusses, AP2 can also express flows in which a person directly approves a transaction at the time of purchase; the stage depends on the actual scope of purchase decisions and execution delegated.

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Existing card networks also separate spending conditions from execution authority. Mastercard's virtual-card platform provides controls for spending limits, transaction limits, and validity periods. In an American Express One AP use case, a company sets spending limits and expiry dates on virtual cards for supplier payments and manages recurring payments.

These functions illustrate a structure in which a person defines the permitted use and a system processes individual payments. Connecting this structure to procedures that record an agent's purchasing authority and verify execution results allows delegation based on the user's tastes and purchase conditions to work alongside spending controls on existing card networks.

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Source: Alibaba

Alibaba's Qwen shows a move from T1 to T2. In a public test in January 2026, users ordered food and booked travel within a conversation but confirmed the final payment themselves. A Taobao integration in May 2026 expanded the scope. Users set a target price and preferred seller in advance, and Qwen placed the order automatically when those conditions were met. Payment authority moved from approval of each transaction to automatic execution under preset terms.

4.1.3 T3: The agent independently plans the purchases needed to achieve a goal

In T3, a person provides a goal, a budget, and acceptance criteria for the result, and the agent independently plans the purchases needed to achieve it. Within its permitted authority, the agent selects resources and suppliers, allocates the budget, and coordinates orders and payments without human approval for each transaction. Delegation expands from executing purchases under specified conditions to making the purchase decisions needed to achieve a goal.

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Procurement of APIs, data, model inference, and computing resources is a representative area where this autonomy may first become useful. The headless merchant endpoint is a supply structure that supports it by providing machine-readable prices, input specifications, and terms of use. An agent can interpret these conditions, compare the resources it needs, and connect calls to payments without passing through product pages or checkout interfaces designed for people.

In this resource procurement, x402 offers a direct advantage by reading prices per call or task and executing payments automatically. When one user request branches into data retrieval, model inference, acquisition of computing resources, and result verification, the agent coordinates the purchases needed for each task. This is a setting in which machine-to-machine economic activity takes shape, because resources can be obtained without repeatedly seeking human judgment and approval.

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4.2 How will agentic commerce develop?

Agentic commerce is spreading quickly at the T1 stage, centered on discovery and recommendations. T2 and T3, in which agents spend funds on behalf of people, remain at an earlier stage. Market forecasts are positive and several experiments are under way, but the pace of development will depend on how much payment authority people are willing to delegate to agents.

Section 4.2.1 uses Gartner's Digital Commerce Hype Cycle to examine technology maturity and the trust infrastructure needed to delegate payments. Gartner's technology categories are not the same as this report's T1, T2, and T3 transaction types. Section 4.2.2 then introduces the author's concepts of trust cost, the Delegation Zone, and net benefit. Section 4.2.3 applies this framework to headless merchant endpoints and presents them as a possible early T3 market.

4.2.1 Technology maturity and transaction-specific delegation conditions must be considered together

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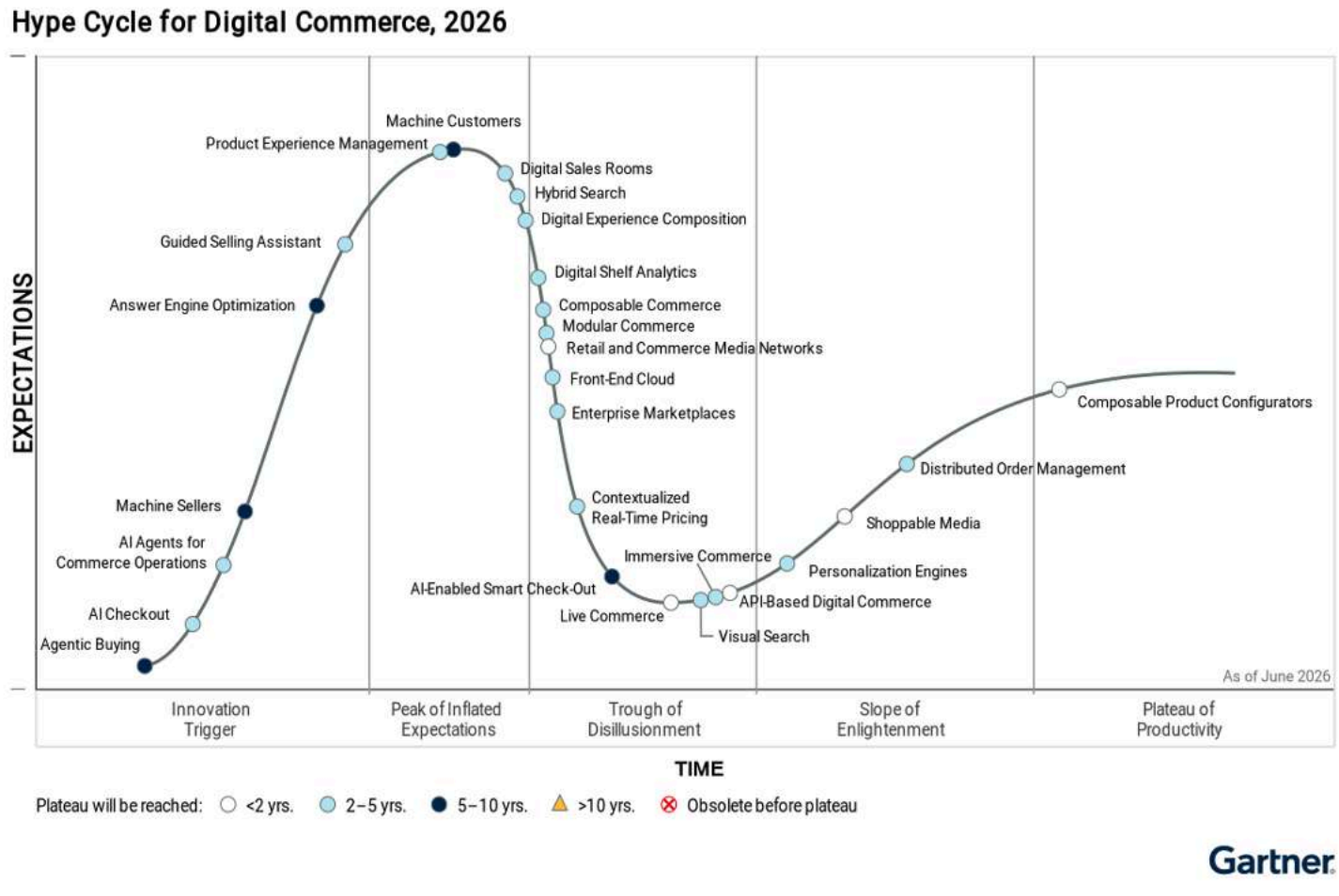

Source: Gartner

Gartner's 2026 Digital Commerce Hype Cycle estimates that AI Checkout and AI Agents for Commerce Operations will take two to five years to reach the Plateau of Productivity. Agentic Buying, Machine Sellers, and Machine Customers will take five to ten years.

> The Plateau of Productivity is the stage at which a technology's business value and conditions for use become clear and mainstream adoption gathers pace. Labels of two to five years or five to ten years indicate the estimated time until the technology reaches this stage.

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AI Checkout and AI Agents for Commerce Operations automate only part of the process while leaving product selection and final approval with a person or seller. Agentic Buying delegates discovery, comparison, and purchase execution to the agent. Machine Customers also determine need and execute transactions without human instructions for each one. Interpreting these differences in maturity requires considering delegated authority, failure costs, and operating conditions alongside technical readiness.

This forecast concerns technology maturity. It does not predict that this report's three transaction types must spread in a fixed sequence. Assisted services retaining approval for each purchase may be adopted first in consumer commerce, while autonomous execution within limited budgets may become useful earlier in purchases of APIs, data, and computing services. Adoption order depends on the transaction target and cost of failure as well as the scope of authority.

The main variable is a trust system that allows users to delegate payment authority with confidence. As transaction amounts rise and purchase terms become more complex, users are more likely to check the terms themselves and retain final responsibility. Even highly accurate recommendations will not lead easily to autonomous payment unless wallet custody, spending limits, permitted counterparties, authority revocation, transaction records, error recovery, and dispute procedures are clear.

For an agent to make purchases through x402 in place of existing payment methods, trust is needed at two levels. The first is trust in the actor: the agent must understand the user's intent and comply with the specified conditions and limits. The second is trust in the infrastructure: the wallet must safeguard assets, the payment infrastructure must verify and settle payments correctly, and transactions must be traceable and actionable when problems occur.

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First, establishing trust in the actor requires evaluating how accurately the agent performs an assigned task under specified conditions. Commerce-agent evaluations consider task success rates, fulfillment of the user's tastes and purchase conditions, total costs including retries, and result quality. The amount of additional human checking or corrective intervention needed to obtain a result of the same quality also matters. As that intervention declines, users can delegate more detailed purchase decisions to the agent.

Next, establishing trust in the infrastructure requires ensuring that granted authority is respected during actual payment execution. Wallets must protect assets and signing keys, while authority-management systems must enforce the permitted transaction scope, spending limits, and validity period. Servers and facilitators must verify and settle payments according to x402 payment terms and retain records that show the processing status at each stage when failures occur. Procedures for revoking authority or recovering access to assets, along with responsibility for incident response, must also be clear.

After payment, users must also be able to verify that the purchased service was delivered under the promised conditions. Operating records of supplier quality and availability can help assess the reliability of the service being called, while the Signed Offers & Receipts in §2.3.5 can provide a basis for comparing transaction terms and processing records after a problem. Managing these transaction records alongside evaluations of agent performance makes it easier to identify where a problem occurred, from purchase decisions through payment and service delivery.

Users can delegate greater autonomy when they can evaluate an agent's performance and the infrastructure can consistently enforce delegated authority. Even if the agent performs well, the scope of delegation will remain limited if asset protection and incident response fall short of existing payment methods. Together, these two levels of trust will determine how far purchase authority can expand.

x402 is a common interface for executing payments within the scope established in this way. If participants agree on how to verify credential issuers, ownership of signing keys, authority scope and validity, and revocation status, and process common payment objects, the burden of connecting agents, wallets, servers, and facilitators can fall. As these integrations become standardized, purchases across services can more easily maintain the same authority and control conditions, supporting the development of agentic commerce.

4.2.2 Trust cost, the Delegation Zone, and net benefit

Section 4.2.1 examined the institutional and technical safeguards required for greater delegation. Those safeguards alone, however, do not explain why users may accept the same level of delegation for one service but reject it for another. This subsection introduces a separate concept, “**trust cost**” to explain that difference.

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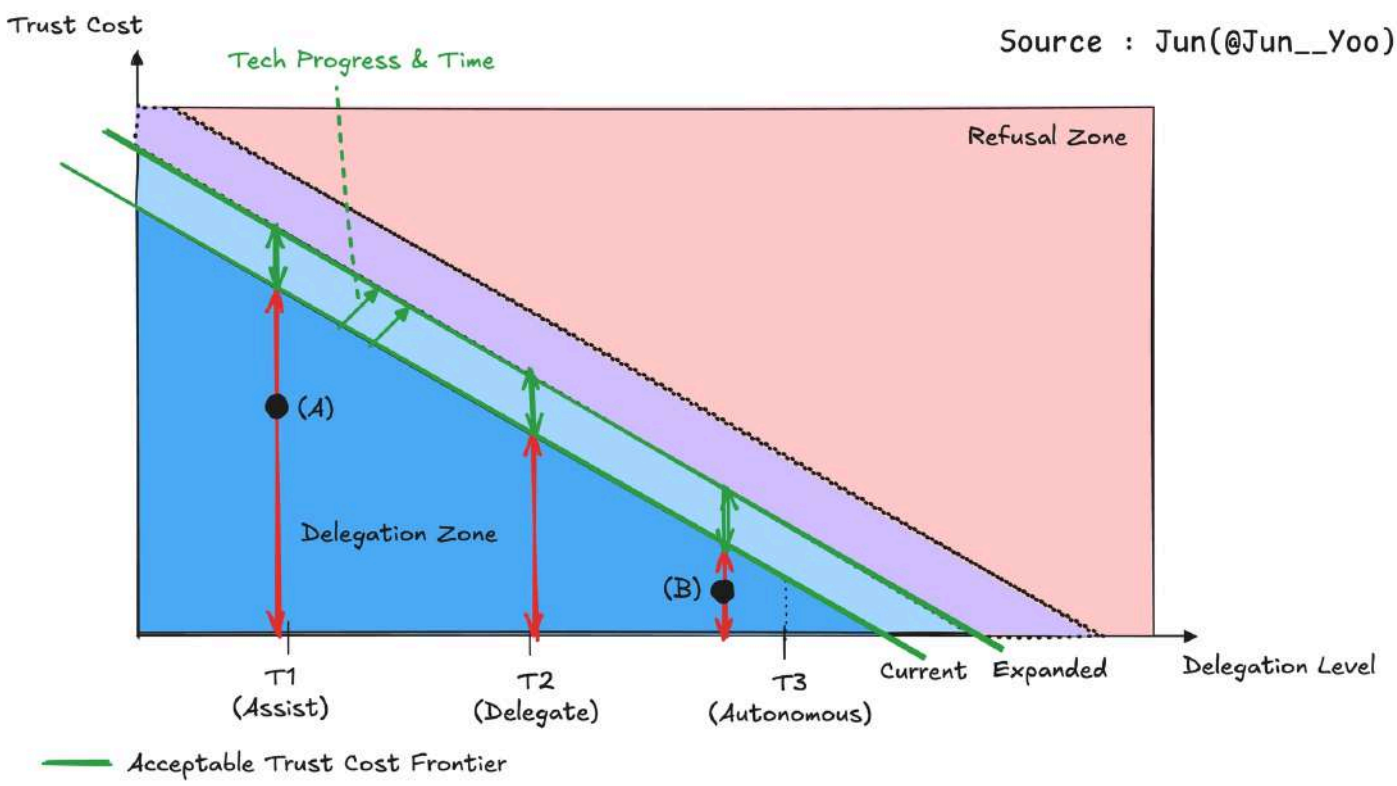
Consider several transaction contexts. For a $1,000 consumer purchase, many users would prefer AI-assisted checkout, in which they retain final approval, over fully autonomous checkout. The answer may change if the transaction involves $10 worth of data from an API endpoint. It may change further at $0.10 or $0.01. The level of delegation can remain the same while the user's willingness to accept it changes substantially.

This report defines the risk and burden accepted by the user as trust cost. Trust cost is the amount of risk and operational burden a user must accept when adopting a particular service under a particular delegation arrangement. It is influenced by the level of delegation, but it is not fixed by that level. Even at the same level of delegation, it varies according to transaction value, the scope of authority, the precision of spending controls, the verifiability of the result, the possibility of recovery, and the maturity of the underlying technology.

The trust mechanisms discussed in §4.2.1 are inputs that can reduce trust cost. They are not the same concept as trust cost itself. Authority records, wallet security, spending limits, revocation, and recovery procedures determine how much risk remains for the user after authority has been delegated.

Reflecting a user's tastes also creates this burden. If preferences change with context or the suitability of a purchase is difficult to judge in advance, the user must specify conditions in detail or check the agent's choices. To the extent that these judgments remain with a person, autonomy in the purchasing process is limited.

As agents become able to reflect preferences and acceptance conditions reliably, users can delegate a wider range of purchase decisions with less involvement in individual choices. Advances in interpreting tastes and evaluating results are conditions for expanding delegation from T2 to T3.



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The Delegation Zone is the range of service-and-delegation combinations that users are willing to accept at a given point in time. The figure's x-axis represents autonomy in purchase decisions and execution, while the y-axis represents the trust cost remaining for an individual service under those conditions. The Acceptable Trust Cost Frontier marks the maximum cost users will accept at each autonomy level. The area below this frontier is the Delegation Zone.

In T1, people intervene before approving the purchase and payment. In T3, they delegate more purchase decisions and execution to agents and may therefore require the remaining risks and burdens to be sufficiently low. Assuming this situation, the report depicts a frontier of acceptable trust cost that declines as autonomy rises. The trust cost of an individual service varies with transaction value, how readily preferences can be interpreted, spending controls, and recovery conditions.

Both the position of an individual service and the boundary of the Delegation Zone can change over time. Improvements in preference interpretation, result evaluation, spending controls, transaction records, authority revocation, and recovery procedures reduce the service's trust cost and move its position downward. A service that was previously outside the Delegation Zone can thereby enter it. As experience and trust accumulate, the maximum trust cost users will accept can also rise, expanding the Acceptable Trust Cost Frontier itself.

Entering the Delegation Zone does not determine adoption by itself. The benefit of automation must exceed the remaining trust cost. This report defines net benefit as benefit minus trust cost. Numerical comparisons require time saved, operating burdens, and residual risks to be converted into the same cost units.

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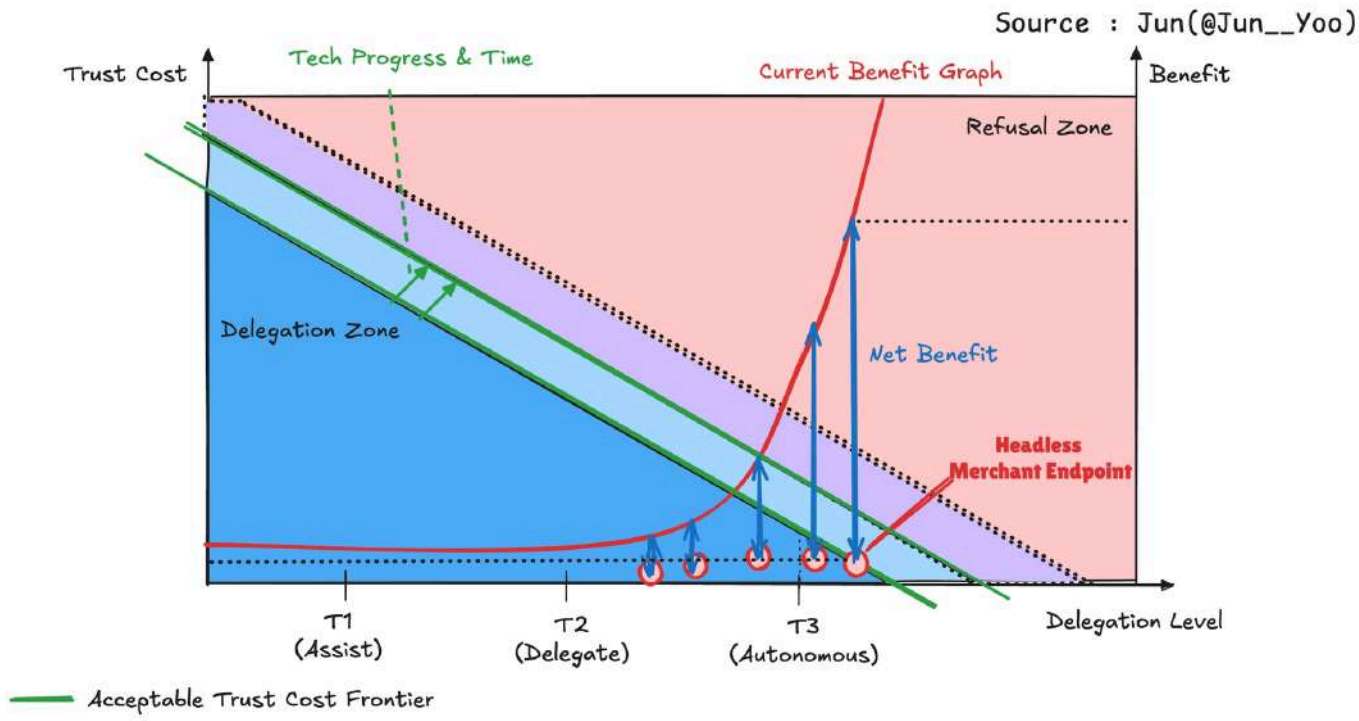

Benefit does not increase linearly with autonomy from T1 to T3. Even if purchase procedures are automated, the burden of judgment and supervision remains when a person must specify tastes and purchase conditions in detail and repeatedly review choices. A workflow that is 90% automated will not save 90% of the user's time if the person must remain available to review and intervene.

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In T3, where purchase planning and supplier selection can also be delegated, this repeated intervention can decline. To illustrate the effect, the report uses a conceptual model in which the current benefit curve increases nonlinearly at higher levels of autonomy.

The relationship between delegation conditions and net benefit gives rise to the following key requirements for service adoption:

- **Entry into the Delegation Zone**: The trust cost of the service must remain within the range users are willing to accept.
- **Positive net benefit**: The benefit of delegation must exceed the remaining trust cost.

An early T3 market is most likely to form where these two conditions overlap: the consequences of each failure are limited, while removing repeated human supervision creates an immediate and substantial benefit. A representative example is the market for headless merchant endpoints.

4.2.3 An early entry point for agentic commerce: T3 headless merchant endpoints

APIs, data, and computing resources often allow usage conditions and acceptance criteria for results to be specified relatively clearly. Once a user sets a goal and budget, an agent can compare the necessary resources and suppliers and plan purchases. The headless merchant endpoints in §4.1.3 are a supply structure well suited to early T3 activity because they connect these decisions to request-level payments.

The key is T3's relative advantage when individual payments are small and frequent. Where the cost of discovery and approval can easily exceed the value of a resource, T3 substantially reduces the human cognitive burden.

In this structure, the user sets a total budget, per-transaction limits, the permitted resource scope, and acceptance conditions for the final output, while the agent coordinates actual purchases within that scope. Limiting spending requires control over the budget for the whole task and costs accumulated through retries and parallel execution, as well as per-transaction limits.

If acceptance conditions for the final result can also be defined in advance, users can delegate resource purchases within an affordable spending range. For transactions meeting these conditions, headless merchant endpoints can reduce trust cost while delivering the benefits of autonomous execution.

Assume that an end user requests an output worth approximately $10. Producing it may require the agent to retrieve data, purchase model inference, acquire computing resources, call a specialized agent, and verify the result.

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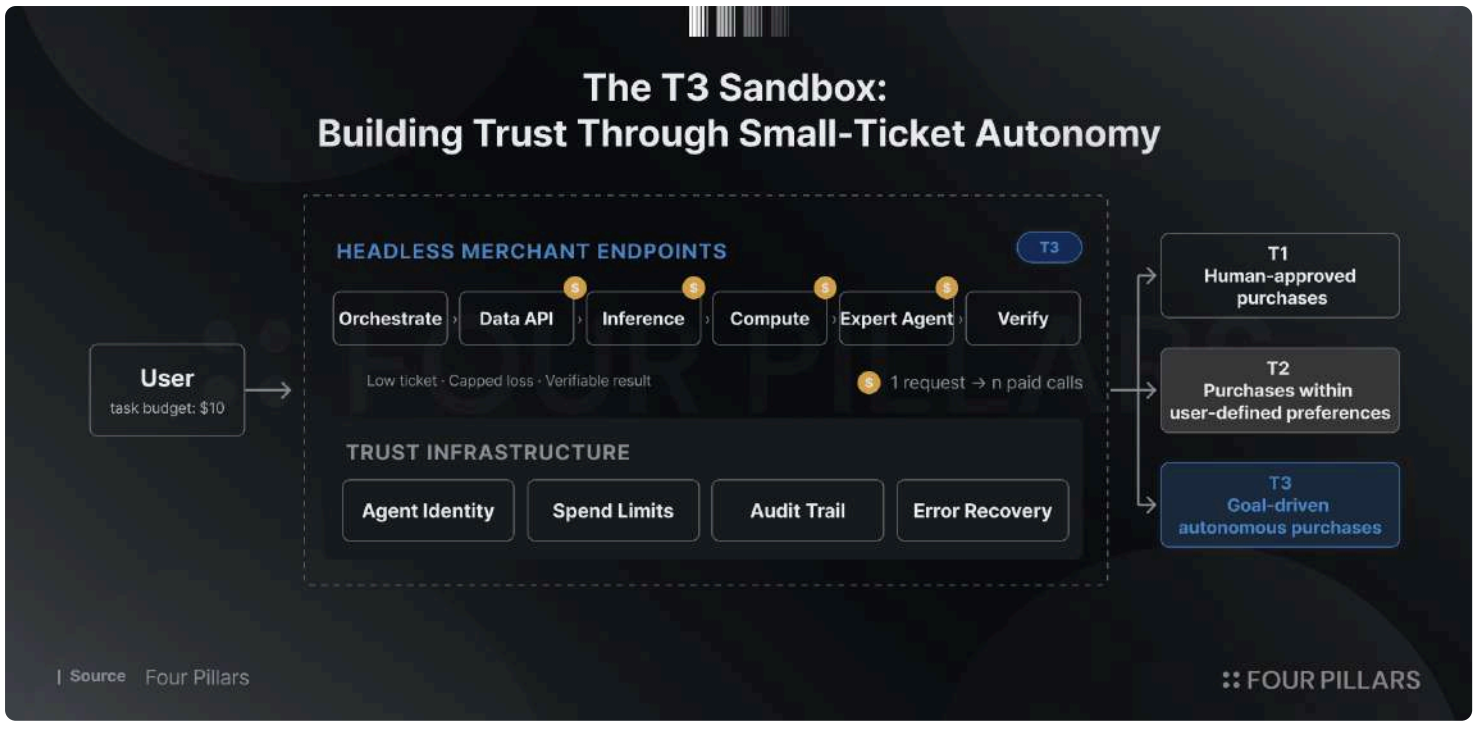

Orchestrator → purchase data API → purchase model inference → purchase compute → call specialized agent → verify result

If a person had to discover every supplier, compare prices, and approve every payment, the cost of search and authorization could exceed the value of the task. A single user request may also divide into several sequential or parallel external calls, with payment to a different supplier at each stage. In a multilayer structure where one agent calls other agents, additional depth and branching can increase the number of potential payment events rapidly.

The relevant point in this subsection is not the exact number of payment events, which §4.4.2 models separately. It is that a multi-step task creates a strong benefit from eliminating human approval at each stage. Full delegation changes the structure of the workflow rather than merely reducing the number of actions performed by a person.

This is also where x402 has its clearest initial fit. A headless merchant endpoint can present machine-readable payment terms and provide a resource within the same request flow. The agent can evaluate the price, authorize payment within its budget, and receive the resource without opening an account or moving through a checkout interface designed for people.

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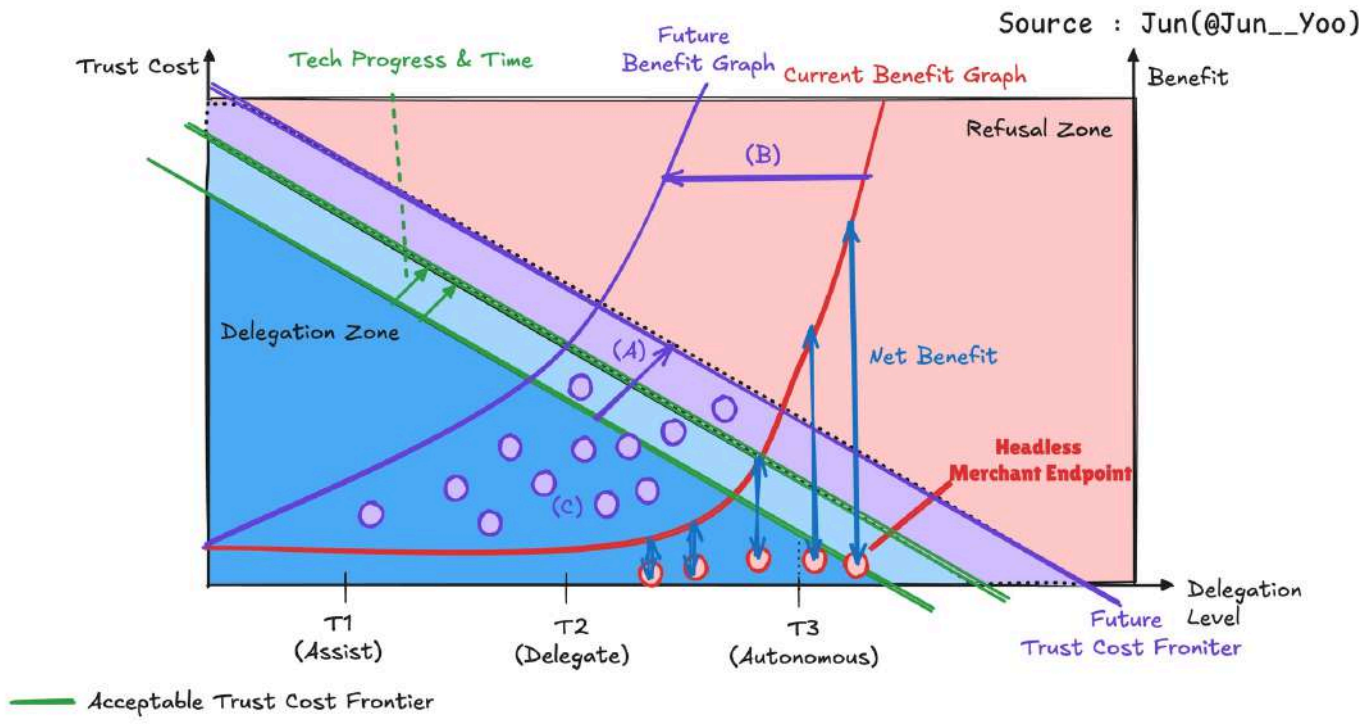

Headless merchant endpoints can therefore operate as an agentic-commerce sandbox. Frequent, low-value transactions allow delegation proofs, spending-limit enforcement, service identity, transaction records, result verification, and error recovery to be tested repeatedly while limiting the loss from individual failures.

Using data from repeated operations to improve preference interpretation, result evaluation, and spending controls can lead to the following changes in the adoption model in §4.2.2:

- **Expansion of the Delegation Zone (A):** As the boundary of acceptable trust cost expands, users can delegate a broader range of purchase decisions and execution.
- **Upward shift in the benefit curve (B):** At the same level of autonomy, a lower burden of specifying conditions, reviewing choices, and redoing work increases the benefit to users.
- **Entry into the Delegation Zone (C):** A decline in a service's residual trust cost or an expansion of the user's acceptance frontier brings previously unacceptable services into the Delegation Zone.

This creates a self-reinforcing cycle in which the T1, T2, and T3 markets grow together.

Early B2C e-commerce followed a similar path. Products such as books, music CDs, and software helped form the early online retail market because they were standardized, comparatively inexpensive, and easy to identify before purchase. ISBNs, titles, and authors allowed consumers to understand what they were buying without inspecting the physical item directly.

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Amazon repeatedly tested the full transaction process within this limited product category. Search and reviews reduced information uncertainty before a transaction. Payment security and order confirmation reduced payment risk. Delivery tracking and refund policies managed failures after the transaction. Trust cost did not disappear. Consumers shifted from trusting each seller directly to trusting the platform's verification process, transaction records, and recovery procedures.

Headless merchant endpoints may play a similar role in agentic commerce. APIs, data, and computing resources provide machine-readable transaction terms, prices, inputs, and outputs, and receipt and format of responses can be checked automatically. Content accuracy and usefulness for the task must be evaluated against separate acceptance criteria, so structured data retrieval and generative-model outputs require different verification procedures.

Repeated T3 transactions equipped with these evaluations and spending controls can build operating evidence for delegating larger amounts and a wider range of purchase decisions to agents in the future.

T1, T2, and T3 then interact and expand together. Their development can proceed in parallel, while evidence accumulated in early T3 markets strengthens the trust infrastructure in §4.2.1, broadens the scope of delegation, and encourages adoption across the agentic-commerce ecosystem.

Section 4.3 translates different rates of this parallel T3 development into payment-volume scenarios. Section 4.4 then examines how the resulting payment flows can be converted into economic value.

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4.3 A scenario model for x402 payment volume in the 2028 AI-intermediated B2B market

This is the report's central quantitative model. It takes Gartner's forecast of $15 trillion in AI-intermediated B2B spending in 2028 as a fixed total, then estimates the changing composition of T1, T2, and T3, x402's share of payment volume within each type, and the resulting payment volume.

By 2028, 90% of B2B buying will be AI agent intermediated, pushing over $15 trillion of B2B spend through AI agent exchanges

In this new ecosystem, verifiable operational data becomes a currency, fueling a data feed economy where digital trust frameworks and verifiability are prerequisites for participation. Products designed with composable microservices, API-first, cloud-native, headless architectures will establish a significant competitive moat. New commercial models will emerge, featuring high-frequency, frictionless sales powered by AI agents that can radically compress the sales cycle for a large range of business and technology purchases.

Source: Gartner

AI-intermediated B2B spending is first divided into T1–T3. The x402 processing shares in §§4.3.1–4.3.3 are then applied to each type and added together.

> The resulting payment volume is also the common baseline for the analysis of ecosystem participants' economic value in §4.4. All market values in §§4.3 and 4.4 refer to this B2B subset.

Gartner forecasts that by 2028, AI agents will intermediate 90% of B2B purchases and more than $15T in B2B spending will pass through AI Agent Exchanges. The model uses $15 trillion of that forecast as a fixed analytical total and calculates how purchasing and payment control might shift among T1–T3 within it.

> As noted above, B2C agentic commerce has a different market total and buyer population and is not added to these results. The output is conditional payment volume within AI-intermediated B2B commerce, rather than an upper bound for the entire x402 market. Any future combination with B2C figures must avoid double-counting consumer purchases and the subsequent B2B spending of merchants and platforms.

The initial allocation uses Gartner's survey of 645 B2B buyers. Of the respondents, 45% used generative AI in a recent purchase, 67% preferred a purchase experience without a sales representative, and 69% preferred validating AI-generated information with a sales representative. The model converts this coexistence of AI-based self-service and human validation into reference shares for T1, T2, and T3.

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Deriving reference shares from the survey

- T2 = 67% preferring a sales-representative-free purchase experience × 69% preferring sales-representative validation of AI-generated information = **46.23%**
- T3 = 45% using generative AI in a recent purchase × 31%, the complement of the preference for sales-representative validation = **13.95%**
- T1 = 100% − 46.23% − 13.95% = **39.82%**

For 2028, the T3 share is set first: 20% under gradual adoption, 25% under rapid adoption, and 35% under high adoption. The remaining share is redistributed between T1 and T2 in the ratio 39.82:46.23.

Converting the baseline into 2028 scenarios

- T1(s) = (100% − T3(s)) × 39.82 ÷ (39.82 + 46.23)
- T2(s) = (100% − T3(s)) × 46.23 ÷ (39.82 + 46.23)
- T3(s) = 20%, 25%, 35%
- Subsequent calculations use unrounded shares and market values.

- **Gradual adoption:** Humans remain involved in the final validation of high-value B2B transactions, while T3 expands mainly in selected digital-resource purchases with low failure costs.
- **Rapid adoption:** Improvements in agents' preference interpretation, result evaluation, and budget controls reduce the extent to which people must specify detailed purchase conditions or review choices. Some purchase flows that previously remained in T1 or T2 consequently move to T3, where agents purchase autonomously toward a goal.
- **High adoption:** Headless merchant endpoints and request-level payments spread widely, making T3 a major transaction type within AI-intermediated B2B spending. This scenario increases T3's share within the same $15 trillion market; it does not add new spending to that total.

The table applies the x402 processing shares in §§4.3.1–4.3.3 to each transaction type's market size. Percentages show the share of total spending within that type processed through x402; amounts show the resulting annual x402 payment volume.

x402 Penetration and Payment Volume Across T1–T3

Transaction type	Gradual adoption	Rapid adoption	High adoption
T1	0% / $0	0% / $0	0% / $0
T2	0.10% / $6.45B	0.50% / $30.22B	1.50% / $78.57B
T3	6.0% / $180.00B	17.5% / $656.25B	35.0% / $1,837.50B

Each cell shows x402 payments as a share of total spending in the corresponding category and the annual x402 payment volume.

Source Four Pillars

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Combined annual x402 payment volume in 2028 reaches **$186.45B** under gradual adoption, **$686.47B** under rapid adoption, and **$1.9161T** under high adoption. T3 accounts for approximately 96% of x402 payment volume in all three scenarios.

In this model, x402's core opportunity is concentrated in T3 flows, where agents handle purchase decisions and execution. Procurement of APIs, data, model inference, and compute represents an early application area. Payment volume is determined by multiplying the size of the T3 market, the share using open request-level payments, and x402 adoption within those transactions.

> Gartner's survey measures buyers' AI use and preferences, but not overlap between responses, actual payment authority, or the distribution of transaction value. The 39.82%, 46.23%, and 13.95% figures are therefore model baselines constructed by mapping survey results to T1–T3. The 31% complement of 69% cannot be interpreted as an autonomous-payment acceptance rate. Using these baselines, the model compares how x402 payment volume changes with delegated authority and payment-method adoption.

4.3.1 T1: Human checkout is excluded from the x402 market

In T1, an agent discovers and compares products and presents purchase options, but a person makes the final decision and payment. The payment is likely to pass through an existing e-commerce checkout and the infrastructure of card networks, banks, or major payment providers.

The model assumes that purchases at this stage are processed primarily through existing payment networks and therefore sets the x402 processing share at 0%.

Scope and assumptions

- Scope: T1 transactions in which a person approves the final payment
- x402 processing share: 0%
- T1 ×402 payment volume = T1 market × 0%

T1: Human-Approved Checkout in the 2028 Model

Scenario	T1 share	T1 market	x402 processing share	x402 payment volume
Gradual adoption	37.02%	$5.5531T	0%	$0
Rapid adoption	34.71%	$5.2060T	0%	$0
High adoption	30.08%	$4.5119T	0%	$0

Source Four Pillars

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This does not mean stablecoins cannot be used in T1. Existing checkouts can use them as a payment asset, a currency-conversion instrument, or a back-end settlement asset. Visa has also presented a B2B payment structure combining its existing merchant network with stablecoin settlement. Even if the payment asset changes to a stablecoin, the model assumes that processing through card networks will remain prevalent.

x402's strengths emerge when an agent reads a price at the request stage and executes payment without human approval for each transaction. T1 users still review a payment screen and complete checkout, providing less incentive to use these functions. To estimate potential x402 payment volume conservatively, the model therefore sets T1's x402 processing share at 0%.

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4.3.2 T2: The scope for x402 within existing payment networks

In T2, a person sets a budget and purchase conditions, and an agent places orders and pays within that scope. Delegating payment authority does not prevent a company from continuing to use its existing cards, bank transfers, platform wallets, or enterprise payment systems. Delegating purchase execution and changing payment methods are separate decisions.

The model reflects this distinction by first setting the share of T2 spending paid in stablecoins, then applying x402's processing share within that amount. This estimates payment volume while distinguishing stablecoin adoption from x402 adoption.

> As explained in §3.2, x402 can also accommodate card-network-based payments. This estimate, however, is limited to x402 payments using stablecoins.

Stablecoin payment shares are set at 1.0%, 2.5%, and 5.0% under gradual, rapid, and high adoption. Within those payments, x402's processing shares are set at 10%, 20%, and 30%, respectively.

> Both sets of percentages are the author's scenario assumptions for comparing payment volume at different adoption levels. They are neither observations of the 2028 B2B market nor direct forecasts from an external institution.

Scope and assumptions

- T2 ×402 payment volume = T2 market × stablecoin payment share × x402 processing share
- Stablecoin payment share: 1.0%, 2.5%, 5.0%
- x402 processing share within stablecoin payments: 10%, 20%, 30%

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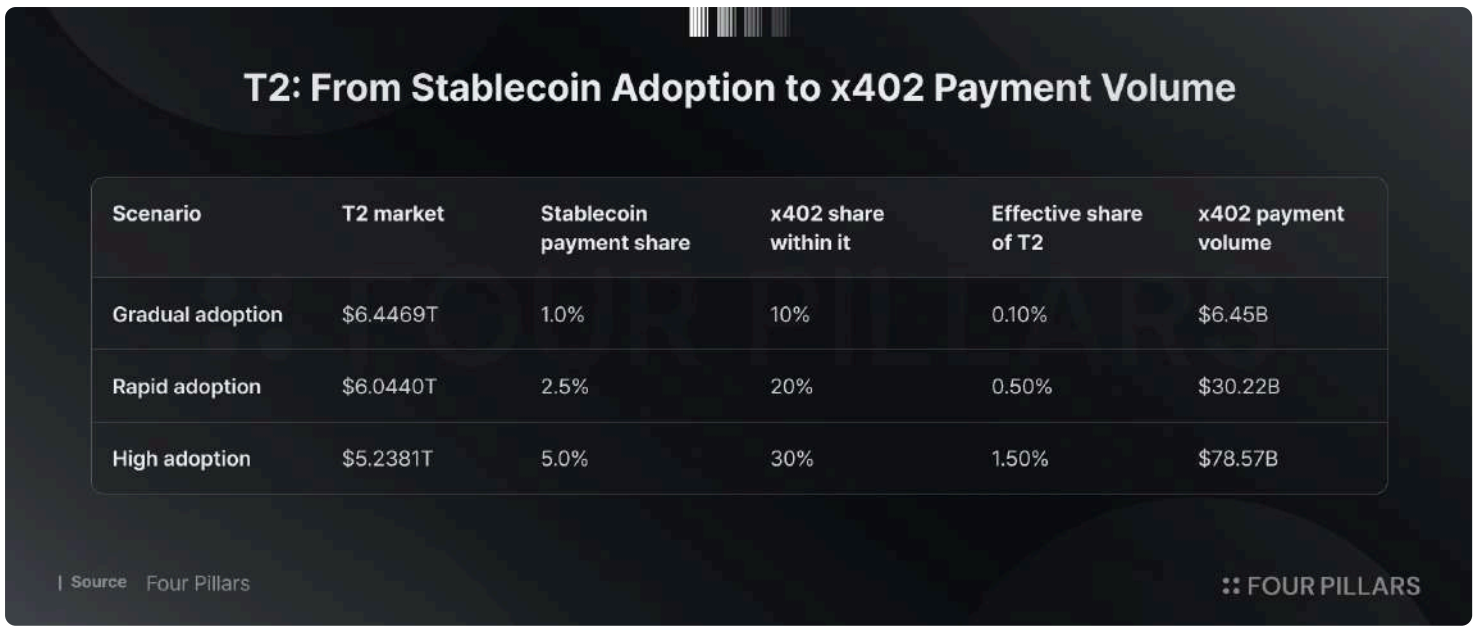

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Deloitte forecasts that stablecoins will support approximately 2.5% of U.S. noncash retail payments by 2030. This includes direct consumer payments and the use of stablecoins to fund or settle card transactions. A consumer may pay with an existing card and the merchant receive fiat, while stablecoins are used at some stage of the transaction.

The relevant insight is that stablecoin use can expand while existing payment networks remain in place. An enterprise agent could similarly place an order through an existing payment provider, which uses stablecoins for fund transfers or settlement. Stablecoin payment volume would grow without the transaction passing through x402.

This is why the model distinguishes the stablecoin share of T2 payments from x402's share within it. Companies must consider credit provision, refunds, dispute handling, accounting, and reconciliation when choosing payment methods. The scope of x402 adoption in enterprise purchasing will likewise depend on operating systems that verify delegated authority, recover from errors, and track transaction histories.

Deloitte's 2.5% forecast concerns U.S. retail payments in 2030 and does not directly support a 2028 global B2B adoption rate. The model draws on Deloitte's description of payment-network coexistence but sets adoption rates as separate scenario assumptions. Applying stablecoin shares of 1.0%, 2.5%, and 5.0%, followed by x402 processing shares of 10%, 20%, and 30%, produces effective x402 shares of T2 spending of 0.10%, 0.50%, and 1.50%.

4.3.3 T3: A market in which agents purchase directly

In this model, T3 is the market in which an agent plans and executes the purchases needed within a goal and budget set by a person. The shares of 20% under gradual adoption, 25% under rapid adoption, and 35% under high adoption represent autonomous purchasing as a share of AI-intermediated B2B spending, corresponding to market sizes of $3.00T, $3.75T, and $5.25T.

Procurement of APIs, data, model inference, and compute represents an area where x402 may first be applied within this market. Since autonomous purchasing can also use long-term contracts, subscriptions, prepaid credits, and internal platform ledgers, the model further distinguishes the portion of total T3 spending paid directly to external suppliers per request.

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For services with predictable demand, companies have incentives to lower prices and stabilize supply through long-term contracts. Individuals can also delegate service use within a prepaid subscription, while suppliers will want to retain subscriptions that provide recurring revenue and stable customers. Human-oriented contracts and subscriptions may therefore continue to account for a substantial share of T3 payments in 2028.

x402 offers its most direct advantage when agents discover external suppliers without prior contracts and pay per call or task. The model reflects this by calculating x402's T3 share as the product of two variables.

The first is the share of T3 payment volume in which agents pay external suppliers directly per request, excluding long-term contracts, subscriptions, and proprietary settlement systems. It is set at 20%, 35%, and 50% under gradual, rapid, and high adoption. The rapid-adoption baseline is 35%, while the high-adoption scenario caps open request-level payments at 50%, reflecting the assumption that contracts and subscriptions may remain mainstream in 2028.

This share represents the scope of open request-level payments across the purchases and trading relationships within T3. The model first identifies the portion of autonomous purchasing, including purchases of APIs, data, and computing services, that uses this payment method, then applies x402's processing share within it.

The second is x402's processing share within open request-level payments: 30%, 50%, and 70%. These scenarios allow for competition from card-based agent payments and other payment protocols, representing cases in which x402 is one of several payment methods, a major standard, or the de facto default.

Scope and assumptions

- x402 share of T3 = open request-level payment share × x402 share within it
- Open request-level payment share: 20%, 35%, 50%
- x402 share within open request-level payments: 30%, 50%, 70%
- T3 ×402 payment volume = T3 market × x402 share of T3

T3: x402's Share of Agent-Native Resource Purchases

Scenario	T3 market	Open request-level payment share	x402 share within it	x402 share of T3	x402 payment volume
Gradual adoption	$3.00T	20%	30%	6.0%	$180.00B
Rapid adoption	$3.75T	35%	50%	17.5%	$656.25B
High adoption	$5.25T	50%	70%	35.0%	$1.8375T

Source Four Pillars

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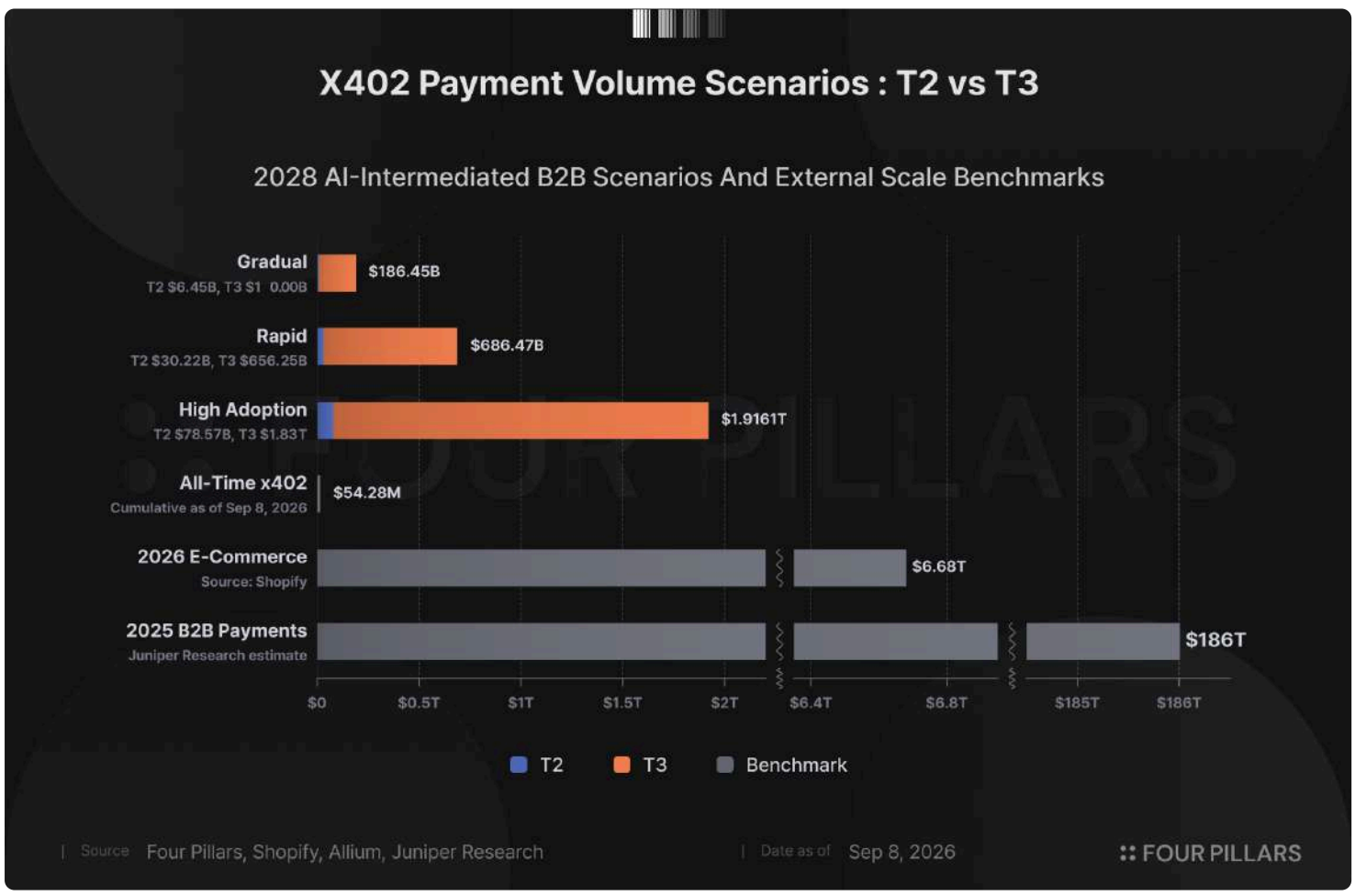
Multiplying the two variables gives effective x402 shares of T3 of 6.0%, 17.5%, and 35.0%. The corresponding payment volumes are $180.00B, $656.25B, and $1.84T.

4.3.4 Combined results: x402's expansion potential in B2B commerce

Scope and assumptions

- Scope: Annual x402 payment volume within 2028 AI-intermediated B2B spending
- Total x402 payment volume in the B2B scenarios = sum of T1, T2, and T3 ×402 payment volume = T2 ×402 payment volume + T3 ×402 payment volume
- Primary comparison benchmark: Juniper Research's estimate of $186T in global B2B payment volume in 2025
- Secondary comparison benchmark: EMARKETER's forecast of $6.88T in global retail e-commerce sales in 2026, cited by Shopify
- Current observed volume: $54.28M in cumulative x402 payment volume as of September 8, 2026



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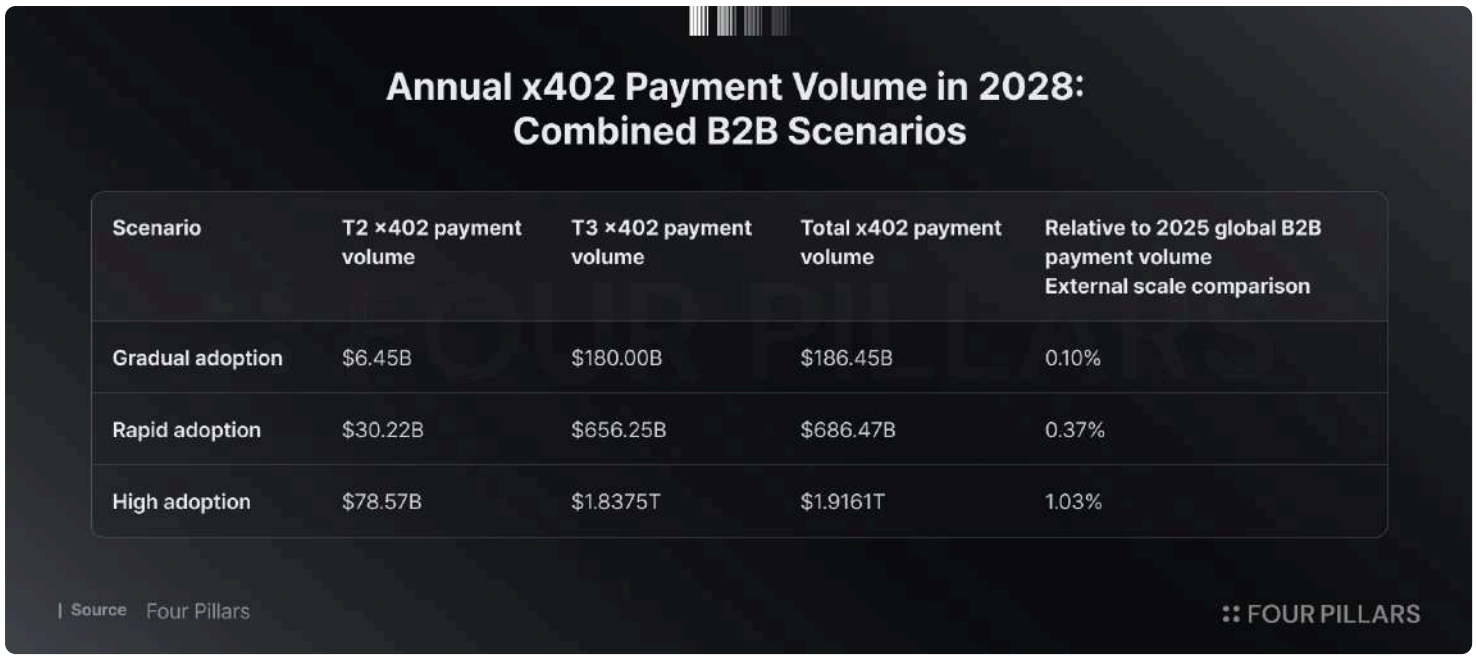

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T3 accounts for 95.6%–96.5% of total x402 payment volume across the three scenarios. This is the model's central conclusion. x402's expansion potential depends more on how quickly T3 markets open for agents to purchase APIs, data, models, and compute per request than on the displacement of existing B2B checkout. Annual payment volume within the B2B scope reaches $186.45B even under gradual adoption and expands to $1.9161T under high adoption.

To compare this scale with the existing B2B payments market, the report uses $186T in global B2B payment volume in 2025 as an external benchmark. The scenarios' x402 payment volumes are equivalent to approximately 0.10%, 0.37%, and 1.03% of that amount.

These amounts represent transaction principal paid to sellers through the protocol, rather than direct revenue for x402 or ecosystem businesses. The next subsection uses total payment volume to estimate how economic value can accrue to different ecosystem layers.

4.4 Which layers convert x402 payment flows into economic value?

Section 4.3 estimates annual x402 payment volume within AI-intermediated B2B commerce at $186.45B under gradual adoption, $686.47B under rapid adoption, and $1.9161T under high adoption. All layer-specific estimates below are derived from these conditional B2B payment volumes. These amounts are transaction principal transferred to sellers, not a revenue pool distributed among participants. Payment volume measures the funds that move through x402. A company's revenue arises when it converts a balance, transaction count, or order path under its control into a billable unit.

The x402 standard has no built-in protocol rent that collects a mandatory percentage of every transaction. Facilitator fees and onchain settlement fees may arise separately, but neither constitutes common protocol-level revenue. Network fees may be burned or distributed to validators, sequencers, or other participants depending on the chain, so they should not be treated as confirmed revenue for a single company.

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Without a built-in protocol fee, the five main layers in the figure convert payment flows into economic value through different control points and economic units.

- Stablecoin issuers earn reserve income from the share of payment demand that remains as a net increase in circulating balances.
- Blockchains charge network fees on the transactions produced when payment events are converted into onchain settlement.
- Facilitators charge for verification and settlement processing of managed payment events.
- Wallets control budget execution and asset paths before and after payment and provide ancillary financial services.
- Discovery and routing services monetize part of the payment volume passing through supplier discovery, selection, and order-allocation paths.

These conversion conditions are first applied to the three scenarios in §4.3. The calculation bases are net additional circulating balances for issuers, onchain settlement transactions for blockchains, processed events for facilitators, and routed payment volume for wallets and discovery and routing services.

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Where Economic Value Accrues in the x402 Ecosystem

Value-capture layer	Gradual adoption	Rapid adoption	High adoption	Output metric
Stablecoin issuers	$25.52M	$94.02M	$262.38M	Pre-expense reserve income on net additional circulating balances, using T2 ×402 dollar-stablecoin payment volume and 95% of T3 payment volume
Blockchains	$8.66M	$31.56M	$88.37M	User-paid network-fee benchmark using Base's average fee and a 500:1 settlement-equivalent ratio
Facilitators: Basic model	$0.76M	$2.77M	$7.75M	Potential revenue from charging for settlement-equivalent transactions
Facilitators: Managed model	$189.84M	$692.14M	$1.938B	Potential service-model revenue from charging for managed payment events
Wallets	$37.29M	$205.94M	$766.43M	Potential ancillary-service revenue from funding, asset exchange, and selection of funding and settlement routes
Discovery and routing	$55.93M	$205.94M	$574.82M	Potential revenue from supplier discovery, selection, and order allocation

| Source Four Pillars

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The annual economic value estimated for each layer in this analysis is derived using different economic units and scopes. Even when the same payment volume is used as the starting point, each layer uses different parameters and revenue-recognition criteria, so the results must be interpreted in light of those structural differences.

- **Stablecoin Issuers:** The model uses the T2 ×402 dollar-stablecoin payment forecast and 95% of T3 payment volume as dollar-stablecoin payment volume, converts it into a three-day average operating balance, and applies reserve yield only to the share representing a net increase in circulating balances.
- **Blockchains:** A user fee benchmark was used by converting 500 payment events into 1 on-chain settlement and applying the Base network's average fee ($0.0057). For the Solana network's exact path, the 500:1 conversion ratio was not applied, and a separate cost structure was used instead.
- **Facilitators:** The basic settlement-fee model charges on onchain settlement-equivalent transactions remaining after batch processing, while the managed-service model treats 50% of T3 payment events as billable. The managed-service estimate is higher, but the two figures represent different business models and should not be read as a single forecast range.
- **Wallets and discovery and routing services:** Of total x402 payment volume, transactions that actually use fee-bearing services for funding and settlement-route selection, supplier discovery, and order allocation are converted into revenue.

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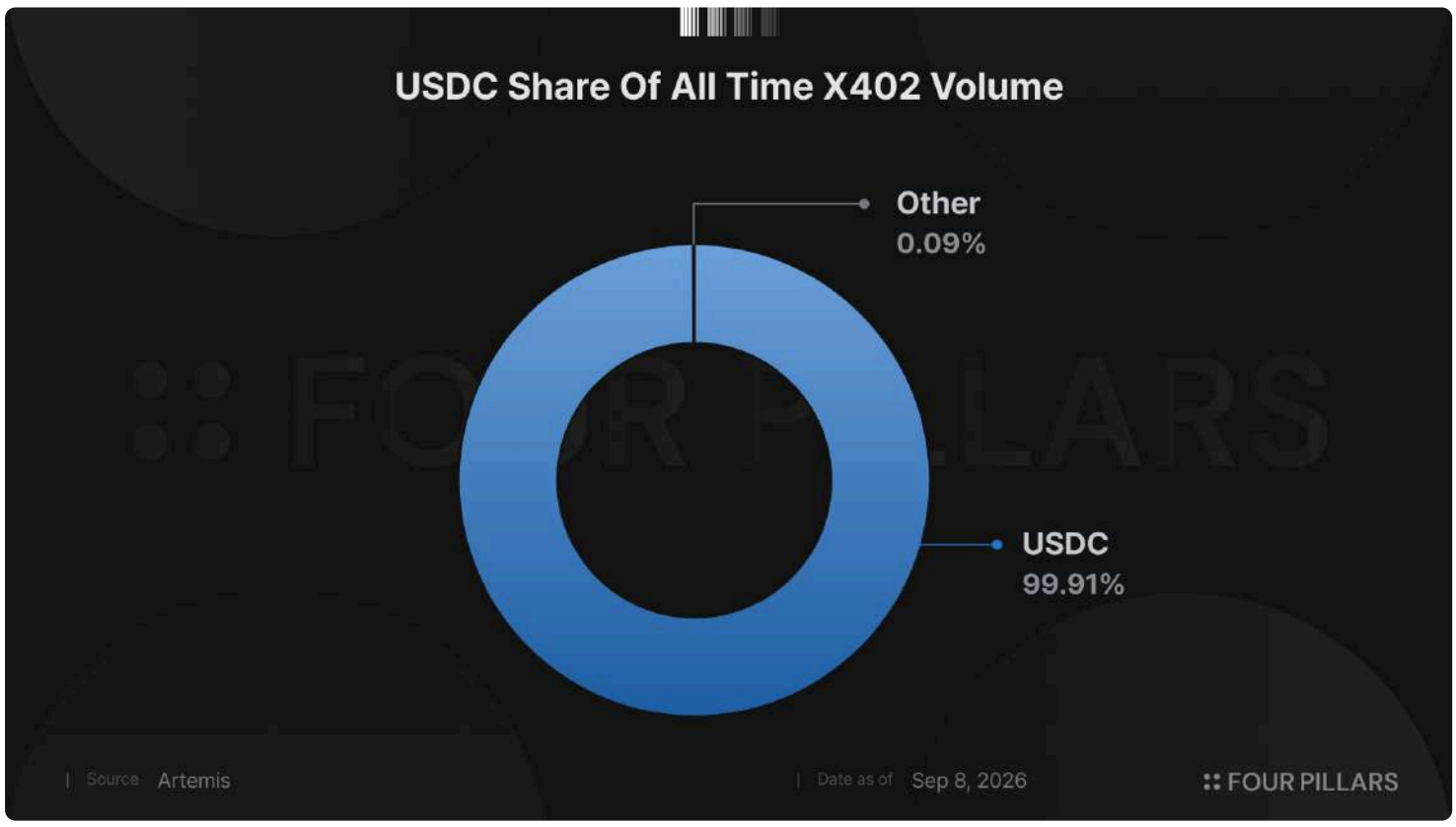
Cautions when interpreting and using the figures

- **Aggregation scope:** The table separately estimates user-paid fees, issuer reserve income, and business revenue under assumed processing shares. Aggregating revenue requires consistent provider boundaries and removal of duplicated pass-through costs and payments between providers.
- **Gross value before costs:** These figures exclude infrastructure and operating costs. Assessing profitability requires additional assumptions about each provider's costs and loss exposure.
- **Baseline dataset and date:** Sections 4.4.2 and 4.4.3 use the Allium data snapshot as of August 4, 2026 to calculate payment unit values and processing counts. The USDC share of payment volume in §4.4.1 uses Artemis observations from the same date.

4.4.1 Stablecoin issuers: The base payment asset and net additional circulating balances

The main value x402 offers a stablecoin issuer is the position of base payment asset. When a stablecoin becomes the reference asset for pricing and settlement, technical integrations and liquidity across wallets, facilitators, discovery services, and other infrastructure also concentrate around that asset. Circle's USDC is the main current example of this market-entry strategy.



Source: Artemis

According to Artemis, 99.91% of cumulative x402 payment volume had settled in USDC as of August 4, 2026. The 99.91% figure is a cumulative observation as of that date, while the 95% used below is a conservative modeling assumption for the dollar-stablecoin share of future T3 payment volume. On the basis of this currency position, Circle has built the Circle Agent Stack, which connects wallets, service discovery, and a nanopayment layer.

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A high payment share does not translate directly into issuer revenue. If one dollar circulates 1,000 times a year, cumulative payment volume reaches $1,000 while the circulating balance remains $1. Only the portion of x402 demand that causes new issuance or delays redemption and thereby increases the average circulating balance provides a basis for reserve interest income.

To estimate how x402 affects circulating balances, the model applies the following key assumptions:

- **Payment-volume scope:** The model uses the T2 ×402 dollar-stablecoin payment forecast calculated in §4.3.2 and treats 95% of T3 payment volume as dollar-stablecoin payment volume.
- **Scheme adjustment:** exact and upto use payer-wallet balances directly, while batch-settlement uses prefunded payment channels. The model recognizes differences in balance creation across schemes but treats the actual scheme mix and replenishment cycles as constrained variables because they cannot yet be measured quantitatively.
- **Effective holding period:** Funds are assumed to remain in the payment path for three days.
- **Effective attribution ratio:** Fifty percent of the total operating balance is attributed to new issuance or delayed redemptions that expand the effective circulating balance.

Scope and assumptions

- Total dollar-stablecoin payment volume = T2 ×402 dollar-stablecoin payment volume + T3 payment volume × 95%
- Average operating balance = annual dollar-stablecoin payment volume × 3 ÷ 365
- Average net additional circulating balance attributable to x402 = average operating balance × 50%
- Annual pre-expense reserve income = average net additional circulating balance attributable to x402 × 3.5%
- Pre-expense reserve income for an individual issuer = average net additional circulating balance attributable to x402 × issuer share × issuer reserve yield

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Stablecoin Issuers: From Payment Flow to Reserve Income

Scenario	Dollar-stablecoin payment volume	Three-day average operating balance	Average net additional circulating balance at 50% attribution	Annual pre-expense reserve income at 3.5%
Gradual adoption	$177.45B	$1.46B	$0.73B	$25.52M
Rapid adoption	$653.66B	$5.37B	$2.69B	$94.02M
High adoption	$1.8242T	$14.99B	$7.50B	$262.38M

Source Four Pillars

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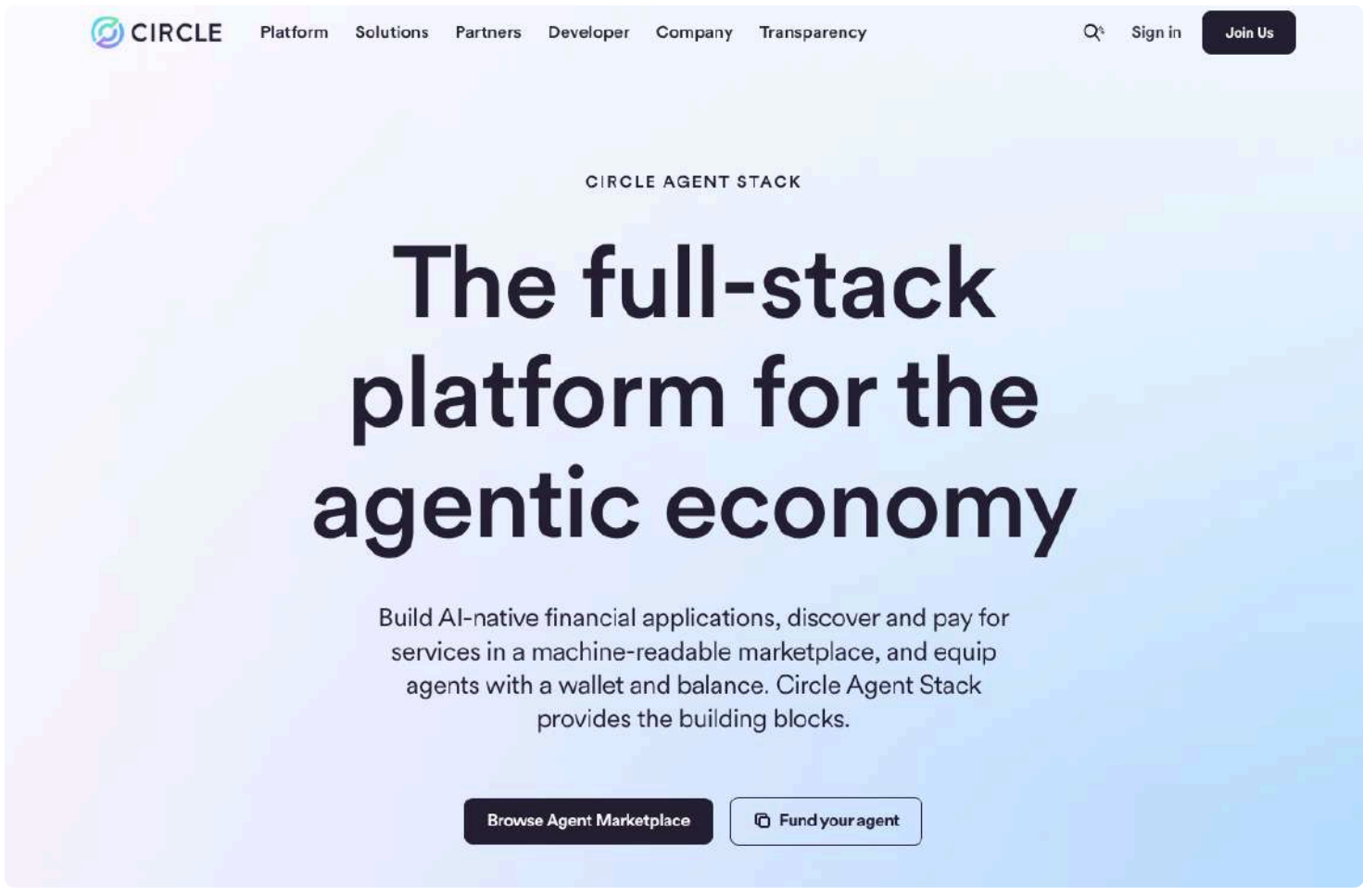
Under the baseline assumptions, the average net additional circulating balance attributable to x402 ranges from $0.73B to $7.50B. Applying Circle's first-quarter 2026 reserve yield of 3.5% produces annual pre-expense reserve income of $25.52M to $262.38M for issuers as a group. Actual attribution depends on each issuer's market share and yield, distribution-partner allocations, and operating costs. The estimate should not be treated as the revenue or net income of an individual issuer.

If the average effective holding period falls to one day, the final estimate falls to one-third of the baseline. If the net additional attribution rate falls to 25%, the estimate is halved. The issuer's economic value therefore depends less on total x402 payment volume than on the rate at which that demand converts into persistent net additional circulating balances. This creates an incentive for the issuer to integrate wallet, discovery, and settlement infrastructure vertically and keep funds within its stablecoin ecosystem, rather than supply only the payment asset.



Source: Circle

The Circle Agent Stack connects budget management in agent wallets, USDC-based service discovery, and gateway nanopayments funded by prepaid USDC in one pipeline. This structure can convert one-time x402 payment demand into a continuously circulating USDC operating balance and delay redemption, thereby helping expand the issuer's effective circulating balance.

4.4.2 Blockchains: Network fees from settlement-equivalent transactions

The economic value associated with a blockchain layer comes from the onchain transactions that record and settle payments, rather than from total payment volume. One $1,000 payment and 10,000 payments of $0.10 produce the same cumulative volume but very different transaction counts and fee demand.

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The model therefore converts T3 payment volume through three steps: payment-size distribution, payment-event count, and onchain settlement method. Case 1 quantifies settlement-equivalent transactions and network fees for Base, while Case 2 separates Solana's currently supported exact path from the draft batch-settlement structure and presents a different calculation method.

- **Case 1: Base settlement-equivalent transactions**

Case 1 calculates settlement-equivalent transactions using Base's user-paid network fees.

x402 Payment Sizes: Transaction Counts Versus Dollar Volume

Transaction size	Share of transaction count	Share of dollar payment volume	Observed average payment by segment
$0.00 to less than $0.01	55.32%	1.43%	$0.0058
$0.01 to less than $0.10	31.47%	7.95%	$0.0569
$0.10 to less than $1.00	12.32%	44.71%	$0.8172
$1.00 to less than $10.00	0.75%	10.18%	$3.0580
$10.00 or more	0.15%	35.72%	$54.7989
Total and overall average	100.00%	100.00%	$0.2251848

Source Four Pillars

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Cumulative transaction data from the Allium x402 dashboard shows a highly asymmetric payment-size distribution. Payments below $0.10 account for 86.78% of transaction count but only 9.39% of dollar volume. Payments of $10 or more account for 0.15% of transaction count and 35.72% of dollar volume.

> Segment shares and aggregate shares are rounded separately, so adding the values displayed in the table may produce slight differences from the aggregate shares in the text. Subsequent calculations use unrounded values.

A single micropayment price therefore cannot convert total payment volume into an event count. The model assumes that this asymmetric payment-size distribution persists in T3 and uses the weighted average of $0.2251848 per payment derived from the August 4, 2026 snapshot.

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Scope and assumptions

- Fixed snapshot as of August 4, 2026: $52.47M in cumulative payment volume and 233,008,604 transactions across 66 weekly observations
- T3 dollar-stablecoin payment volume = T3 ×402 payment volume × 95%
- Observed average payment in each segment = $52.47M × segment share of payment volume ÷ (233,008,604 transactions × segment share of transaction count)
- Intermediate calculations use unrounded source values; only displayed results are rounded
- Paid events by segment = T3 dollar-stablecoin payment volume × segment share of payment volume ÷ observed average payment in the segment
- Annual Base settlement-equivalent transactions = all T3 paid payment events ÷ 500
- Daily Base settlement-equivalent transactions = annual Base settlement-equivalent transactions ÷ 365
- Total fees at Base's average fee = annual Base settlement-equivalent transactions × $0.0057 (BaseScan, August 4, 2026)

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From T3 Payment Events to Base Settlement-Equivalent Transactions

Scenario	T3 dollar-stablecoin payment volume	Annual paid payment events	Annual Base settlement-equivalent transactions	Daily Base settlement-equivalent transactions	Relative to Base's 7.05M daily transactions	Total fees at Base's average fee
Gradual adoption	$171.00B	759.38B	1.519B	4.16M	0.59×	$8.66M
Rapid adoption	$623.44B	2.769T	5.537B	15.17M	2.15×	$31.56M
High adoption	$1.7456T	7.752T	15.504B	42.48M	6.03×	$88.37M

Source Four Pillars

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For Base, the model assumes that high-frequency payments on the EVM path are aggregated through offchain vouchers and then redeemed in a batch under the batch-settlement scheme, reducing settlement cost per request. To compare settlement efficiency, the model converts 500 payment events into one onchain settlement transaction on Base. This ratio is a scenario variable set to reflect both network load and economic efficiency.

The following table presents estimated network fees for each adoption scenario under alternative settlement ratios.

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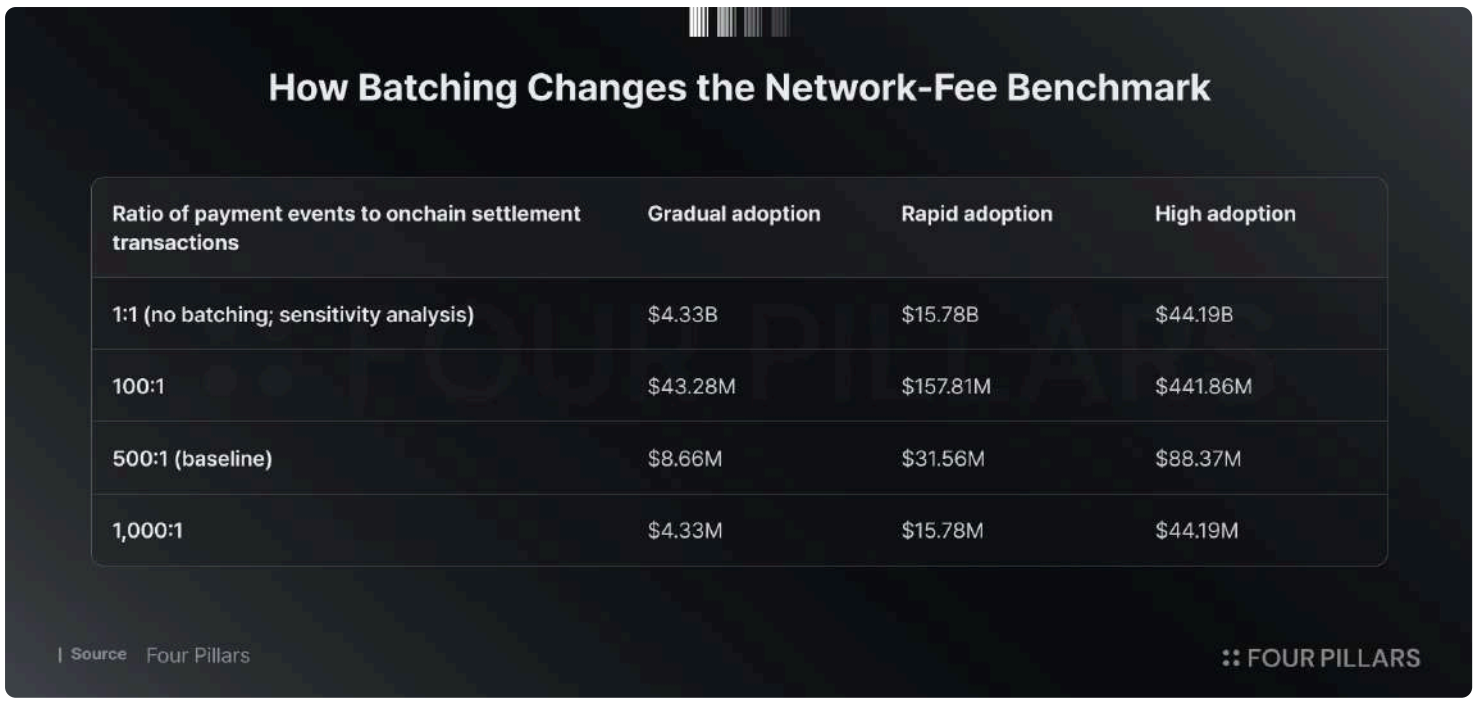

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Because the settlement ratio is an assumption, the results are sensitive to it. As the table shows, lowering the ratio to 100:1 increases estimated fees fivefold relative to the baseline, while raising it to 1,000:1 cuts them in half.

Case 2: Solana settlement structure

Under the exact scheme currently supported by the Coinbase CDP Facilitator on Solana, the buyer constructs and partially signs the token-transfer transaction. The seller or a third-party facilitator then adds the final signature as feePayer and submits it to the Solana network. Each accepted payment maps to one onchain transaction, so network fees for this path equal the number of exact settlements multiplied by the applicable per-transaction fee on Solana.

The draft SVM batch-settlement specification proposes a different settlement path. A client prefunds an escrow channel and signs a cumulative Ed25519 voucher for each paid request. The server verifies and stores the vouchers offchain, then submits the latest voucher to advance the channel's settled watermark and calls distribute to transfer the newly settled amount to payTo.

Under the current draft, a single claim transaction can include the latest vouchers from one to four channels. Individual payment requests therefore do not themselves create onchain transactions, while onchain transactions arise separately at the main stages of the channel lifecycle, including channel opening, top-ups, claims, distributions, and closure.

Accordingly, Solana's annual network fees cannot be calculated by uniformly applying the Base formula of payment events ÷ 500 × $0.0057. They must instead be estimated from the onchain transactions generated across the channel lifecycle. This section therefore presents the accounting model without directly estimating a total amount.

Solana annual network fees = (exact settlements × exact transaction fee) + Σ(channel-lifecycle transactions × applicable transaction fee)

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4.4.3 Facilitators: Monetizing verification throughput and operating functions

Facilitators provide services that verify payment proofs and submit onchain settlements. As card merchants use payment processors, sellers can outsource network connectivity, fee sponsorship, and settlement confirmation. Coinbase's CDP Facilitator is a representative example.

As interfaces become standardized, simple signature verification and settlement submission alone are unlikely to sustain high margins. A facilitator can use extensions to provide integrated enterprise B2B infrastructure, including high-performance bulk processing, key management, compliance, audit logs, and uptime SLA guarantees. This can reduce merchants' operating burden while differentiating the service and supporting higher-value revenue models.

To reflect these differences in structure and function, this analysis distinguishes a basic settlement fee model from a managed service model and applies the following assumptions.

- **Basic settlement fee model:** The model applies Coinbase Developer Platform's published pricing. The first 1,000 onchain facilitator transactions each month are free, each additional onchain transaction costs $0.001, and signature verification is free. The model uses Base as its benchmark and assumes a batch-settlement structure in which offchain voucher verification is free and multiple payment events are aggregated into one onchain transaction. It therefore applies $0.001 to the 500:1 Base settlement-equivalent transactions calculated in §4.4.2.
- **Managed service model:** The model assumes a charge of $0.0005 per payment event for policy checks, verification logs, compliance, and monitoring. This is a scenario variable used to test the economics of a value-added infrastructure service, not an observed market price.
- **Processing share and billing basis:** Both models assume that the facilitator under analysis handles 50% of T3 paid payment events and Base settlement-equivalent transactions. The basic model uses post-batch onchain settlement transactions as its billing base, while the managed model uses individual pre-batch payment events.

Scope and assumptions

- Share handled by the facilitator under analysis = 50% of T3 paid payment events and Base settlement-equivalent transactions
- Base settlement-equivalent transactions = T3 paid payment events ÷ 500
- Managed payment events = T3 paid payment events × 50%
- Basic settlement fees = annual Base settlement-equivalent transactions × 50% × $0.001
- Gross revenue from managed event processing = managed payment events × $0.0005
- Assumed Base network-fee sponsorship = the Base fee benchmark in §4.4.2 × 50%
- Amount remaining after Base network fees = gross revenue from managed event processing − assumed Base network-fee sponsorship

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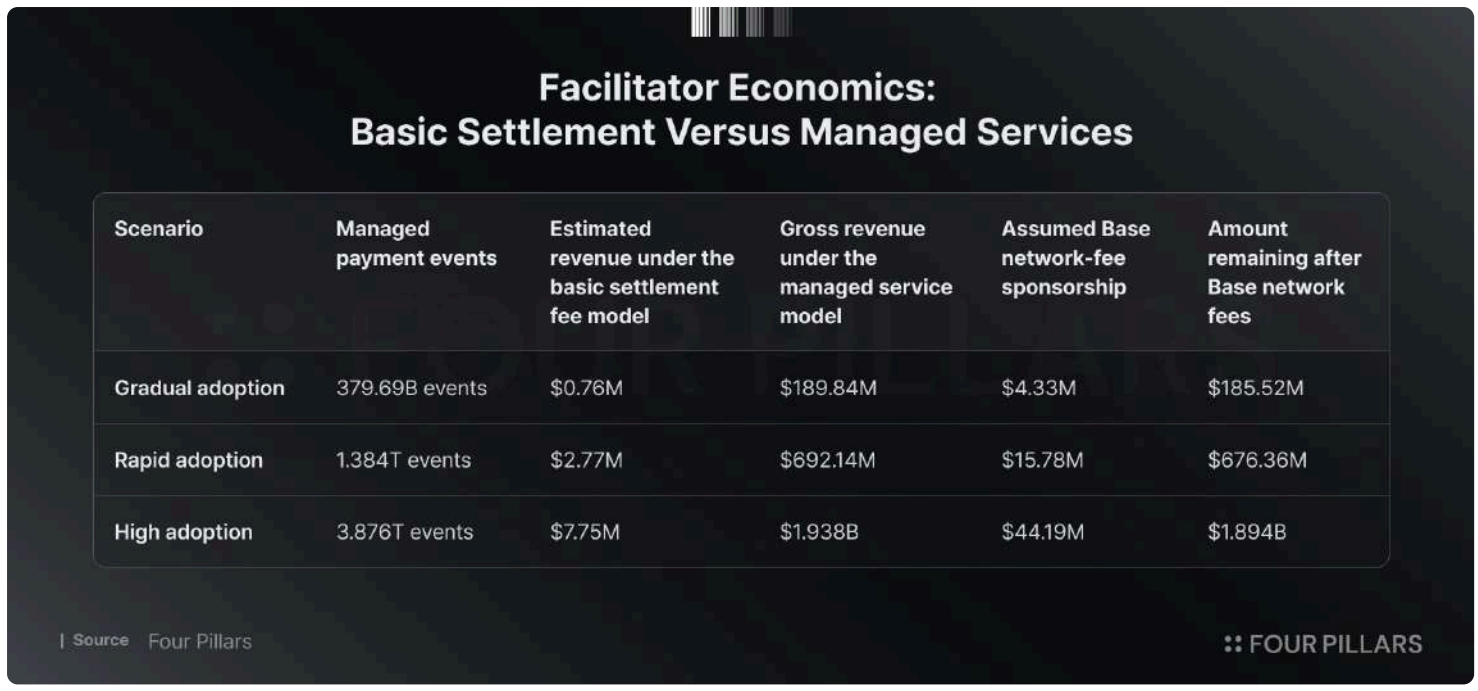

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Under the basic settlement fee model, estimated revenue is $0.76M to $7.75M. As batch settlement becomes more common and more sellers operate their own infrastructure, external facilitator revenue is likely to move toward this model.

Gross revenue under the managed service model is $189.84M to $1.938B. This estimate assumes that policy checks, compliance, audit records, and monitoring are billable for each pre-batch T3 payment event. The two models use different billing units and represent separate business models.

The basic model uses post-batch onchain settlement transactions as its billing base, while the managed model uses individual pre-batch payment events. Better batching therefore reduces the basic model's fee base, while the managed model depends on seller willingness to pay and the actual cost of providing the service.

Both models also use the payment-event counts and settlement-equivalent transactions derived from the chain-mix proxy in §4.4.2. If the future chain mix or chain-specific payment-size distributions change, the billing base for both models will change as well. These figures should therefore be read as scenario values that share the same proxy, not as chain-specific forecasts.

If the facilitator sponsors all Base network fees for the transactions it handles, the assumed cost is $4.33M to $44.19M, leaving $185.52M to $1.894B after network fees. These amounts exclude labor, security infrastructure, compliance, outage credits, failed-payment costs, and customer acquisition costs, so they are not net profit. Maintaining a workable balance between these operating costs and revenue while providing a stable managed service is therefore the central challenge of this model.

Solana requires a separate calculation model that reflects transaction fees, ATA creation costs, priority fees, and which party bears each cost.

A facilitator's long-term competitiveness depends less on raw /verify and /settle throughput than on reliable multichain policy enforcement, key management, settlement-failure handling, and audit accountability. Sellers and platforms can internalize these functions, so independent facilitators will need differentiated operations and pricing rather than settlement submission alone.

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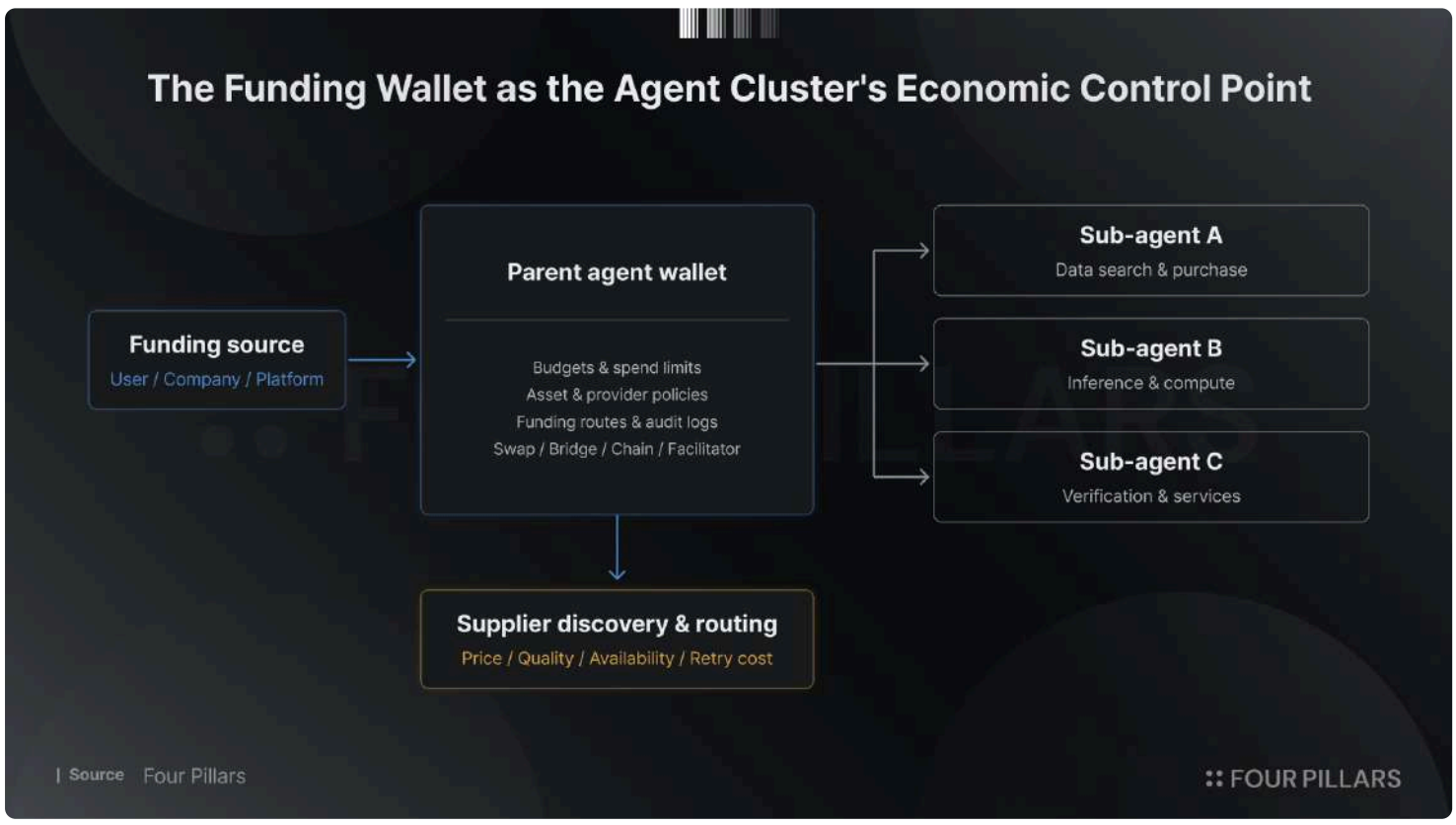
4.4.4 Wallets: Monetizing funding and payment routes



When a company operates multiple agents, it must allocate budgets by task and control which assets may be paid to whom. Policy-based wallets connect several agents' permissions to one funding source or divide budgets among subwallets, checking spending conditions for every payment request. The wallet connects an agent's payment authority with actual use of funds.

The spending limits and permitted-protocol settings in MetaMask Agent Wallet, together with automated x402 payments and support for up to 50 subwallets in OKX Agentic Wallet, illustrate this operating model. Companies can define each agent's permitted spending scope and manage fund use through the wallet.

Once spending conditions have been defined, funds must be prepared for payment. Crossmint supports this process by connecting wallet funding with delegated payment authority. A user funds the wallet and defines its permitted use; the agent then executes x402 payments within the authority granted.

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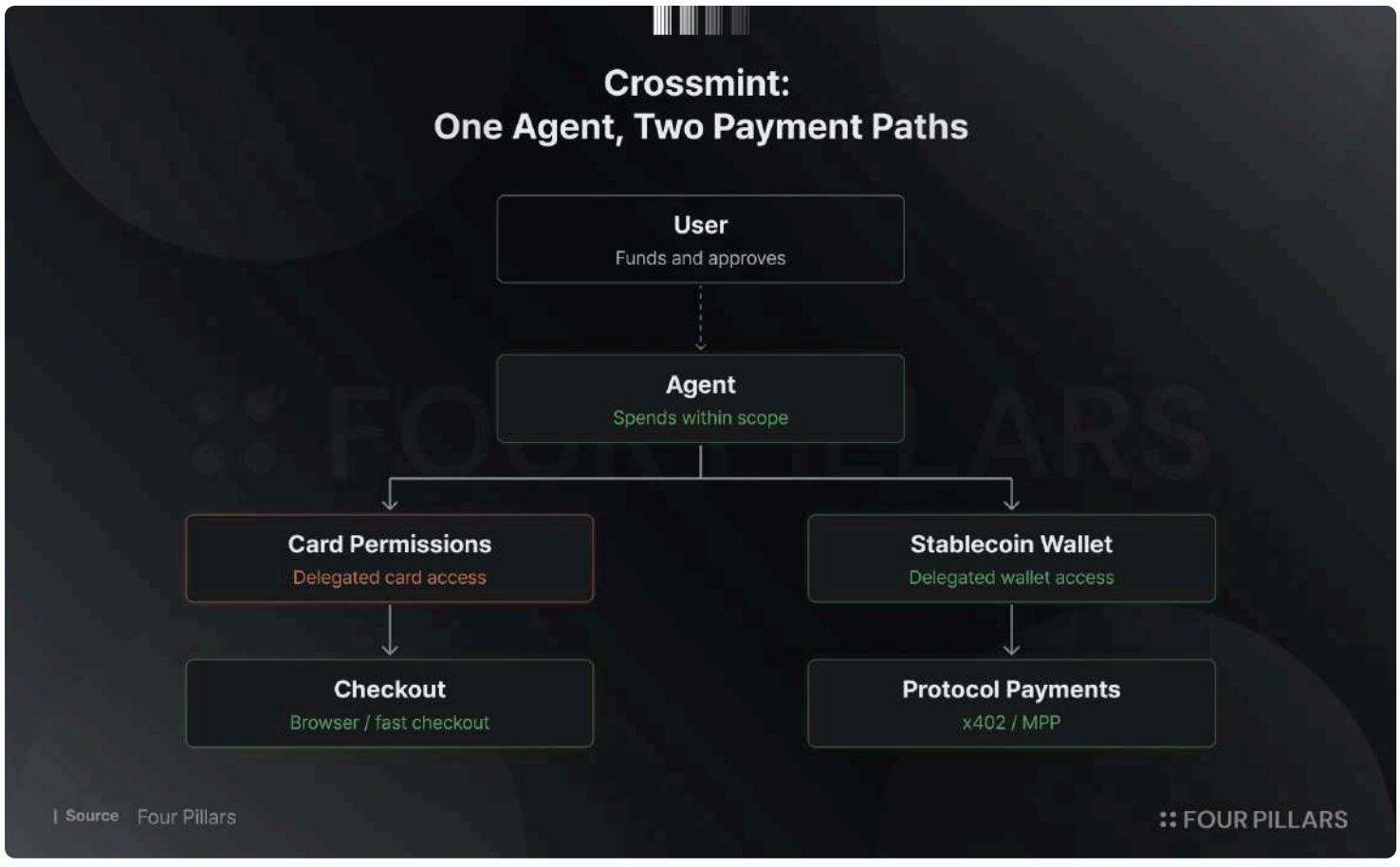

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When wallets support the process from authority management through funding and payment, providers can monetize each function according to its pricing model. Budget controls and audit records can support enterprise subscriptions or usage-based fees, while asset exchange, top-ups, and services for selecting funding and settlement routes can generate transaction fees.

This analysis models **fee-based ancillary-service revenue** that can be estimated from the payment volume in §4.3, treating enterprise subscriptions and data-service revenue separately.

The calculation starts with payment volume routed through managed wallets offering these services. Funding a wallet once and reusing the stablecoins does not incur a funding or exchange fee on every payment. The model therefore **separates service transaction volume that actually incurs fees from total wallet-routed payment volume and applies an average fee rate only to that amount**.

The managed-wallet routing share is set at 40% under gradual adoption, 60% under rapid adoption, and 80% under high adoption. Higher scenarios assume broader enterprise use of wallet providers' fund-management and ancillary services. In each scenario, swaps, bridges, onramps, or services for selecting funding and settlement routes occur on an amount equivalent to 20% of routed payment volume, at an average fee of 25 bp (0.25%).

The routing share, monetization share, and average fee rate are all the author's scenario assumptions. The product examples illustrate wallet capabilities; they are not evidence for those percentages or fee rates.

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Scope and assumptions

- Share routed through managed wallets offering billable services = 40%, 60%, 80% under gradual, rapid, and high adoption
- Payment volume routed through those wallets = total x402 payment volume × managed-wallet routing share
- Monetized transaction volume = managed-wallet-routed payment volume × 20%
- Potential ancillary-service revenue = monetized transaction volume × 25 bp

Wallet Economics: Monetizing Funding and Payment Routes

Scenario	Total x402 payment volume	Managed-wallet routing share	Managed-wallet-routed payment volume	Monetized transaction volume	Potential annual ancillary-service revenue
Gradual adoption	$186.45B	40%	$74.58B	$14.92B	$37.29M
Rapid adoption	$686.47B	60%	$411.88B	$82.38B	$205.94M
High adoption	$1.9161T	80%	$1.5329T	$306.57B	$766.43M

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Under these assumptions, potential annual ancillary-service revenue reaches **$37.29M**, **$205.94M**, and **$766.43M**. Combining the managed-wallet routing share with the monetization share produces effective revenue rates of 2 bp, 3 bp, and 4 bp against total x402 payment volume.

Sustaining this revenue requires a reason for companies to entrust funding and fund movement to that wallet. When agents can compare costs across routes, competitive costs for top-ups, exchange, and selection of funding and settlement routes become important. Integrating budget policies and transaction records can also reduce a company's operating burden; switching wallets then requires considering the cost of migrating those policies and records.

Operating capabilities give companies reasons to choose and keep using a wallet, while ancillary services generate fee revenue from its fund flows. Wallet monetization depends on growth in both routed payment volume and actual use of paid services.

4.4.5 Discovery and routing: Monetizing supplier selection and order allocation

Discovery finds the suppliers an agent can purchase from and the conditions they offer. A router compares candidates against the user's budget and policies, as well as price, quality, and availability, to select the supplier that will actually be called. Discovery builds the set of tradable candidates, while routing allocates individual orders. The two functions may be provided in the same product or split between an application and external services.

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While a facilitator verifies and settles a selected payment, a supplier router decides where to send the order beforehand. This layer influences purchase outcomes through supplier connections and order allocation and provides services that can be monetized in the process.

Discovery and routing structures fall into two groups based on how much of the order flow they internalize.

First, an integrated gateway handles supplier search, credentials, request transformation, payment, and API calls through one route. Because the platform directly intermediates orders and calls, it can generate direct revenue through per-transaction margins, supplier distribution fees, and enterprise B2B contracts.

Second, an open index and router catalogs supplier information and prices and recommends the best route, while the client makes the payment and API call directly to the supplier. This model cannot easily add a markup to every transaction, but it can charge for supplier verification, performance-based routing, policy enforcement, audit logs, and service-level agreement (SLA) guarantees.

The pricing power of both structures depends on their ability to connect supplier information with actual purchase outcomes. They must observe unit prices, call success rates, latency, availability, and result quality among suppliers offering the same service, compare costs including retries, and reallocate orders accordingly. Combining these supplier observations with the agent task evaluations in §4.2.1 provides more evidence for choosing call routes that meet the user's conditions.

As this data accumulates, suppliers gain an incentive to pay distribution costs to attract orders, while buyers use managed routing services to reduce search costs and the burden of failed calls.

This analysis assumes that 30% of total x402 payment volume passes through external discovery and routing services. The remaining 70% comes from precontracted suppliers, catalogs embedded in applications, and direct request routes. Internal supplier selection along those paths is distinguished from payment volume routed through external services.

Of payment volume routed through external discovery and routing, 20% is assumed to generate charges through platform fees, API margins, managed routing, and supplier distribution agreements. An average fee rate of 50 bp (0.50%) is applied to those transactions. Fixed enterprise subscriptions and data-sales revenue are excluded because they do not scale with transaction volume.

Scope and assumptions

- Payment volume routed through external discovery and routing = total x402 payment volume × 30%
- Monetized transaction volume = payment volume routed through external discovery and routing × 20%
- Potential distribution revenue = monetized transaction volume × 50 bp

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Under these assumptions, the potential annual distribution revenue of discovery and routing is $55.93M under gradual adoption, $205.94M under rapid adoption, and $574.82M under high adoption. The blended revenue rate against total x402 payment volume is 3 bp. This applies 50 bp only to the 20% of payment volume routed through external discovery and routing that is actually billable.

Actual revenue is determined by the product of the share routed through external discovery and routing, the monetizable transaction share, and the average rate. As supplier information becomes standardized and agents can move easily among multiple indexes, prices for basic search and listing services may fall. In contrast, a single route that provides supplier identity verification, real-time pricing and availability, call success rates, failure recovery, and enterprise audit functions can support a wider range of charges.

The long-term value of discovery and routing ultimately comes from controlling which suppliers enter the comparison set and where orders are sent. As x402 standardizes payment execution, differentiation through execution alone becomes harder. The accuracy of supplier selection and order-performance data then determine this layer's pricing power.

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Chapter 05

x402 under the Linux Foundation: Governance and Accountability Structure

5.1 The x402 Foundation separates business and technical authority

5.2 Accountability layers compete above a neutral core

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5. x402 under the Linux Foundation: Governance and Accountability Structure

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As discussed in §1.4, x402 moved from a company-led open-source protocol in 2025 to a standardization effort managed through joint governance in 2026. The previous section examined where value is captured in the payment flow. This section addresses control over the specification and the allocation of incident responsibility, both of which become more important as the market grows.

The main issue is not whether the code is public. It is how authority over specification changes, the reference implementation and trademarks, budgets, conformity testing, and incident response is divided.

5.1 The x402 Foundation separates business and technical authority

The x402 Foundation strictly separates business operations from technical decision-making. Legally, the foundation is organized as a Linux Foundation Directed Fund, while the technical project operates as x402, a Series of LF Projects, LLC.

- **Governing Board:** Oversees business operations, including budget execution, membership management, external activities, legal coordination, and conformity-program administration. LF Projects, LLC also owns and centrally manages the official trademarks.
- **Technical Steering Committee (TSC):** Manages specification development, code, release approval, subprojects, security-reporting procedures, and working groups. Under the funding charter, the Governing Board cannot set the direction of the technical project at its own discretion.

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- **Participation and voting structure:** Technical contributions and eligibility to serve on the TSC are separate from membership tiers. Anyone may contribute under the project rules. New TSC members are elected based on their record as contributors or maintainers and require approval from a majority of the existing TSC.

The TSC roster in the official repository currently consists of representatives from Coinbase, Cloudflare, and Stripe. The committee prioritizes consensus and handles matters requiring a vote under the requirements of its technical charter. Its roles and meeting procedures are available in the TSC operations repository.

5.2 Accountability layers compete above a neutral core

A payment system must secure technical neutrality and clear operational accountability together. Several reports offer relevant insights:

- An IMF research note from April 2026 describes the tension between probabilistic AI behavior and payment infrastructure requiring deterministic processing. It examines intent, authorization, and settlement separately, arguing that the performance of autonomous execution depends on institutional design and governance as well as technology.
- March 2026 guidance from the UK's Competition and Markets Authority (CMA) explains that existing consumer-protection duties also apply to businesses using AI. Companies must test systems before deployment, monitor actual performance, and establish procedures to intervene and improve them when problems arise.
- A March 2026 research brief from the Korea Financial Telecommunications & Clearings Institute (KFTC), which operates Korea's interbank payment network, frames agent-payment legal issues through three questions: to which user a payment request should be attributed; under what conditions advance delegation qualifies as an individual transaction instruction; and how responsibility for agents' private keys and wallets should be allocated.

When a person sets a budget and conditions while software executes individual payments, payment networks need records linking the user's delegation to the actual payment request. Limits, purposes, validity periods, revocation, and subsequent records must therefore be designed together. Applied to x402 services, this creates the operational task of tracking wallet authorization policies, facilitator settlement records, and merchant delivery results within the same transaction context.

Building on this discussion, the author proposes an operating structure that manages agent permissions and spending histories together and blocks further execution when anomalies appear. Tracking and revoking redelegated authority can strictly limit the scope of an incident and reduce trust costs across the system.

The following table sets out the operational responsibilities proposed in this report, based on the functions each participant controls. Wallets manage spending authority and cumulative budgets; discovery services and routers manage supplier selection and order allocation; merchants manage resource delivery and access conditions; and facilitators manage verification and settlement.

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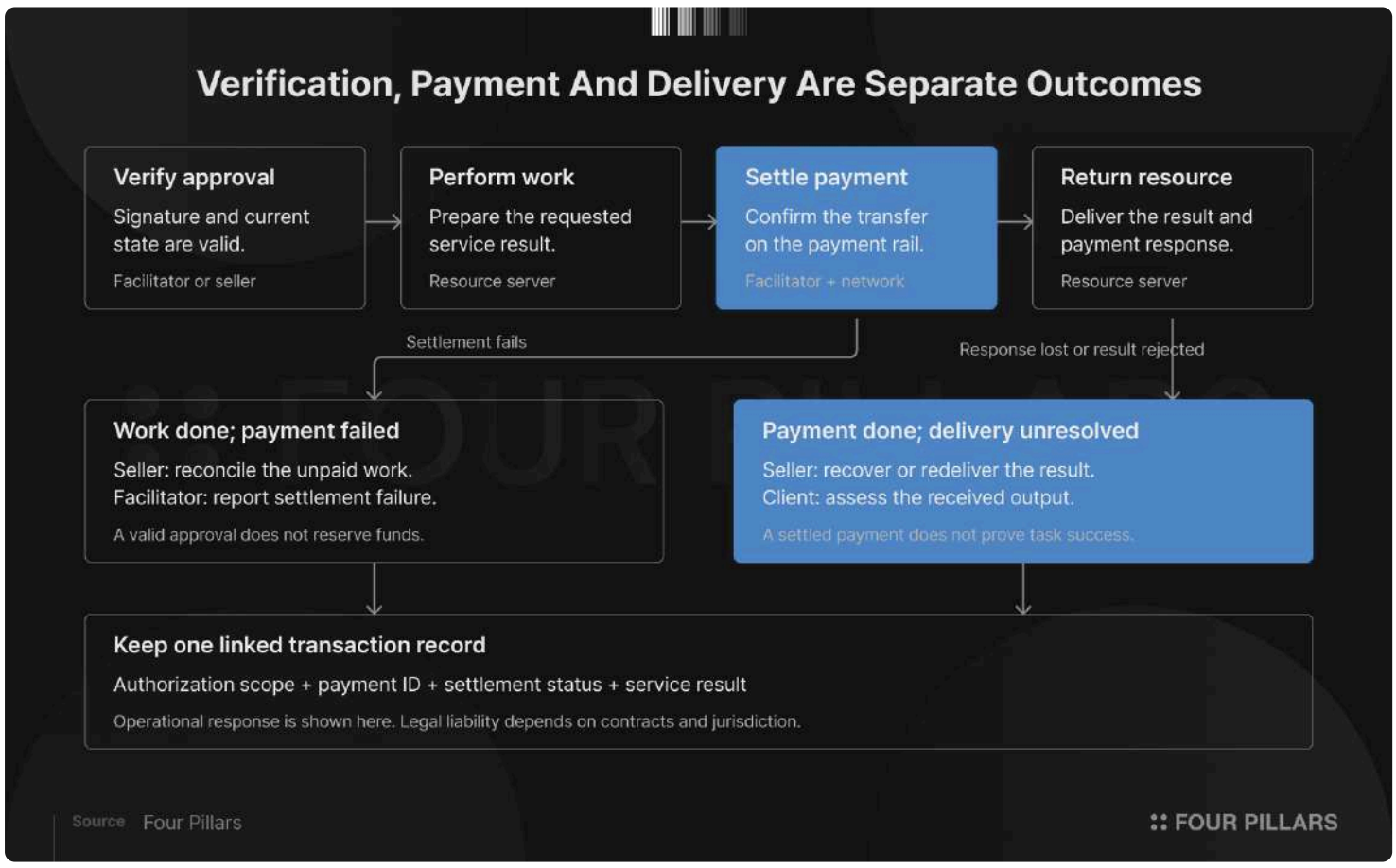
To examine this division of responsibility more closely, consider an agent whose API purchase fails. In this situation, payment-authorization verification, funds settlement, and delivery of the service result must each be checked separately.



First, the server may complete the API work but settlement may fail. The balance or authorization state can change even after verification. The facilitator must therefore return the reason for failure and the actual settlement status, while the merchant reconciles the work already performed with the unpaid balance. Checking the transaction state first reduces the risk of repeating the same work or payment during a retry.

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Second, settlement may complete but the response may be lost or the result may fail to meet the purchase conditions. The merchant handles redelivery of stored results and service fulfillment, while the client evaluates the received result against the user's task requirements. In this process, the payment identifier in §2.3.3 can be used to look up payment and processing records for the same request and decide whether to redeliver the result. The signed transaction records in §2.3.5 can also provide evidence for comparing the agreed conditions with what was delivered.

Using this structure therefore requires linking the scope of delegation, payment identifier, settlement status, and service result in the same transaction record. Operational responses are allocated according to the functions each participant controls, while the final allocation of losses depends on contracts and applicable law. These records can establish the authority under which a purchase was executed and the stage at which it failed.

For example, in high-value B2C and B2B transactions, card networks and existing payment service providers (PSPs) provide tokenization, issuer authorization, and dispute handling. In low-value M2M transactions, specialized facilitators and cloud gateways manage wallet-policy controls, gas sponsorship, retries, and batch settlement. Even with the same x402 interface, the scope of responsibility and cost structure will differ according to transaction size and risk.

In the token-transfer paths described in this report, the blockchain records the execution and order of fund movements. That record alone does not automatically establish the user's purchase intent, actual product delivery, or grounds for a refund—a limitation visible in x402 transaction records onchain. The parties responsible for delivering contracted services and handling exceptions and disputes must therefore be defined alongside those executing payments.

Regulation provides guardrails that allow innovation to accelerate without losing direction. Clear accountability and institutional guidance are needed promptly to turn AI-agent payments from a technical possibility into infrastructure the market can trust.

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Chapter 06

Where infrastructure platforms connect x402

6.1 Cloudflare: Connecting merchant charging with agent spending

6.2 AWS: Bringing payment authority and budgets into agent operations

6.3 Google: Standardizing purchases and approvals while keeping payment methods open

6.4 Fastly: Enforcing access policies along the request path

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An open payment standard needs paths into existing service ecosystems to spread. In practical terms, top-down adoption led by major infrastructure platforms is the most dependable route for introducing new technologies and standards, more so than bottom-up adoption alone.

These companies are not acting out of charity. They have adopted x402 early because they see opportunities to build technical advantages and address pain points in their existing infrastructure. Understanding their rationale and direction is therefore important during the market's formative stage.

Agent developers need wallet management and budget controls; resource sellers need pricing and access controls. Transactions also require a way to verify that a delegated agent acted within predefined conditions. Cloudflare, AWS, Google, and Fastly are combining these requirements with payment infrastructure at different layers:

The four companies differ in how they use x402 through their existing customer touchpoints and the stages they directly control.

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Agentic Commerce Strategies of Major Infrastructure Providers

Comparison criterion	Cloudflare	AWS	Google	Fastly
Existing customer touchpoint	The edge that intermediates web requests, and content sellers	AgentCore execution environments and CloudFront/WAF customers	Commerce participants and Google Cloud API users	Application operators using Compute and the edge
Main stage of control	Merchant pricing and access conditions; buyer wallets and spending policies	Task-specific payment sessions and budgets; merchant access policies for AI traffic	UCP purchase workflows and the exchange of AP2 authorization conditions	Checking requester identity and payment signals; allowing access and forwarding requests
Payments and asset management	Separate account and virtual wallets connected to supported payment rails	Coinbase and Stripe wallet integrations; merchant settlement connected to external facilitators	Purchase and authorization specifications separated from payment execution; Pay.sh connects Solana wallets with API purchases	External payment services handle wallets and settlement; the edge enforces access policies
Availability in announcement materials	Monetization Gateway waitlist and wallet-handle reservations opened	AgentCore Payments preview; WAF monetization available	AP2 and UCP specifications published; Pay.sh API purchasing service announced	Technical series and joint implementation example with Skyfire published
Link to monetization (this report's interpretation)	Connect request processing and wallet operations to paid services for existing customers	Expand cloud usage, including agent execution and WAF; WAF charges no separate fee on content revenue	Broader integration of purchase and authorization specifications, and greater use of cloud APIs for models, data, and other services	Connect demand for policy checks and request processing to edge and Compute usage

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Cloudflare and Fastly connect payments through request paths, AWS through agent execution and security environments, and Google through purchase and authorization specifications. Comparing their adoption scope and revenue opportunities therefore requires looking beyond payment volume to growth in the requests, execution tasks, and API usage each company handles. Let us examine each company's approach in detail.

6.1 Cloudflare: Connecting merchant charging with agent spending

Cloudflare designs merchant charging policies and agent spending policies together. Sellers define resource prices and access conditions, while buyers define agent budgets and counterparties. The former is the role of Monetization Gateway; the latter belongs to Cloudflare Wallets.

Together, the products create a meeting point between buyers and sellers on the agentic web. When the buyer's spending policy matches the seller's access conditions, an agent can request a resource and pay for it.

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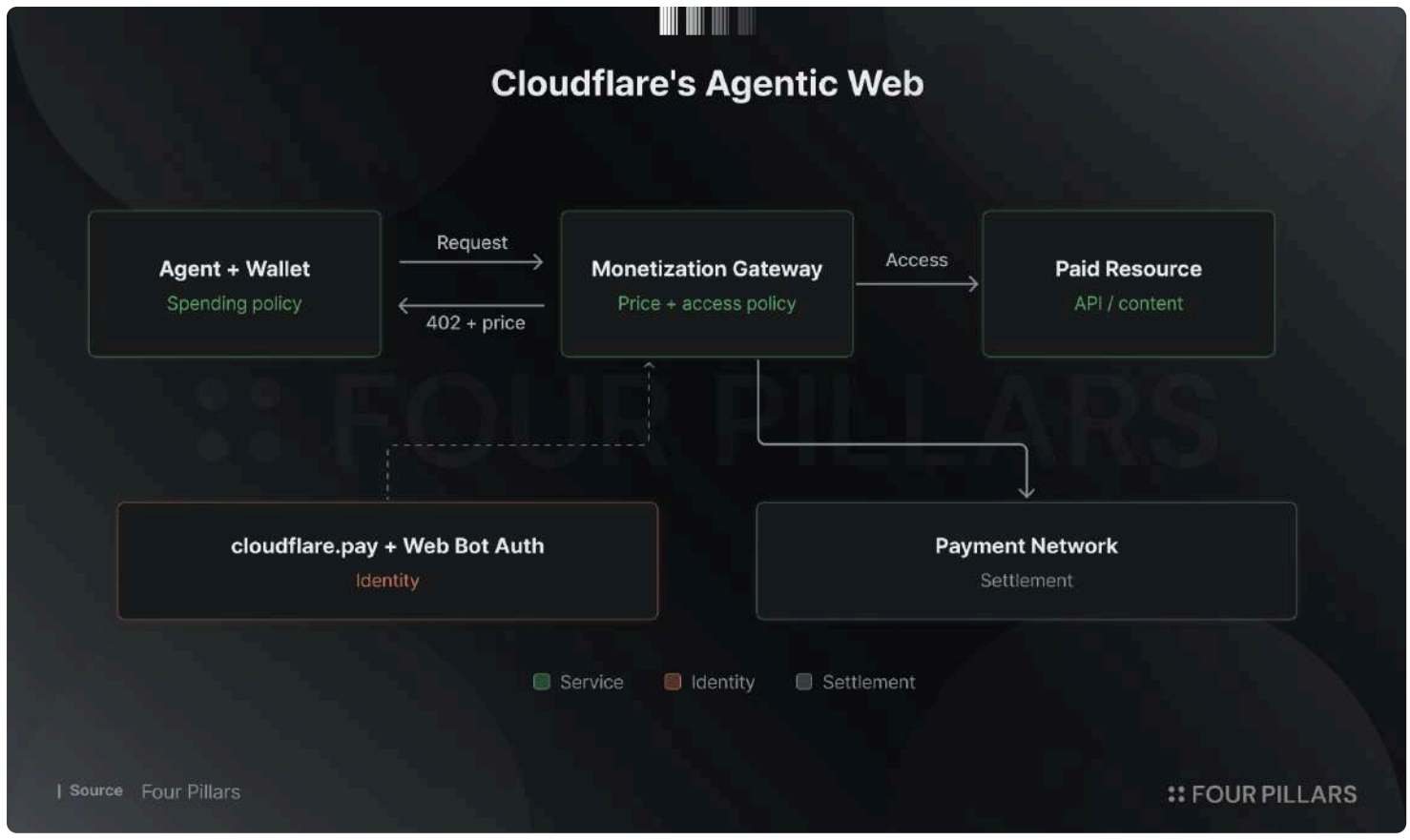

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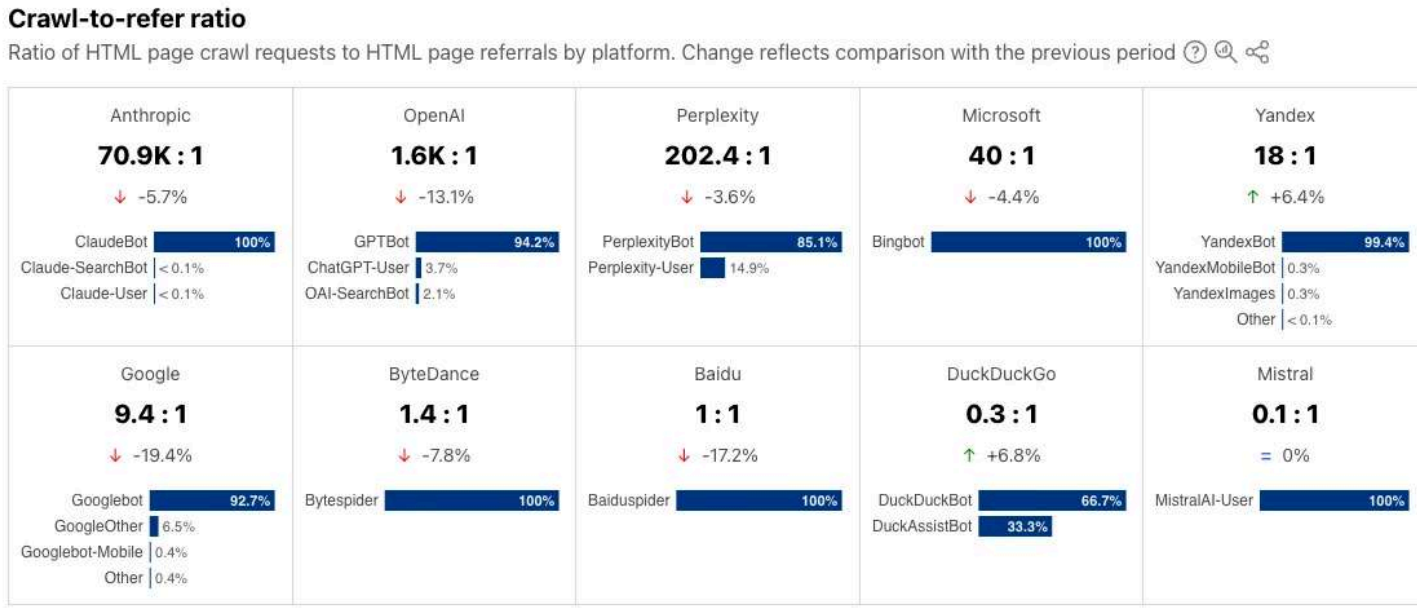
6.1.1 The need for an agent checkout

Behind Cloudflare's particularly active approach to x402 is the imbalance between content collection and referral visits. Its crawl-to-referral metric measures content-collection requests per visit referred to the original website. A higher ratio means fewer visits return to the source relative to the content a platform collects.



Source: Cloudflare

If growing AI data-collection traffic does not produce visits to the original site, content providers face higher server loads while losing advertising exposure and subscription-conversion opportunities. Blocking agents indiscriminately, however, means giving up substantial potential traffic generated at machine speed and future customer relationships. This dilemma creates demand for an agent checkout that charges and settles directly for agents' consumption of data.

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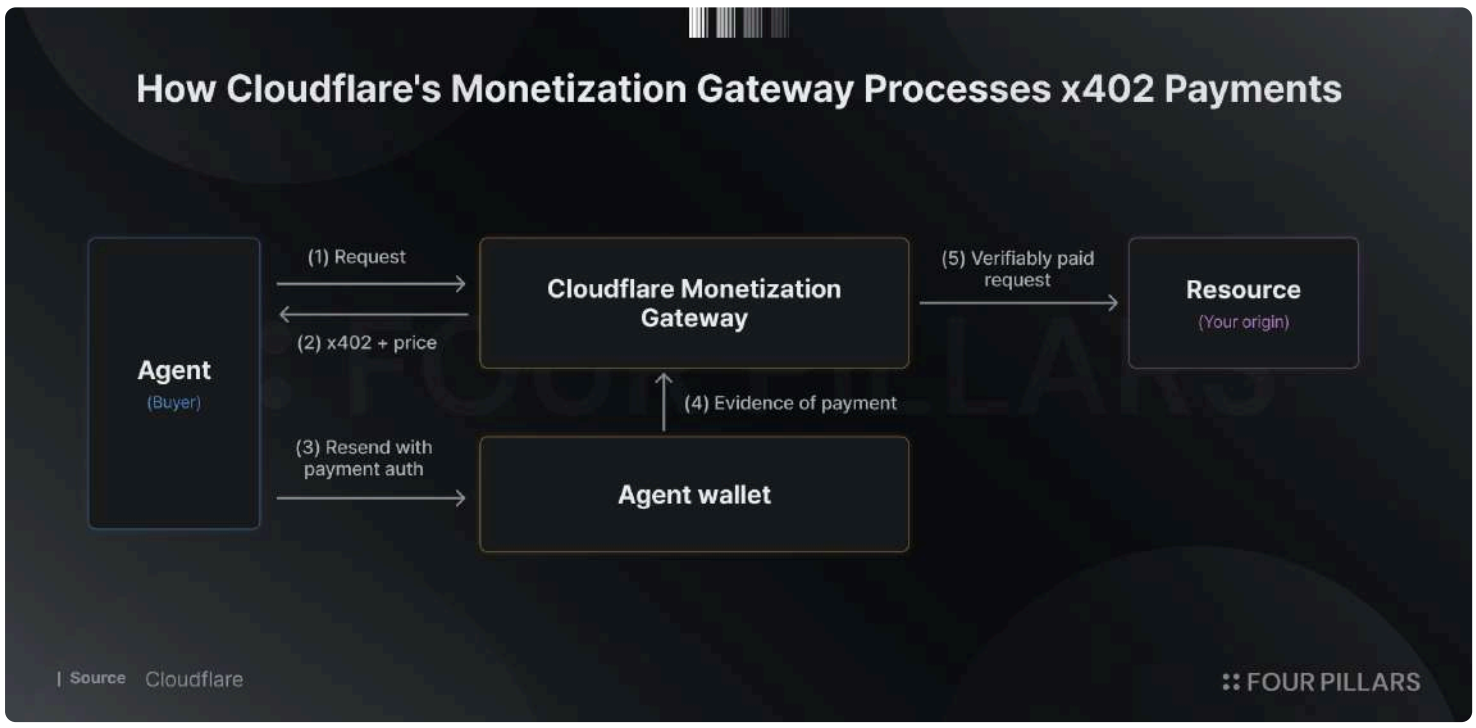
6.1.2 An x402-based Monetization Gateway



Source: Cloudflare

Monetization Gateway, announced in July 2026, is designed to apply pricing and access policies in front of an origin server. Sellers can price particular API paths or resources, and the gateway presents the required payment terms. The buyer repeats the request with authorization data and gains access after the payment conditions are checked.

The edge applies pricing and access policies and connects payment confirmation to resource delivery. Funds move to the seller's wallet through a supported payment network. The merchant's application receives the approved request and provides data or a service.

> The origin is the seller's main server hosting the actual API or data. The edge is the front layer of the network, closer to buyers, that processes requests first.

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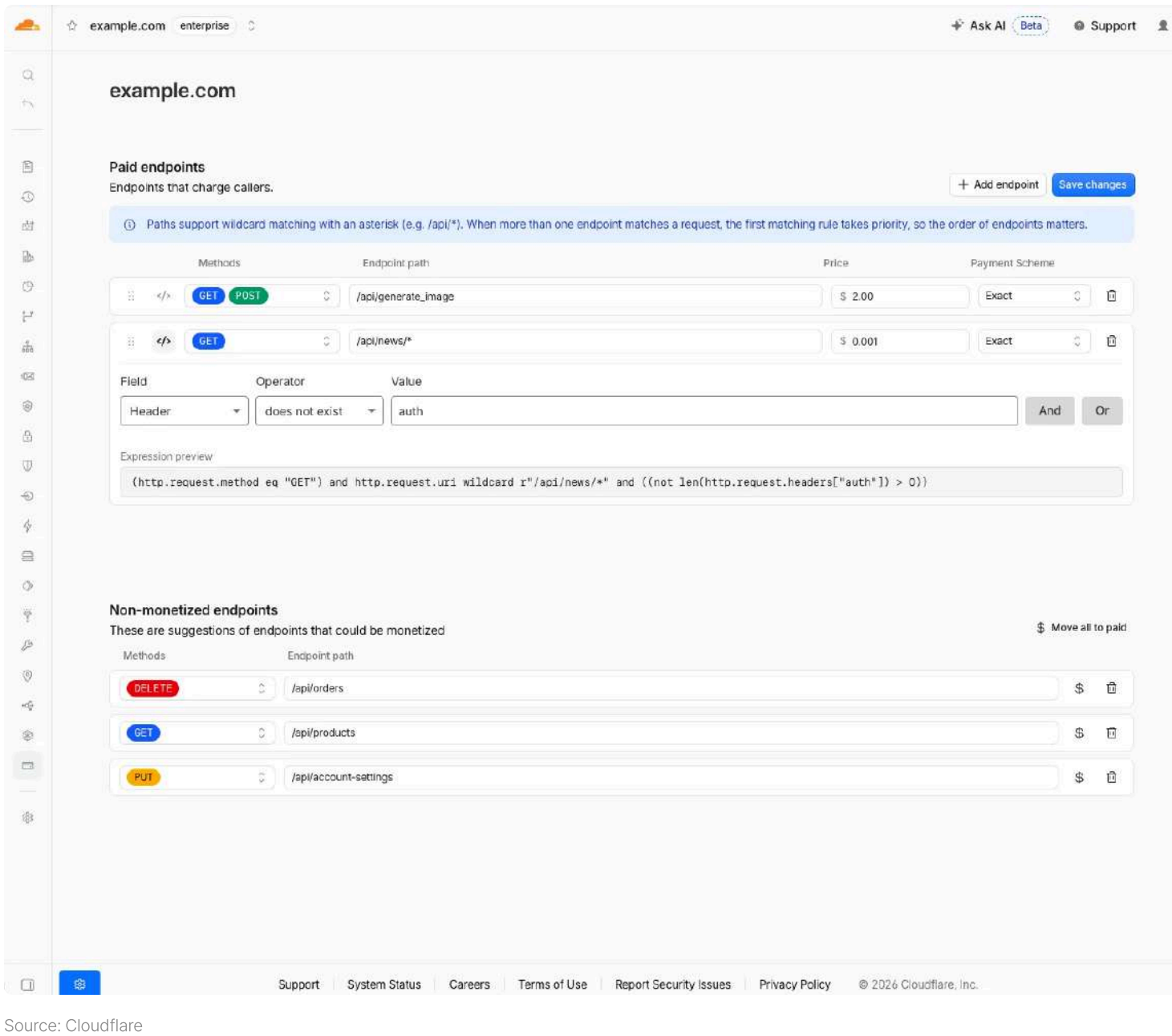

Source: Cloudflare

Cloudflare provides for pricing and access-policy configuration through a dashboard and Terraform. This reduces the work of building buyer-specific prepaid accounts and billing flows separately for each application, making it easier for sellers to define prices, delivery terms, and exception policies.

Sellers can also adapt access conditions to their resources. They can require payment for public paid data and add identity and permission checks for personal accounts or confidential resources.

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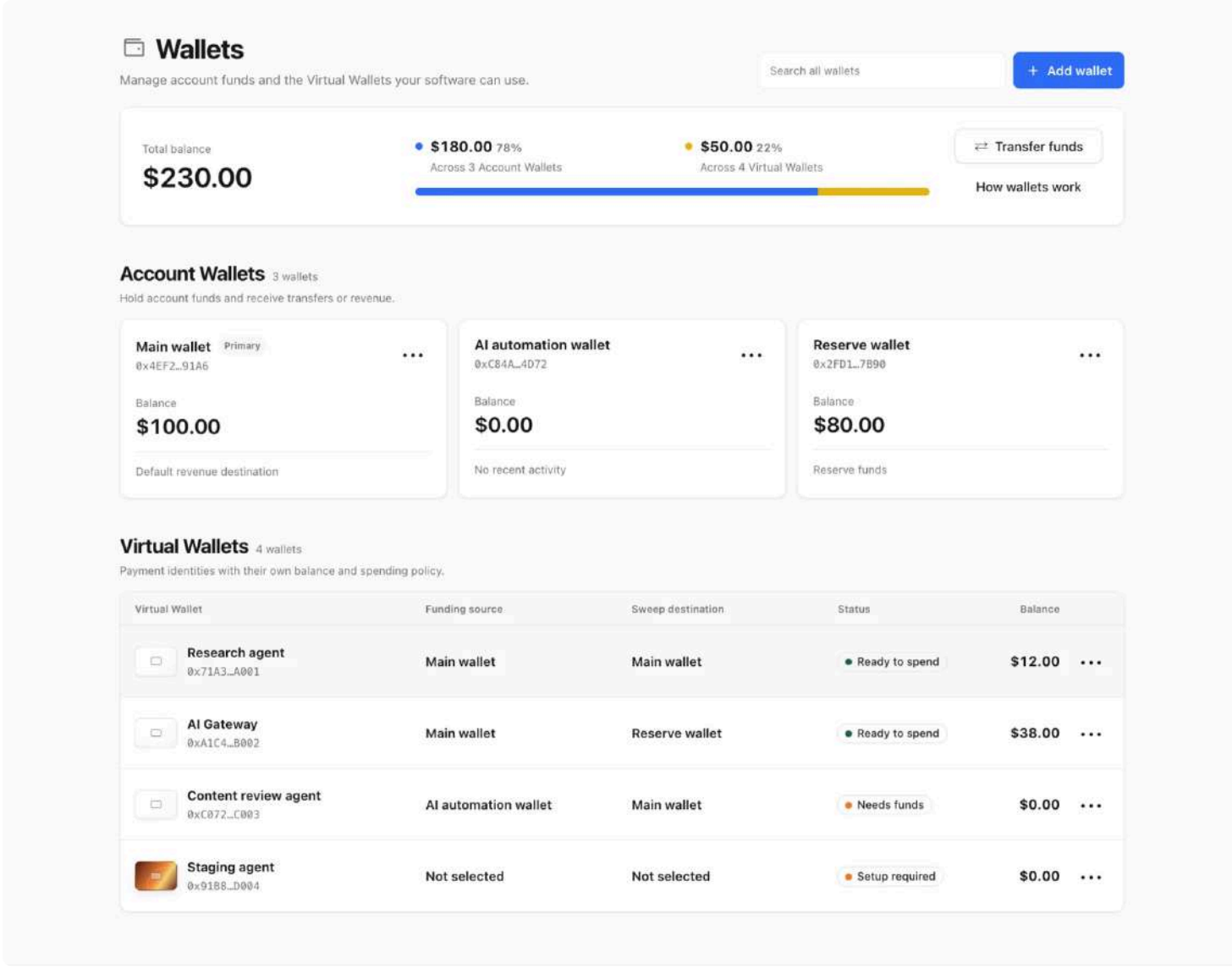

Source: Cloudflare

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6.1.3 Account wallets control funds; virtual wallets spend with limited authority

Merchants' ability to accept payment is insufficient if agents cannot access funds. Wallets announced in August 2026 divide this buyer-side problem between two wallet types. An Account Wallet is the unit in which a person manages funds and policies. A Virtual Wallet is the unit through which an agent spends under those policies.

Account owners add or withdraw funds and grant spending limits to agents. Virtual wallets can specify permitted merchants, spending limits, and maximum amounts per transaction. Agents access virtual wallets with API keys and are designed to spend within the account owner's boundaries.

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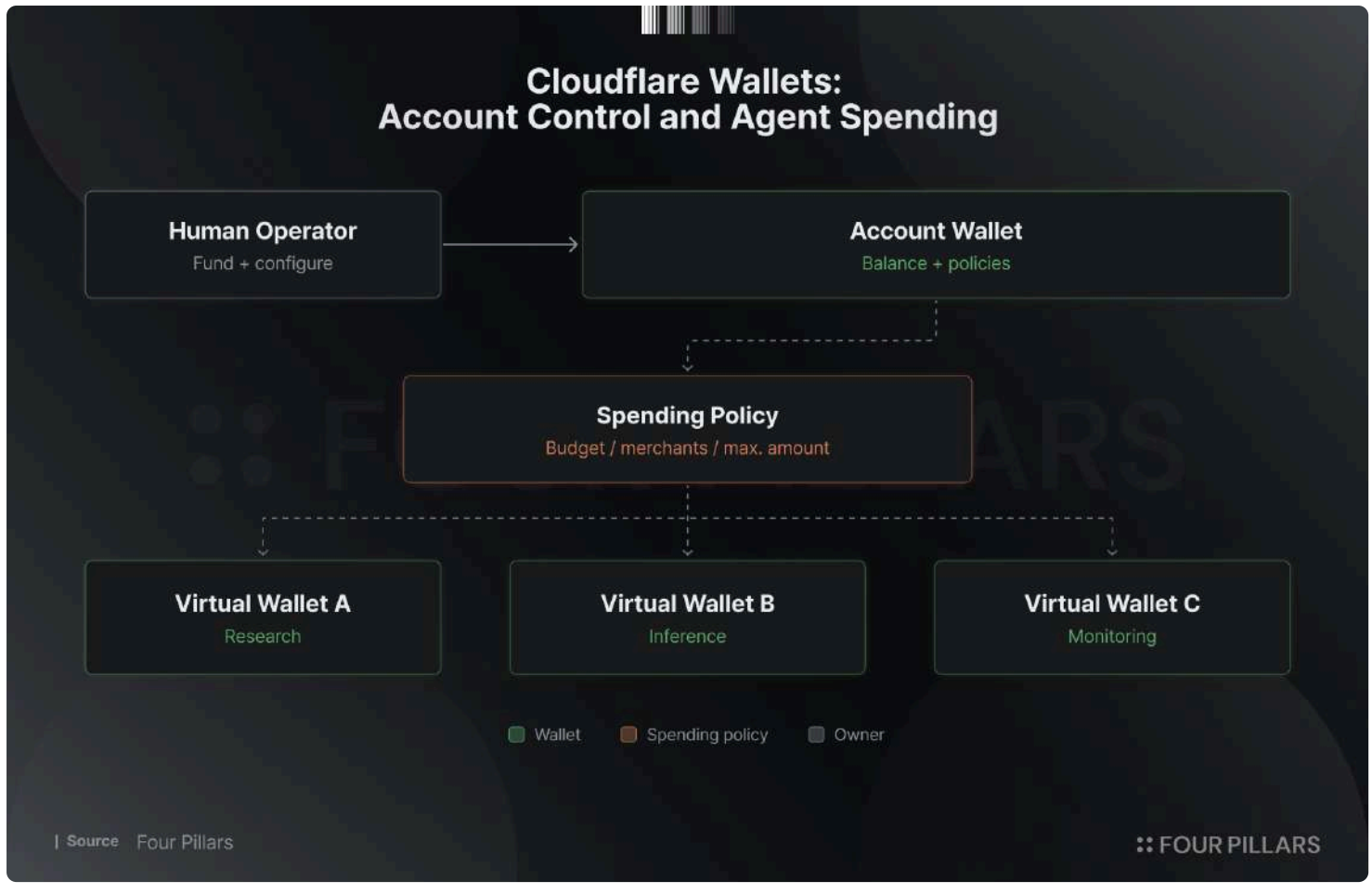

In a hypothetical deployment, a research agent could receive a $10 budget per task and a $0.25 limit per transaction. An inference agent could have a weekly budget and a list of permitted model providers. The operator allocates the purchasing authority each agent needs within one fund-management system.

This is a concrete way to reduce the trust costs discussed earlier. Per-transaction caps limit the cost of a single mistaken purchase, while task budgets limit accumulated spending from repeated calls. Operators can permit autonomous purchasing within an affordable budget.

The wallet value discussed in §4.4 therefore extends beyond asset storage. Allocating budgets among agents and managing permitted counterparties and spending limits can become a basis for operations. A wallet determines what transactions an agent may attempt before payment execution; Cloudflare's programmable wallets illustrate this direction.

The accompanying cloudflare.pay is an identifier through which counterparties reference accounts. Web Bot Auth identifies the requester, wallet policies define permitted spending, and payment networks move the funds. Connecting these functions to the request path allows sellers to verify both the requester and the payment terms.

Buyers can access multiple merchants using virtual-wallet API keys and a common payment interface. This can reduce the work of registering payment methods and connecting billing accounts whenever they add a supplier.

What matters is that these functions connect to existing request paths. Sellers define the conditions for providing a resource, and buyers define the conditions under which agents may buy it. Cloudflare aims to move request infrastructure beyond allowing or blocking traffic to governing the terms on which the two parties transact.

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The July 2026 Monetization Gateway announcement opened a waitlist. The August wallet announcement opened handle reservations, with deposits, withdrawals, and virtual-wallet issuance presented as forthcoming features.

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6.2 AWS: Bringing payment authority and budgets into agent operations

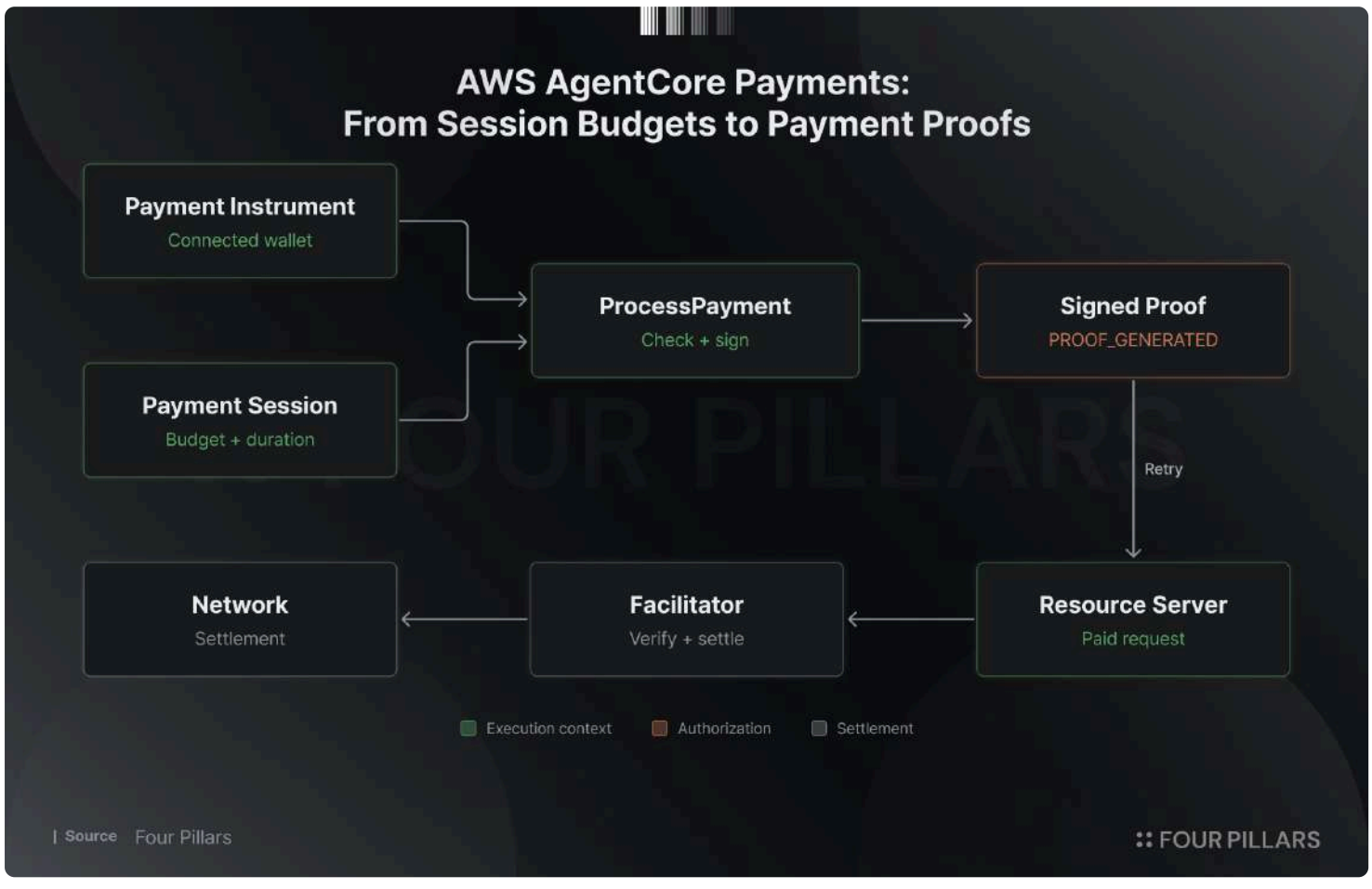
AWS starts with the question of how to connect payments to enterprise-operated agents. Understanding an external API's price does not itself execute a transaction. Operators must determine which wallet to use, how much this task may spend, and how payment requests and results will be recorded.

AgentCore Payments, announced in preview on May 7, 2026, connects Coinbase and Stripe wallet infrastructure to Amazon Bedrock, allowing agent payments to operate with existing identity-management and observability tools. x402 was the initially supported protocol; additional payment standards were presented as a subsequent expansion.

6.2.1 Separating the payment instrument from permission to use it for a task

AWS defines Payment Instruments and Payment Sessions as separate resources. A Payment Instrument represents a connected embedded wallet. A Payment Session defines the period during which that wallet may be used for payment and an optional spending budget.

The same wallet can therefore have different budgets and validity periods for different tasks. AgentCore applies session conditions to each payment request, controlling fund use within the scope assigned to that task.



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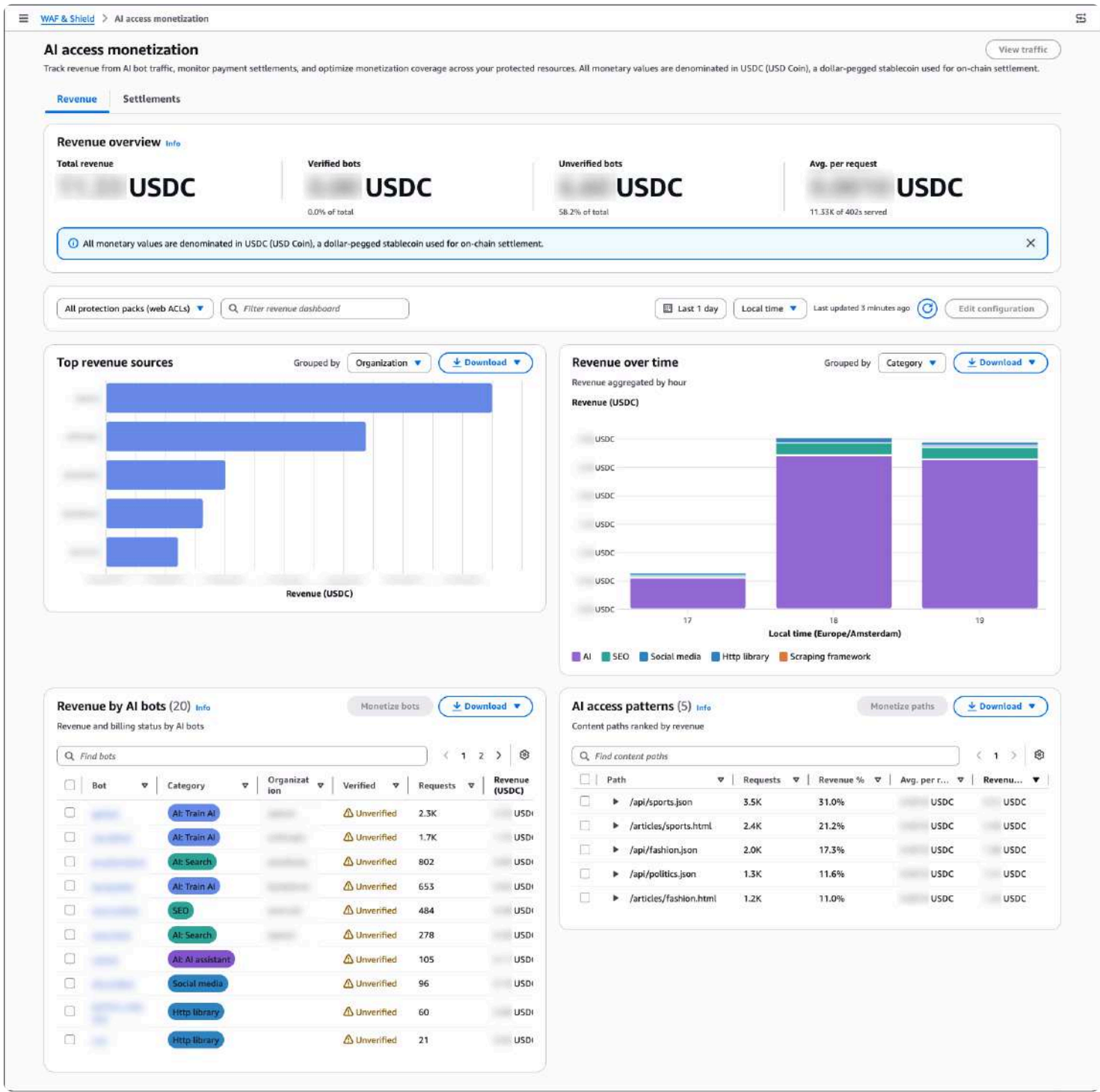
When an agent receives a 402 response, it passes the payment terms, instrument, and session to ProcessPayment. The service checks the request against the configured budget and returns signed payment proof. The client attaches this proof and repeats the request for the paid resource.

Once PROOF_GENERATED is returned, the client includes the generated authorization proof in its repeated request. The merchant side then performs verification and settlement and returns the result. Operators record proof generation, settlement, and resource-delivery states to determine which stage should be retried.

AWS also connects service discovery to payments. By supporting Coinbase's x402 Bazaar MCP server through AgentCore Gateway, it creates a path for agents to discover paid services needed for a task and proceed to payment. This connects the service discovery in §2.3.1 and limited autonomous purchasing in §4.2 within one operating environment.



Source: AWS

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6.2.2 Sellers add pricing policies to existing security configurations

AWS began supporting content-provider charging by adding AI traffic monetization to WAF Bot Control on June 15, 2026. Sellers can define prices and access rules by path and agent type in a Web ACL attached to CloudFront, returning a 402 response when payment is required.

In the announced x402 path, Coinbase's facilitator handles verification and settlement and supports USDC payments on Base and Solana. Buyers can pay through an x402-compatible runtime, so a seller's use of WAF does not require buyers to adopt AgentCore. Each side can adopt the functions appropriate to its operating environment and connect through x402.

WAF integrates external payment services and does not impose an additional fee on content revenue generated by this feature. AWS thus expands the operating scope of its cloud by supporting existing customers' content charging and agents' external-resource purchases.

Where Cloudflare connects merchant access conditions and buyer spending policies along the request path, AWS includes payment instruments and sessions in an agent's task context and supports merchant charging through WAF. Its approach integrates buying and selling functions into existing cloud services and connects them through a common payment standard.

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6.3 Google: Standardizing purchases and approvals while keeping payment methods open

Google starts by expressing purchase procedures and user approvals in a common format. Counterparties must be able to verify not only what an agent intends to buy but also what conditions the user approved. Google's AP2 and UCP exchange commerce and authorization information, while x402 connects payment execution in supported configurations.

6.3.1 AP2 records the conditions approved by the user

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The Agent Payments Protocol (AP2), announced in September 2025, expresses user intent and purchase and payment authorization as signed mandates. Counterparties can use these records to verify what was approved and whether the records have been altered.

When a user approves a transaction directly, authorization is tied to the confirmed product and price. For purchases delegated in advance, conditions such as a maximum price and timing are recorded first, and the agent executes purchases satisfying them. The approval record remains available afterward as evidence of what was permitted.

In the same announcement, Google introduced the A2A x402 extension as a configuration applicable to actual products. AP2 records the authorization conditions; wallets and services verify signatures and validity periods and execute payment through x402. The buyer's delegation thus extends to an individual service's payment request.

This structure represents who authorized a transaction and under what conditions separately from the payment asset. Participants verify the signer and approved terms, then pay through a supported method. Each participant's implementation and operating system handles identity verification, signing-key attribution, actual fund movement, and dispute resolution.

AP2 also supports **Bring Your Own Identity (BYO ID)**, connecting external identity systems to authorization information. The official delegation documentation describes verifying approval and delegation using user credentials from external issuers trusted by the verifier. Connecting Google or another ecosystem's identity system this way lets participants establish who authorized a purchase using identity evidence they trust. They can apply consistent authority controls and payment-approval policies across platforms without separate, fragmented arrangements.

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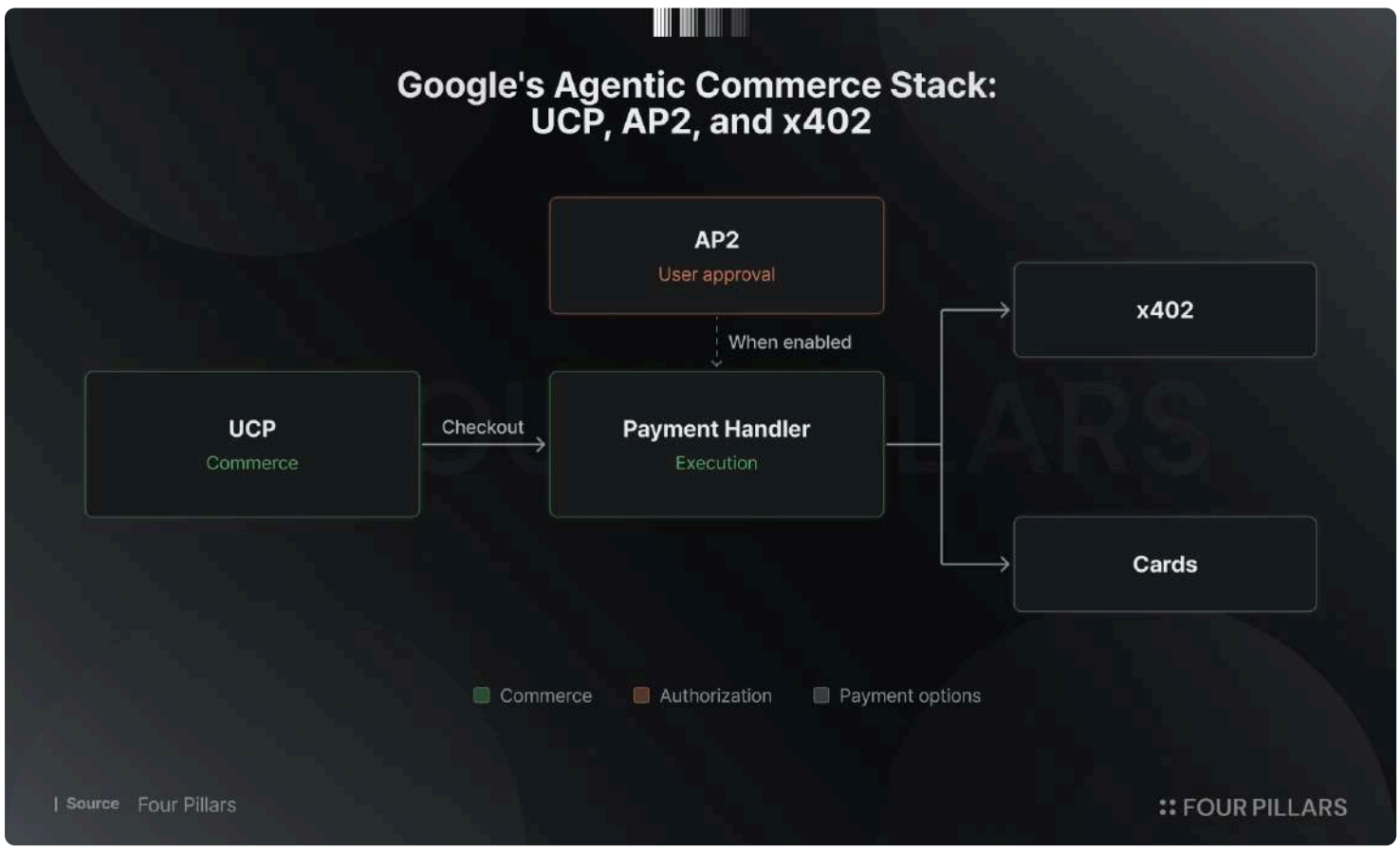
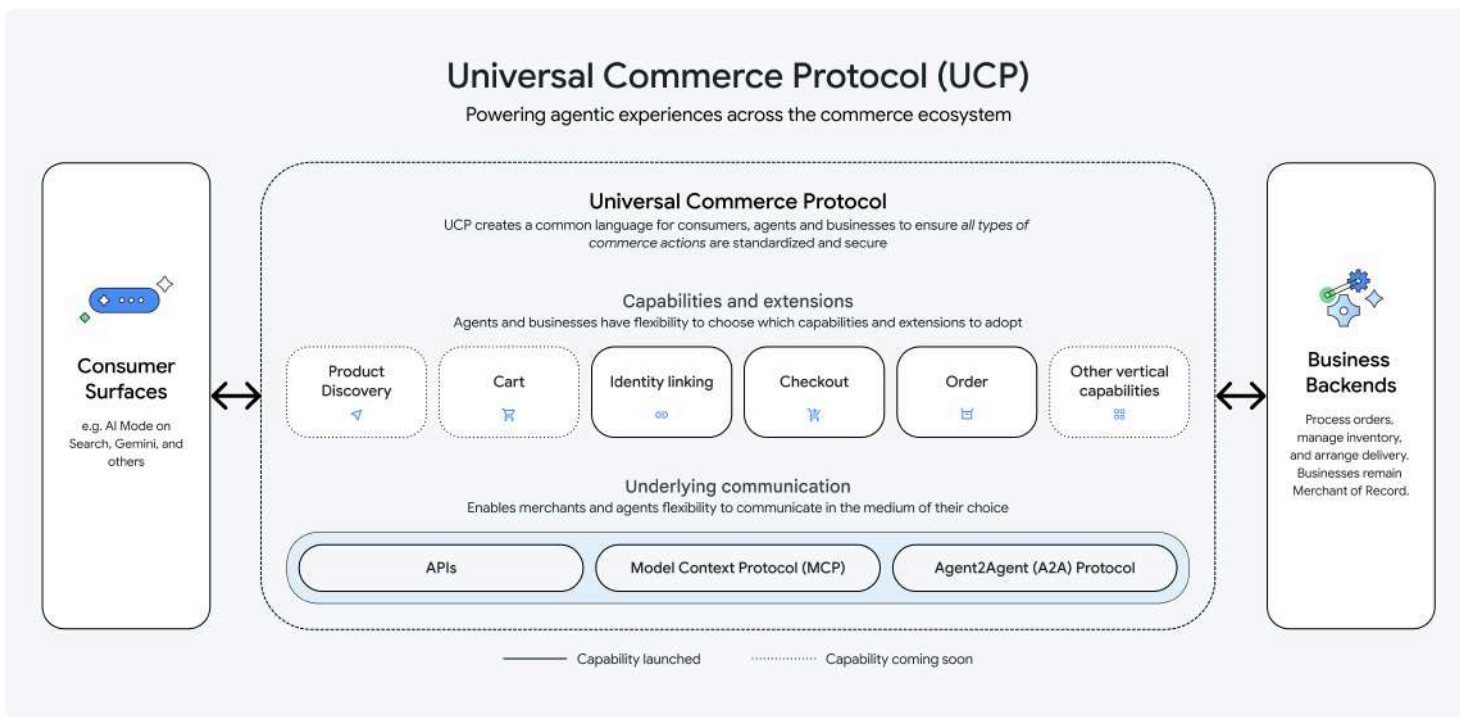
6.3.2 UCP connects purchase procedures and supported capabilities

UCP's AP2 mandate extension is enabled through negotiation between merchants and platforms. Participants decide which extensions to use when starting a session and require the agreed authorization proofs throughout that session.

UCP connects purchase procedures and supported capabilities; an enabled AP2 extension exchanges authorization conditions; and the selected payment method executes the payment. Compatible methods can include cards and x402.





The Universal Commerce Protocol (UCP) is a common standard for exchanging commerce capabilities and checkout state between services. Merchants publish their supported capabilities, platforms inspect them, and both parties determine which purchase procedures and extensions to use.

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Combining UCP's purchase procedures with AP2's authorization information reduces the work merchants and payment providers face integrating different purchase and approval formats for each service. Merchants and wallets supporting x402 can connect payment execution to this procedure, while Google's commerce services offer a distribution path for those integrations.

Pay.sh, announced by the Solana Foundation in collaboration with Google Cloud in May 2026, supports agents' API purchases. Agents can purchase paid API resources, including Google Cloud services, through x402 and MPP. Developers can incorporate resource discovery and payment into an agent's workflow.

The significance is that agents can extend existing developer-oriented cloud services to nonspecialists. The announced Pay.sh API proxy intermediates access and payment for services including BigQuery, Gemini, and Cloud Run. As agents handle discovery, calls, and payment, users may face less work managing each service's accounts and billing systems.

As more users build software with coding agents, procuring the paid resources those projects require can create new cloud demand. Google has an opportunity to expand purchase- and authorization-standard integrations while broadening its cloud user base.

Google's strategy can be interpreted as extending AP2 and UCP to integrate with x402 and connecting those integrations to the use of its AI models, commerce, and cloud services. Common standards help different participants exchange purchase conditions and approval records, while procurement through agents broadens access to existing developer tools. x402 is a flexible payment-execution option within both flows.

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6.4 Fastly: Enforcing access policies along the request path

In a technical series published in July 2026, Fastly described the edge as a policy-enforcement point for agentic commerce. Its focus is checking identity, authority, and payment signals before requests reach an application to determine access, with wallets and payment networks provided externally.

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After applying policy to a request, the edge chooses whether to respond, block, require further checks, or forward it. Incorporating payment confirmation into this decision allows paid-access conditions to be checked before a request reaches the seller's application. The selected external payment service handles wallet operations and settlement.

Connecting x402 terms and authorizations to request policies lets the edge quote prices to unpaid requests and pass authorizations in repeated requests to external verification and settlement services. It applies the returned payment result to the access policy and forwards permitted requests to the resource server.

Here, request forwarding sends a request that has passed access policies to the origin server. The supplier router in §4.4.5 compares price, quality, and availability across sellers to select what to purchase. These functions handle request forwarding and order allocation in the market, respectively.

A joint example with Skyfire illustrates this access-policy approach. Fastly Compute verifies agent-identification credentials and payment signals supplied by Skyfire, then applies the result to service-access conditions.

Fastly helps existing edge customers connect their chosen payment services to request policies. Customers select wallets and settlement providers, while Fastly handles identity checks, payment-condition checks, and request forwarding. As payments are added, the range of policies enforced at the edge expands.

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Chapter 07

Ecosystem perspectives: x402 and agentic commerce

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- **Jing Yi Tan (a.k.a Cookies)** — AI Ecosystem, Monad Foundation
- **Michael Blau** — Founder of dripstack.com
- **Jesse Chung** — Co-Founder of Seal

Q1 In which market will x402 first find its moat?

Kevin Leffew

Real time data access via Data APIs, specifically financial data for trading use cases.

x402 turns HTTP into the tool routing substrate for LLMs. Once an endpoint quotes its own price in-band and accepts payment in the same request, you don't need a platform sitting between a model and the tool it calls: the URL is the tool, the 402 is the price, and the payment authorization is the credential. That's why the first moats we're seeing are forming in GPU inference and data APIs.

Richard Widmann

x402 will likely find its footing in the "code to cloud" revolution. Developers using agents to build applications need to deploy the software they've built - and code generation is only part of that. Providing an agent a way to provision the horizontal cloud tooling required to deploy an application such as Cloud Run, BQ, and storage is a natural first place for agents to leverage x402 primitives without having to go through the traditional sign up flows. This would accelerate any developers ability to ship applications built with their agents - but it is especially impactful for the newer wave of nontechnical developers coding with agents who are not familiar with developer tool flows and credentialing.

Michael Blau

I think agentic investing and trading is one of the most interesting markets. So much of the best financial data is locked behind expensive API subscriptions. "Go pull the best data you can find and help me build an investment strategy" feels like a killer use case for agents.

With x402, you could give an agent one balance and let it pay for the data it needs as it goes. That's a much better UX than subscribing to a bunch of financial data providers, managing API keys, and wiring everything together before you can even ask the question.

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Jing Yi Tan (a.k.a Cookies)

x402's first moat will be agent-to-API micropayments for inputs: inference, data feeds, specialized tools. Traditional account / API-key rails assume a human who creates an account, pass KYC verification, add a credit card into the system and gets API-key access. These are bottlenecks for agents. Given that agents are highly versatile and requires access to various types of APIs, using x402 allows pay-as-you-go when they discover services required and manage cost without having to do monthly / annually subscriptions for each service.

Jesse Chung

Specialized, precise data. Others have given good examples: realtime, business-critical data like a closing price in a trading context. I'd go further. Almost any genuinely precise data is worth paying for, because discovery of that data has never been a properly functioning market. Producing it is difficult, and demand for it has never been expressible at the right granularity. Nobody takes out a subscription to retrieve one number, so the data either gets bundled into something much larger or it doesn't get sold at all. Meanwhile the signal-to-noise ratio of the open internet is extremely low, and at the speed agents operate, every wasted retrieval compounds. Paying a few cents for an answer you can trust beats spending ten times that in tokens and latency looking for it free.

Q2 Which layer will capture sustainable economic value?

Kevin Leffew

Facilitation looks like the obvious value-capture layer today and is probably the weakest one long term. Published facilitator pricing runs on onchain settlement transactions, verification is free, failed settles aren't billed, and under batch-settlement a single onchain transaction can settle thousands of individual payments. So the per-transaction fee base shrinks in direct proportion to how well the ecosystem batches. Facilitation is a competitive operational service with real costs and thin, compressing margins.

The durable position is whoever controls order flow, discovery, routing, and the policy layer deciding which supplier gets called. Verification is a commodity; supplier selection is not. Ranking that weights real economic usage over a rolling window creates a genuine incumbency advantage, because a new endpoint can't buy its way to the top of a usage-weighted index.

Q3 What must be solved first to achieve trust and interoperability?

Raj Parekh

Financial services have always been built off compounded trust that builds over years. X402 has built that trust and support with the x402 foundation members with the types of organizations/enterprises that building projects to support x402 further. As trusted organizations such as Monad Foundation, Cloudflare, Visa, Coinbase, and more start to build products/services to enable x402 support the trust will compound. In fact, x402 foundation further instills a greater trust and neutrality that will help proliferate x402 further as an open source project similar to what Linux has done to software broadly.

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Jesse Chung

Service provider quality. Half the services listed in the directory today are failing or gone. The supply and demand curve hasn't matured yet, and because these are micropayments, there isn't enough revenue behind any individual provider to justify holding a high quality bar.

Both sides have to move together: enough genuinely useful products on the provider side, and clients that actively consume them and drive real volume. That lifts the whole landscape.

Q4 Thoughts on headless merchant endpoints

Kevin Leffew

Headless merchant endpoints are really about enabling client-authorized payments outside of a UX construct. Headless endpoints win on token usage/consumption cost, while also providing a better UX for LLM interfaces.

A traditional browser checkout puts DOM and screenshots in the model's context on every step; a 402 and a JSON schema is a few hundred tokens end to end, and the payment authorization is the credential, so there's no account to open first.

Jesse Chung

Beyond the benefits of redesigning the API surface, headless merchants solve the discovery problem. Every service can be rated individually against the specific need at hand. That shifts the product maturity lever from "builders must ship meaningfully useful features covering a sizable customer base" to "builders provide a narrow, tailor-made feature set, and the customer discovers it on their own."

That's probably the most efficient way to interact with anything. The only reason it hasn't been the norm is that humans have a limited effort-to-gain ratio bounded by time. Agents don't have that constraint, which makes this the natural evolution rather than a workaround.

Q5 Can x402 move beyond its early concentration on Base and USDC to reach true chain and currency neutrality?

Kevin Leffew

Yes, and it's an explicit design goal rather than an aspiration. A payment standard that only works on one chain isn't a standard, it's a product, so the protocol was built neutral from the start: CAIP-2 network identifiers, any ERC-20 on EVM through EIP-3009 or Permit2, any SPL token on Solana. Nothing in the specification privileges one chain or one dollar token, and the Foundation's membership reflects that. Linux Governance firmly establishes credible neutrality.

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Jesse Chung

By design it can, and it may. But that only happens on social consensus. The ecosystem has to want to grow beyond Base and USDC. x402 only exploits the payment rail and the facilitation stack, so the pull of network effects is far weaker than in traditional crypto products. "My money needs to be here" is a weak claim in this context.

Technically, x402 services can live anywhere and settle in any currency. I'm not convinced the general use case inspires neutrality on its own, and pushing for neutrality for its own sake is the crypto mistake we shouldn't repeat.

Q6 Micropayments have failed repeatedly. What grounds are there to believe the agent era will be different?

Kevin Leffew

The standard explanation for why micropayments kept failing is Szabo's: “mental transaction costs”. The barrier was never the fee, it was that deciding whether a $0.002 purchase is worthwhile costs more than $0.002 of human attention. No amount of fee reduction fixes that human cognitive bias, which is why every attempt died at the same place regardless of the rail.

Agents don't have that cognitive bias. They evaluate and execute in milliseconds, they carry no reluctance about small purchases, and their switching cost between two equivalent suppliers is effectively zero. An ideal agent is a homo economicus construct.

Raj Parekh

Micropayments failed historically due to different set of factors and structures. For example, if I bought a song on the Itunes store for 99 cents via my card and then needed to return or claim it was fraudulent the chargeback/dispute process for 99 cent transaction is cost prohibitive for all parties involved.

A new class of services are also now available as pay per use software which enables a new class of pricing options that companies have not been able.

Before, there were three options

- Free tier
- Starter
- Enterprise

Now, with x402, you can add a pay-per-use option, as companies like Soar are doing.

Jing Yi Tan (a.k.a Cookies)

Traditionally, micropayments failed on human UX (metering payments is tedious) and card economics. Agents x x402 tackle both. Agents can manage the various services they are paying for, and stablecoins make sub-cent settlement viable. x402 compresses the payment process from create account → get key → top up credits to just wallet → pay-per-call.

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Jesse Chung

Speed, and agentic token economics. This runs neck and neck with the headless merchant thesis. Agents don't have the bandwidth problem humans do, so their most efficient mode of operation is "do this right now." If the fastest path to done is paying a few cents for the exact information needed at that moment, agents will take it, and the human operator should want them to. If not, why use agents at all?

Q7 What is the largest regulatory bottleneck holding back x402 adoption?

Jing Yi Tan (a.k.a Cookies)

A key constraint is 'who is liable for mistakes' when agents make payments. When an agent overpays, buys the wrong thing or a merchant disputes fulfillment, who is at fault? Is it the user behind the agent, the agent itself, the merchant or the facilitator? Currently there have yet to been regulations around delegated spending authority, with reference to agent wallets funded by users. I'm excited to see how this develops with the x402 adoption growing.

Q8 What are the main metrics that will determine x402's success or failure?

Kevin Leffew

Total Payment Volume, Total Transaction volume

Q9 For nondeterministic agents, in what unit should trust be measured?

Jing Yi Tan (a.k.a Cookies)

Since there is no pre-defined path in which nondeterministic agents take, they should be measured against a policy envelope. This can include the following (not exhaustive):

a) Budget adherence: Did the agent sped remain under the pre-defined cap (how much paid per task, who is being paid)

b) Task success: Was the agent able to achieve the outcome desired by the user (this will likely require human-in-the-loop for grading and improvement of the agent policy itself)

c) Path variance: Does the agent adhere to the policy consistently, or are there volatile periods where spend spikes?

Q10 How are card networks approaching agentic commerce and x402?

Kevin Leffew

All major card providers are participating in the x402 Foundation

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